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GENE ACTIVATION TARGETS FOR ENHANCED HUMAN T CELL FUNCTION

Abstract

Described herein are regulators of T cells as well as methods of modulating such T cell regulators, and methods of identifying new agents that modulate the T cell regulators. Modification of such T cell regulators in lymphoid and/or myeloid cells can provide lymphoid/myeloid cells that can be administered to subjects in need thereof, for example, subject suffering from immune disorders, cancer and other diseases and conditions.

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Background/Summary

PRIORITY [0001] This application claims the benefit of priority of U.S. Provisional Patent Application No. 63/138,841, filed on Jan. 19, 2021, the benefit of priority of which is claimed hereby, and which is incorporated by reference herein in its entirety.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0003] This application contains a Sequence Listing which has been submitted electronically in ST25 format and is hereby incorporated by reference in its entirety. Said ST25 file, created on Apr. 5, 2024, is named “3730183WO1.txt” and is 314,677 bytes in size.

BACKGROUND

[0004] Examples of cellular therapeutic agents that can be useful as anticancer therapeutics include CD8+ T cells, CD4+ T cells, NK cells, macrophages, dendritic cells, and chimeric antigen receptor (CAR) T cells. Use of patient-derived immune cells can also be an effective cancer treatment that has little or no side effects. NK cells have cell-killing efficacy and have several side effects due to not having antigen specificity. Dendritic cells are therapeutic agents belonging to the vaccine concept in that they have no function of directly killing cells and are capable of delivering antigen specificity to T cells in the patient's body so that cancer cell specificity is imparted to T cells with high efficiency. In addition, CD4+ T cells play a role in promoting productive, antigen-dependent immune responses, and CD8+ T cells are known to have antigen specificity and cell-killing function.

[0005] However, most cell therapeutic agents, which have been used or developed to date, have major clinical limitations. For example, cancer cells, on their own, secrete substances that suppress immune responses in the human body, or do not present antigens necessary for production of antibodies against such cancer cells, thereby preventing an appropriate immune response from occurring.

SUMMARY

[0006] Regulators of T cell function are described herein as well as methods of using such regulators. Genome-wide CRISPR activation (CRISPRa) and CRISPR interference (CRISPRi) screens were performed in primary human T cells to identify genetic regulators of therapeutically relevant T cell phenotypes. These screens identified 1074 genes exhibiting significant responses to those phenotypes. The screen identified known genes involved in T cell function, showing that the screens reliably identified genes that actually do affect T cell function. However, the screens also identified novel genes involved in T cell function.

[0007] Methods are described herein that involve ex vivo modification of any of the regulator genes listed in Tables 1-7 or FIGS. 1-4 within at least one lymphoid or myeloid cell, or a combination thereof to generate at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells. For example, the modification can be one or more deletions, substitutions or insertions into one or more endogenous genomic sites of any of the genes listed in Tables 1-7 or FIGS. 1-4. The modification can be reduction of expression or translation of any of the genes listed in Tables 1-7 or FIGS. 1-4. The reduction of

expression or translation can be by an inhibitory nucleic acid (e.g., RNAi, shRNA, siRNA). The modification can be increased expression of any of the genes listed in Tables 1-7 or FIGS. 1-4. For example, the increased expression can be by modification of one or more promoters of any of the genes listed in Tables 1-7 or FIGS. 1-4. The modification can be one or more CRISPR-mediated modifications or activations of any of the genes listed in Tables 1-7 or FIGS. 1-4. The modification can involve transformation of at least one lymphoid or myeloid cell, or a combination thereof, with one or more expression cassettes comprising a promoter operably linked to a nucleic acid segment comprising a coding region of any of the genes listed in Tables 1-7 or FIGS. 1-4.

[0008] The methods can also include administering at least one of the modified lymphoid cells, at least one of the modified myeloid cells, or a mixture of modified lymphoid and modified myeloid cells to a subject.

[0009] In some cases, the method can include incubating the at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to form a population of modified cells. Such a population of modified cells can be administered to a subject. In some cases, the subject can have a disease or condition. For example, the disease or condition is an immune condition or cancer.

[0010] Also described are methods that involve contacting at least one test agent with test cells to provide a test assay mixture, and measuring: [0011] cellular proliferation of the test cells, cytokine release by the test cells, or a combination thereof; [0012] activation of the test cells; [0013] expression or activity of any of the regulators listed in Tables 1-7 or FIG. 1-4 in the cells; or [0014] a combination thereof.

[0015] The methods can also include comparing the measured results to control results. The control results can be results of the test cells measured without any of the test agents.

[0016] For example, the test cells can include lymphoid and/or myeloid cells. Examples of the test cells can include cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.

[0017] The results so measured can be compared to results of a control cell mixture that includes the T cells and test cells measured without any of the test agents.

DRAWINGS

[0018] FIGS. 1A-E. Genome-wide CRISPRa screens for cytokine production in stimulated primary human T cells. (A) Schematic of CRISPRa screens. (B) sgRNA log.sub.2-fold changes for genes of interest in IL-2 (left) and IFN- γ (right) screens. Bars represent the mean log.sub.2-fold change for each sgRNA across two human blood donors. Density plots above represent the distribution of all sgRNAs. (C and D) Scatter plots of median sgRNA log.sub.2-fold change (high/low sorting bins) for each gene, comparing screens in two donors, for IL-2 (C) and IFN- γ screens (D). (E) Comparison of gene log.sub.2-fold change (median sgRNA, mean of two donors) in IL-2 and IFN- γ screens.

[0019] FIGS. 2A-1H. Integrated CRLISPRi and CRISPRi screens map the genetic circuits underlying T cell cytokine response in high resolution. (A and B) Median sgRNA log.sub.2-fold change (high/low sorting bins) for each gene, comparing CRISPRi screens in two donors, for IL-2 (A) and IFN- γ screens (B). (C) Distributions of gene mRNA expression for CRISPRa and CRISPRi cytokine screen hits in resting CD4-T cells (this study). (D) Comparison IL-2 CRISPRi and CRISPRa screens with genes belonging to the T cell receptor signaling pathway (KEGG pathways) indicated in colors other than gray. (E) Comparison IFN- γ CRISPRi and CRISPRa screens with manually selected NF- κ B pathway regulators labeled. All other genes are shown in gray. (F) Map of NF- κ B pathway regulators labeled in (D). (G) Map of screen hits with previous evidence of defined function in T cell stimulation and costimulation signal transduction pathways. Genes shown are significant hits in at least one screen and were selected based on review of literature and

pathway databases (e.g., KEGG and Reactome). Tiles represent proteins encoded by indicated genes, with the caveat that due to space constraints, subcellular localization is inaccurate, as many of the components shown in the cytoplasm occur at the plasma membrane. Tiles are colored according to log.sub.2-fold change Z-score as shown in the sub-panel, with examples of different hits. Large arrows at the top represent stimulation/costimulation sources. (H) Select screen hits with less well-described functions in T cells in the same format as (G). For (H), only significant hits from the top positive and negative ranked genes by log₂-fold change for each screen were candidates for inclusion.

[0020] FIGS. 3A-1. Characterization of CRISPRa screen hits by arrayed profiling. (A) Schematic of arrayed experiments. (B) Comparison of IL-2 (in CD4 T cells) and IFN- γ (in CD8_{sup}.+ T cells) CRISPRa screens, with genes targeted by the arrayed sgRNA panel indicated, as well as their screen hit categorization. Paralogs of arrayed panel genes that were also highly ranked hits are additionally indicated. (C) Representative intracellular cytokine staining flow cytometry for indicated cytokines in control (NO-TARGET_1 sgRNA) or VAV1 (VAV1_1 sgRNA) CRISPRa T cells after 10 hours of stimulation. (D) Intracellular cytokine staining of full arrayed sgRNA panel, showing percent of cells gated positive for indicated cytokines in CD4⁺ or CD8^V T cells. Points represent the mean value of four donors, with and without stimulation. Dashed vertical lines represent the mean no-target control sgRNA control value with stimulation. * $q < 0.05$, ** $q < 0.01$, Mann-Whitney U test followed by q-value multiple comparison correction. Medium-stimulation dose is shown for IL-2 and IFN- γ and low-dose stimulation is shown for TNF- α . (E) Scatter plot comparison of log.sub.2-fold changes in percent cytokine positive cells for arrayed panel sgRNAs versus the mean of no-target control sgRNAs in stimulated CD4_{sup}.+ and CD8_{sup}.+ cells, using the same data from (D). (F) Secreted cytokine staining arrayed panel grouped by indicated gene categories, with sgRNAs targeting IL2 and IFNG genes removed. Points represent a single gene and donor measurement. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, Mann-Whitney U test. (G) Principal component analysis of secreted cytokine measurements resulting from indicated CRISPRa sgRNAs. (H) Heatmap of selected secreted cytokine measurements grouped by indicated biological category. Values represent the median of four donors, followed by Z-score scaling for each cytokine.

[0021] FIGS. 4A-J. CRTSPRa perturb-seq captures diverse T cell states driven by genome-wide cytokine screen hits. (A) Schematic of CRISPRa Perturb-seq experiment. (B) Categorical breakdown of genes targeted by sgRNA library, with the library comprising hits from our primary genome wide CRISPRa cytokine screens as indicated. Genes with a summed log.sub.2-fold change < 0 across both screens (diagonal line) are categorized as negative regulators. (C) UMAP projection of post-quality control filtered restimulated T cells, colored by blood donor. (D) Distribution of CD4 and CD8 T cells across restimulated T cell UMAP projection. Each bin is colored by the average log.sub.2(CD4/CD8) transcript levels of cells in that bin. (E) Restimulated T cell UMAP colored by average cell activation score in each bin. (F) Boxplots of restimulated T cells' activation scores grouped by sgRNA target genes. Dashed line represents the median activation score of no-target control cells. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ Mann-Whitney U test with Bonferroni correction. (G) Restimulated T cell UMAP with cells colored by cluster. (H) Heatmap of differentially expressed marker genes in each cluster. The top 50 statistically significant (FDR < 0.05) differentially upregulated genes for each cluster are shown, with genes that are upregulated in multiple clusters being given priority to the cluster with the higher log.sub.2-fold change for the given gene. The top marker genes by log.sub.2-fold change in each clusters' section are listed to the right. Top overrepresented sgRNAs in each cluster by odds ratio are listed to the next right. Top differentially upregulated cytokine genes in each cluster are listed to the next right. Mean cell log.sub.2(CD4/CD8) cell transcript values in each cluster are shown on the far right. (I) Restimulated T cell UMAP with the expression of indicated genes shown. (J) Contour density plots of restimulated cells assigned to indicated sgRNA targets in UMAP space. The no-target control

contour is shown in grayscale underneath. “Perturbed Cells” represents all cells assigned a single sgRNA other than no-target control sgRNAs.

[0022] FIG. 5 provides In vitro data using the identified hits for T cell cancer therapies.

Description

DETAILED DESCRIPTION

[0023] Methods and compositions are described herein for modulating T cell responses. The T cells can be modulated in vivo or ex vivo. T cells modulated ex vivo can be administered to a subject who may benefit from such administration. Methods are also described herein for evaluating test agents and identifying agents that are useful for modulating T cell functions.

[0024] Regulation of cytokine production in stimulated T cells can be disrupted in autoimmunity, immunodeficiencies, and cancer. Systematic discovery of stimulation-dependent cytokine regulators requires both loss-of-function and gain-of-function studies, which have been challenging in primary human cells. We now report genome wide CRISPR activation (CRISPRa) and interference (CRISPRi) screens in primary human T cells to identify gene networks controlling interleukin 2 and interferon gamma production. Arrayed CRISPRa confirmed key hits and enabled multiplexed secretome characterization, revealing reshaped cytokine responses. Coupling CRISPRa screening with single-cell RNA-seq enabled deep molecular characterization of screen hits, revealing how perturbations tuned T cell activation and promoted cell states characterized by distinct cytokine expression profiles. Together, these screens reveal genes that reprogram immune cell functions.

Modulating T Cell Responses

[0025] Lists of negative and positive regulators of T cells are provided in Tables 1-7 or FIGS. 1-4. Such regulators can modulate gamma interferon (IFN- γ) production, interleukin 2 (IL2) production, cellular proliferation of T cells, or a combination thereof. Any of the regulators of T cells can be used in the methods and compositions described herein. Agents that modulate the listed regulators can also be used in the methods and compositions described herein. For example, to positively regulate T cells one or more expression cassettes encoding one or more positive T cell regulators, one or more agents that increase the expression or activity of such positive regulator, or agents that inhibit negative regulators of T cells can be used. To negatively regulate T cells, for example, antibodies, one or more expression cassettes encoding one or more negative T Cell regulators, one or more agents that increase the expression or activity of such negative T cell regulator, or agents that inhibit positive regulators of T cells can be used. Agents that can modulate the T cell regulators can include expression vectors, inhibitory nucleic acids, antibodies, small molecules, guide RNAs, nucleases (e.g., one or more Cas nucleases), nuclease-dead Cas variants (e.g., dCas9-VP64, dCas9-KRAB), or a combination thereof.

[0026] For example, T cells and other types of cells can be modified ex vivo to increase or decrease any of the T cell regulators listed in Tables 1-7 or FIGS. 1-4, and the modified cells can be administered to a subject that may benefit from such administration. In another example, the expression or activity of any of the T cell regulators listed in Tables 1-7 or FIGS. 1-4 can be modulated by in vivo administration of expression vectors, virus-like particles (VLP), CRISPR-related ribonucleoprotein (RNP) complexes, and combinations thereof that include or target any of the regulators listed in Tables 1-7 or FIGS. 1-4. The regulator nucleic acids, regulator protein, regulator guide RNAs and CRISPR nucleases can be introduced via one or more vehicles such as by one or more expression vectors (e.g., viral vectors), virus like particles, ribonucleoproteins (RNPs), nanoparticles, liposomes, or a combination thereof. The vehicles can include components or agents that can target particular cell types (e.g., antibodies that recognize cell-surface markers), facilitate cell penetration, reduce degradation, or a combination thereof.

[0027] In addition, new agents can be identified by screening methods described herein that include, for example, evaluating assay mixtures containing one or more test agents and a population of T cells after incubation of the assay mixtures for a time and under conditions sufficient for determining whether the test agent can modulate the expression or activities of any of the regulators described herein. In some cases, the assay mixtures can include T cells and other types of cells, for example, other immune cells such as those that can interact with T cells. Useful test agents identified by such methods can, for example, increase or decrease the expression or activities of any of the regulators listed in any of Tables 1-7 or FIGS. 1-4.

[0028] Hence, any of the regulators of T cells, as well as agents that can modulate those regulators (i.e., modulators), can be used in the methods and compositions described herein.

[0029] The T cell regulators were identified by detecting altered IL-2 cytokine production, 1IFN- γ production, and cell proliferation of T cell receptor (TCR) stimulated primary T cells isolated from two different donors that were subjected to CRISPR-mediated genetic modification. Both positive and negative regulators of T cells were identified.

[0030] The agents that can modulate T cells or the T cell regulators described herein can be expression systems encoding a regulator or modulating agent, antibodies, small molecules, inhibitory nucleic acids, peptides, polypeptides, guide RNAs, cas nucleases (e.g., a cas9 nuclease), nuclease-dead cas variants (e.g., dCas9-VP64, dCas9-KRAB), and combinations thereof. Examples of such agents are described hereinbelow.

[0031] The regulators and/or the agents that modulate the regulators can be evaluated by various assay procedures. Such assay procedures can also be used to identify new T cell regulators. In some cases, the assay procedures can be used to evaluate the utility of a type (positive or negative effect), quantity, or extent of a regulator or modulating agent activity on T cell activity or T cell numbers.

[0032] For example, the methods for evaluating Applicants' regulators/agents or new regulators/agents can involve contacting one or more T cells (or a T cell population) with a test agent to provide a test assay mixture, and evaluating the test assay mixture for at least one of:

[0033] Detecting and/or quantifying cytokine (e.g., interferon- γ , (IFN- γ , interleukin-2(IL-2)) production; [0034] Quantifying the numbers of T cells within the test assay mixture; [0035] Detecting proliferation via quantification of a dye that dilutes with cell divisions; [0036] Detecting whether T cells in the test assay mixture express one or more of the positive or negative regulators described herein; [0037] Quantifying the number of cells that express one or more of the positive or negative regulators expressed by a population of T cells; or [0038] A combination thereof.

[0039] The T cells or T cell populations that are contacted with the test agent/test regulator can also include a variety of lymphoid and/or myeloid immune cells. For example, test agents can be introduced into an assay mixture that contains cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof,

[0040] Test agents that exhibit in vitro activity for modulating the T cells or for modulating the amount or activity of any of the regulators described herein can be evaluated in animal disease models. Such animal disease models can include cancer disease animal models, immune system disease models, or combinations thereof.

Positive T Cell Regulators

[0041] The following genes are positive regulators of T cells as detected by interferon- γ production (see Table 1): APOBEC3C, APOBEC3D, APOL2, ASB12, BACE2, BCL9, BICDL2, C15orf52, C1orf94, CD2, CD247, CD28, CNGB1, CTSK, DEAF1, DEF6, DEPDC7, DKK2, EMIP1, EOMES, EP300, FLT3, FOSL1, FOXQ1, GINS3, GLMN, GNA11, HELZ2, -RASLS5, IFNG, IL1IR, IL9R, KLHDC3, KLRC4, LAT, LCP2, LDB2, LTBR, MVB12A, NBPFF6, NIT1, NLRC3, ORCI, OTUD7A, OTUD7B, PIK3AP1, PLCG2, PRDMI, PRKD2, PROCAI, RELA, RNF217, SAFB2, SLC16A1, SLC5AI0, SLC7A3, SPPL2B, TAGAP, TBX21, TMEM150B, TMILGD2,

TNFRSF12A, TNFRSF14, TNFRSF1A, TNFRSF1B, TNFRSF8, TNFRSF9, TOR1A, TPGS2, TRADD, TRAF3TP2, TRIM21, VAV1, WT1, ZNF630, and ZNF717. Example 2 provides additional positive regulators T cells that were detected by interferon- γ production. [0042] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases, which are incorporated by reference.

[0043] A few examples of protein sequences encoded by some of the genes detected as positive regulators of T cells by interferon- γ production are provided. For example, an amino acid sequence for the protein encoded by the human BICDL2 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. A1A5D9, shown below as SEQ ID NO:1.

TABLE-US-00001		10	20	
30	40	50	MSSPDGSPFP	SGPLSGGASP
SGDEGFFPFV	LERRDSFLGG	GPGPEEPEDL		60
70	80	90		100 ALQLQQKEKD
LLLAAELGKM	LLERNEELRR	QLETLSAQHL	EREERLQQEN	
110	120	130	140	150
HELRRGLAAR	GAEWEARAVE	LEGDVEALRA	QLGEQRSEQQ	DSGRERARAL
	160	170	180	
190	200	SELSEQNLRL	SQQLAQASQT	EQELQRELDA
LRGQCQAQAL	AGAE LRTRLE		210	
220	230	240		250 SLQGENQMLQ
SRRQDLEAQI	RGLREEVEKG	EGRLQTTHEE	LLLLRRERRE	
260	270	280	290	300
HSLELERARS	EAGEALSALR	RLQRRVSELE	EESRLQDADV	SAASLQSELA
	310	320	330	
340	350	HSLDDGDQGG	GADAPGDTPT	TRSPKTRKAS
SPQSPPEEI	LEPPKKRTSL		360	370
380	390		400 SPAEILEEKE	VEVAKLQDEI
SLQQAELQSL	REELQRQKEL	RAQEDPGEAL		410
420	430	440		450 HSALSDRDEA
VNKALELSLQ	LNRVSLERDS	LSRELLRAIR	QKVALTQELE	
460	470	480	490	500
AWQDDMQVVI	GQQLRSQRQK	ELSASASSST	PRRAAPRFSL	RLGPGPAGGF
LSNLFRRT				

A cDNA and a chromosomal sequence encoding the BICDL2 protein is available from the NCBI database as accession no. AL833749 and ACI08134, respectively.

[0044] An amino acid sequence for the protein encoded by the human C1orf94 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q6P1IW5, shown below as SEQ ID NO:2.

TABLE-US-00002		10	20	
30	40	50	MRGGGGCVLA	LGGQRGFQKE
RRRMASGNGL	PSSSALVAKG	PCALGPFPRY		60
70	80	90		100 IWIHQDTPQD
SLDKTCHEIW	KRVQGLPEAS	QPWTSMEQLS	VPVVGTLRGN	
110	120	130	140	150
ELSFQEEALE	LSSGKDEISL	LVEQEFLSLT	KEHSILVEES	SGELEVPGSS
	160	170	180	
190	200	PEGTRELAPC	ILAPPLVAGS	NERPRASIIV
GDKLLKQKVA	MPVISSRQDC		210	

220	230	240	250	DSATSTVTDI	
LCAA EVKSSK	GTEDRGRILG	DSNLQVSKLL	SQFPLKSTET		
260	270	280	290		300
SKVPDNKNVL	DKTRVTKDFL	QDNLFSGPGP	KEPTGLSPFL	LLPPRPPPAR	
	310	320	330		
340	350	PDKLPELPAQ	KRQLPVFAKI	CSKPKADPAV	
ERHHLMEWSP	GTKEPKKGQG		360		
370	380	390	400	SLFLSQWPQS	
QKDACGEEGC	CDAVG TASLT	LPPKKPTCPA	EKNLLYEFLG		
410	420	430	440		450
ATKNPSGQPR	LRNKVEVDGP	ELKF NAPVTV	ADKNNPKYTG	NVFTPHEPTA	
	460	470	480		
490	500	MTSATLNQPL	WLNLNYP PPP	VFTNHSTFLQ	
YQGLYPQQAA	RMPYQQALHP		510		
520	530	540	550	QLGCYSQQVM	
PYNPQQMGQQ	IFRSSYTPLL	SYIPFVQPNY	PYPQRTPPKM		
560	570	580	590	SANPRDPPLM	
AGDGPQYLFP	QGYGFGSTSG	GPLMHSPYES	SSGNGINF		

A cDNA and a chromosomal sequence encoding the Q6P1W5 protein is available from the NCBI database as accession no. AK123355 and AC115286, respectively.

[0045] An amino acid sequence for the protein encoded by the human CNGB1 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q14028, shown below as SEQ ID NO: 3.

TABLE-US-00003		10	20		
30	40	50	MLGWVQRVLP	QPPGTPRKTK	
MQEEEEVEPE	PEMEA EVEPE	PNPEEAETES		60	
70	80	90	100	ESMPPEESEK	
EEEVAVADPS	PQETKEAALT	STISLRAQGA	EISEMNSPSR		
110	120	130	140		150
RVL TWLMKGV	EKVIPQPVHS	ITEDPAQILG	HGSTGDTGCT	DEPNEALEAQ	
	160	170	180		
190	200	DTRPGLRLLL	WLEQNLERVL	PQPPKSSEVW	
RDEPAVATGA	ASDPAPPGRP		210	220	
230	240	250	QEMGPKLQAR	ETPSLPTPIP	
LQPKEEPKEA	PAPEPQPGSQ	AQTSSLPPTR		260	
270	280	290	300	DPARLVAVVL	
HRLEMALPQP	VLHGKIGE QE	PDSPGICDVQ	TISILPGGQV		
310	320	330	340		350
EPDLVLEEVE	PPWEDAHQDV	STSPQGTEVV	PAYEEENKAV	EKMPRELSRI	
	360	370	380		
390	400	EEEEKEDEEEE	EEEEEEEEEEE	EVTEVLLDSC	
VVSQVGVGQS	EEDGTRPQST		410		
420	430	440	450	SDQKLWEEVG	
EEAKKEAE EK	AKEEAEEVAE	EEAEKEPQDW	AETKEEPEAE		
460	470	480	490		500
AEAASSGVPA	TKQHPEVQVE	DTDADSCPLM	AEENPPSTVL	PPPSPAKSDT	
	510	520	530		
540	550	LIVPSSASGT	HRKKLPSEDD	EAEELKALSP	
AESPVVAWSD	PTTPKDTDGQ		560		
570	580	590	DRAASTASTN	SAIINDRLQE	

LVKLEKERT	KVKEKLIDPD	VTSDEESPKP	610	
620	630	640	650	SPAKKAPEPA
PDTKPAAEAP	VEEEHYCDML	CCKFKHRPWK	KYQFPQSIDP	
660	670	680	690	700
LTNLMYVLWL	FFVVMAWNWN	CWLIPVRWAF	PYQTPDNIHH	WLLMDYLCDL
	710	720	730	
740	750	IYFLDITVFQ	TRLQFVRGGD	IITDKKDMRN
NYLKSRRFKM	DLLSLLPLDF	760		
770	780	790	800	LYLKVGVNPL
LRLPRCLKYM	AFFEFNSRLE	SILSKAYVYR	VIRTTAYLLY	
810	820	830	840	850
SLHLNSCLYY	WASAYQGLGS	THWVYDGVGN	SYIRCYYFAV	KTLITIGGLP
	860	870	880	
890	900	DPKTLFEIVF	QLLNYFTGVF	AFSVMIGQMR
DVVGAAATAGQ	TYYRSCMDST	910		
920	930	940	950	VKYMNFYKIP
KSVQNRVKTW	YEYTWHSQGM	LDESELMVQL	PDKMRLDLAI	
960	970	980	990	1000
DVNYNIVSKV	ALFQGCDRQM	IFDMLKRLRS	VVYLPNDYVC	KKGEIGREMY
	1010	1020	1030	
1040	1050	IIQAGQVQVL	GGPDGKSVLV	TLKAGSVFGE
ISLLAVGGGN	RRTANVVAHG	1060	1070	
1080	1090	1100	FTNLFILDKK	DLNEILVHYP
ESQKLLRKKA	RRMLRSNNKP	KEEKSVLILP	1110	
1120	1130	1140	1150	PRAGTPKLFN
AALAMTGKMG	GKGAKGGKLA	HLRARLKELA	ALEAAAKQQE	
1160	1170	1180	1190	1200
LVEQAKSSQD	VKGEEGSAAP	DQHTHPKEAA	TDPPAPRTPP	EPPGSPPSSP
	1210	1220	1230	
1240	1250	PPASLGRPEG	EEEGPAEPEE	HSVRICMSPG
PEPGEQILSV	KMPEEREKEA	E		

[0046] A cDNA and a chromosomal sequence encoding the Q14028 protein is available from the NCBI database as accession no. UI8945 and L 15296, respectively,

[0047] An amino acid sequence for the protein encoded by the human DEPDC7 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q96QD5. shown below as SEQ ID NO. 4

TABLE-US-00004	10	20		
30	40	50	MATVQEKAAA	LNLSALHSPA
HRPPGFSVAQ	KPFGATYVWS	SIINTLQTQV	60	
70	80	90	100	EVKKRRHRLK
RHNDCFVGSE	AVDVIFSHLI	QNKYFGDVDI	PRAKVVRVCQ	
110	120	130	140	150
ALMDYKVFEA	VPTKVFGKDK	KPTFEDSSCS	LYRFTTIPNQ	DSQLGKENKL
	160	170	180	
190	200	YSPARYADAL	FKSSDIRSAS	LEDLWENLSL
KPANSPhVNI	SATLSPQVIN	210	220	
230	240	250	EVWQEETIGR	LLQLVDLPLL
DSLLKQQEAV	PKIPQPKRQS	TMVNSSNYLD	260	
270	280	290	300	RGILKAYSDS
QEDEWLSAAI	DCLEYLPDQM	VVEISRSFPE	QPDRTDLVKE	

310	320	330	340	350
LLFDAIGRYY	SSREPLLNHL	SDVHNGIAEL	LVNGKTEIAL	EATQLLLKLL
	360	370	380	
390	400	DFQNREEFRR	LLYEMAVAAAN	PSEFKLQKES
DNRMVVKRIF	SKAIVDNKNL		410	
420	430	440	450	SKGKTDLLVL
FLMDHQKDVF	KIPGTLHKIV	SVKLMAIQNG	RDPNRDAGYI	
460	470	480	490	500
YCQRIDQRDY	SNNTEKTTKD	ELLNLLKTLD	EDSKLSAKEK	KKLLGQFYKC
	510	HPDIFIEHFG	D	

A cDNA and a chromosomal sequence encoding the Q96QD5 protein is available from the NCBI database as accession no. AJ245600 and AC107939, respectively.

[0048] An amino acid sequence for the protein encoded by the human HRASLS5 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q96KN8, shown below as SEQ ID NO:5.

TABLE-US-00005	10	20		
30	40	50	MGLSPGAEGE	YALRLPRIPP
PLPKPASRTA	STGPKDQPPA	LRRSAVPHSG		60
70	80	90		100
ESVGFAALVQ	LPAKQPPPGT	LEQGRSIQQG	EKAVVSLETT	
110	120	130	140	150
PSQKADWSSI	PKPENEGKLI	KQAAEGKPRP	RPGDLIEIFR	IGYEHWAIYV
	160	170	180	
190	200	EDDCVVHLAP	PSEEFVGS	TSIFSNRAVV
KYSRLEDVLH	GCSWKVNNKL		210	
220	230	240	250	DGTYLPLPVD
KIIQRTKKMV	NKIVQYSLIE	GNCEHFNGL	RYGVPRSQQV	
260	270	EHALMEGAKA	AGAVISAVVD	SIKPKPITA

A cDNA and a chromosomal sequence encoding the HRASLS5 protein is available from the NCBI database as accession no. AB298804 and AP000484, respectively.

[0049] An amino acid sequence for the protein encoded by the human KLHDC3 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BQ90, shown below as SEQ ID NO: 6.

TABLE-US-00006	10	20		
30	40	50	MLRWTVHLEG	GPRRVNHAHV
AVGHRVYSFG	GYCSGEDYET	LRQIDVHIFN		60
70	80	90		100
PVKS AIRGQA	PVVPYMR YGH	STVLIDDTVL	LWGGRNDTEG	
110	120	130	140	150
ACNVLYAFDV	NTHKWFTPRV	SGTVPGARDG	HSACVLGKIM	YIFGGYEQQV
	160	170	180	
190	200	DCFSNDIHL	DTSTMTWTLI	CTKGSPARWR
DFHSATMLGS	HMYVFGGRAD		210	
220	230	240	250	RFGPFHSNNE
IYCNRI RVFD	TRTEAWLDCP	PTPVLPEGRR	SHSAFGYNGE	
260	270	280	290	300
LYIEGGYNAR	LNRHEHDLWK	FNPVSFTWKK	IEPKGKGPCP	RRRQCCCIVG
	310	320	330	
340	350	DKIVLFGGTS	PSPEEGLGDE	FDLIDHSDLH
LCKLAVIQYN		360	370	380
				ILDFSPSLKT

LDQSCLPDI RWELNAMTIN SNISRPVSS HG

A cDNA and a chromosomal sequence encoding the KLHDC3 protein is available from the NCBT database as accession no. AB055925 and AL136304, respectively,

[0050] An amino acid sequence for the protein encoded by the human NBPF6 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q5VWK0, shown below as SEQ ID NO: 7.

TABLE-US-00007		10	20	
30	40	50	MVVSADPLSS	ERAEMNILEI
NQELRSQLAE	SNQQFRDLKE	KFLITQATAY		60
70	80	90		100 SLANQLKKYK
CEEYKDIIDS	VLRDELQSME	KLAEKLRQAE	ELRQYKALVH	
110	120	130	140	150
SQAKELTQLR	EKLREGRDAS	RWLNKHIKTL	LTPDDPDKSQ	GQDLREQLAE
	160	170	180	
190	200 GHRLAEHLVH	KLSPENDEDE	DEDEDDKDEE	
VEKVQESPAP	REVQKTEEKE	210		
220	230	240	250	VPQDSLEECA
VTCSNSHNPS	NSNQPHRSTK	ITFKEHEVDS	ALVVESEHPH	
260	270	280	290	300
DEEEEALNIP	PENQNDHEEE	EGKAPVPPRH	HDKSNSYRHR	EVSFLALDEQ
	310	320	330	
340	350 KVCSAQDVAR	DYSNPKWDET	SLGFLEKQSD	
LEE VKGQETV	APRLSRGPLR	360		
370	380	390	400	VDKHEIPQES
LDGCCLTPSI	LPDLIPSYHP	YWSTLYSFED	KQVSLALVDK	
410	420	430	440	450
IKKDQEEIED	QSPPCPRLSQ	ELPEVKEQEV	PEDSVNEVYL	TPSVHHDVSD
	460	470	480	
490	500 CHQPYSSTLS	SLEDQLACSA	LDVASPTAA	
CPQGTWSGDL	SHHRSEVQIS	510		
520	530	540	550	QAQLEPSTLV
PSCLRLQLDQ	GFHCGNGLAQ	RGLSSTTCSF	SANADSGNQW	
560	570	580	590	600
PFQELVLEPS	LGMKNPPQLE	DDALEGSASN	TQGRQVTGRI	RASLVLILKT
	610	620	630	IRRRLPFSKW
RLAFRFAGPH	AESAEIPNTA	ERMQRMIG		

A cDNA and a chromosomal sequence encoding the Q5VWK0 protein is available from the NCBT database as accession no. BC125161 and AL390038, respectively.

[0051] An amino acid sequence for the protein encoded by the human OTUTD7B gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q5VWK0, shown below as SEQ HD NO:8.

TABLE-US-00008		10	20	
30	40	MVVSADPLSS	ERAEMNILEI	NQELRSQLAE
SNQQERDLKE		50	60	
70	80	KFLITQATAY	SLANQLKKYK	CEEYKDIIDS
	90	100	110	120
KLAEKLRQAE	ELRQYKALVH	SQAKELTQLR	EKLREGRDAS	
130	140	150	160	RWLNKHLKTL
LTPDDPDKSQ	GQDLREQLAE	GHRLAEHLVH	170	
180	190	200	KLSPENDEDE	DEDEDDKDEE

VEKVQES PAP	REVQKTEEKE	210	
220	230	240 VPQDSLEECA	VTCSNSHNPS
NSNQPHRSTK	ITFKEHEVDS	250	
260	270	280 ALVVESEHPH	DEEEEEALNIP
PENQNDHEEE	EGKAPVPPRH	290	
300	310	320 HDKSNSYRHR	EVSFLALDEQ
KVCSAQDVAR	DYSNPKWDET	330	
340	350	360 SLGFLEKQSD	LEEVKGQETV
APRLSRGPLR	VDKHEIPQES	370	380
390	400 LDGCCLIPSI	LPDLIPSYHP	YWSTLYSFED KQVSLALVDK
	410	420	430 440
IKKDQEEIED	QSPPCPRLSQ	ELPEVKEQEV	PEDSVNEVYL
450	460	470	480 TPSVHHDVSD
CHQPYSSTLS	SLEDQLACSA	LDVASPTCAA	490
500	510	520 CPQGTWSGDL	SHHRSEVQIS
QAQLEPSTLV	PSCLRLQLDQ	530	
540	550	560 GFHCGNGLAQ	RGLSSTICSE
SANADSGNQW	PFQELVLEPS	570	
580	590	600 LGMKNPPQLE	DDALEGSASN
TQGRQVTGRI	RASLVLILKT	610	620
630 IRRRLPESKW	RLAFRFAGPH	AESAEIPNTA	ERMQRMIG

A cDNA and a chromosomal sequence encoding the Q5VWK0 protein is available from the NCBI database as accession no. BC125161 and AL390038, respectively.

[0052] An amino acid sequence for the protein encoded by the human TPGS2 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q68CL5, shown below as SEQ ID NO:9.

TABLE-US-00009	10	20	
30	40 MEEEASSPGL	GCSKPHLEKL	TIGITRILES SPGVTEVTII
	50	60	70 80
EKPPAERHMI	SSWEQKNNCV	MPEDVKNFYL	MINGFHMTWS
90	100	110	120 VKLDEHIPL
GSMAINSISK	LTQLTQSSMY	SLPNAPTLAD	130
140	150	160 LEDDTHEASD	DQPEKPHEDS
RSVIFELDSC	NGSGKVCIVY	170	180
190	200 KSGKPALAED	TEIWELDRAL	YWHFLIDTET AYYRLLLITHL
	210	220	230 240
GLPQWQYAFT	SYGISPQAKQ	WESMYKPITY	NTNLLTEETD
250	260	270	280 SFVNKLDPSK
VEKSKNKIVI	PKKKGPVQPA	GGQKGPSGPS	290 300
GPSTSSTSKS	SSGSGNPTRK		

[0053] A cDNA and a chromosomal sequence encoding the TPGS2 protein is available from the NCBI database as accession no AK295817 and AC009854, respectively.

[0054] An amino acid sequence for the ZNF630 protein encoded by the human ZNF630 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q2M218, shown below as SEQ ID NO:10.

TABLE-US-00010	10	20	
30	40 MIESQEPVTF	EDVAVDFTQE	EWQQLNPAQK
TLHRDVMLET	50	60	
70	80 YNHLVSVGCS	GIKPDVIFKL	EHGKDPWIIIE SELSRWIYPD
	90	100	110 120

RVKGLESSQQ	IISGELLFQR	EILERAPKDN	SLYSVLKIWH	
130	140	150		160 IDNQMDRYQG
NQDRVLRQVT	VISRETLTDE	MGSKYSAFGK		170
180	190	200 MENRCTDLAP		LSQKFHKEDS
CENSLKSNSD	LLNYNRSYAR	210		
220	230	240 KNPTKRERCG		RPPKYNASCS
VPEKEGFIHT	GMEPYGDSQC	250		
260	270	280 EKVISHKQAH		VQYKKFQARE
KPNVCSMCGK	AFIKKSQLII	290		300
310	320 HQRIHTGEKP	YVCGDCRKAF		SEKSHLIVHQ
RIHTGEKPYE	330	340		
350	360 CTKYGRAFSR	KSPFTVHQRV		HTGEKPYECE
ECPKAFSQKS	370	380		
390	400 HLIHQRVHT	REKPFECSEC		RKAFCEMSHL
	410	420		430
KPYECTECGK	TFPRKTQLII	HQRTHTGEKP		440
450	460	470		480 CQQSHLIGHQ
RIHTGEKPYPV	CTDCGKAFSQ	KSHLTGHQRL		490
500	510	520 HTGEKPYMCT		ECGKSFSQKS
PLIIHQRIHT	GEKPYQCGEC	530		540
550	560 GKTFSQKSLI	IIHLRVHTGE		KPYECTECGR
	570	580		590
HQRGHTGEKP	YECSECGKAF	CGKSPLIIHQ		600
610	620	630		640 CAESGMTFEW
KSQMITYQRR	HTGEKPSRCS	DCGKAFCQHV		650 YFTGHQNPYR
KDTLYIC				

A cDNA and a chromosomal sequence encoding the Q2M218 protein is available from the NCBI database as accession no. BC112139 and Z98304, respectively.

[0055] An amino acid sequence for the ZNF717 protein encoded by the human ZNF717 gene that is a positive regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BY31, shown below as SEQ ID NO:11.

TABLE-US-00011	10	20	
30	40 MLETYNLSVS	LQELVSFEEV	AVHFTWEEWQ
DLDDAQRTLY	50	60	
70	80 RDVMLETYSS	LVSLGHCITK	PEMIFKLEQG
	90	100	110
NLRLSAVQII	DDLIERSHES	HDRFEWQIVI	TNSNTSTQER
130	140	150	160 VELGKTENLN
SNHVLNLIIN	NGNSSGMKPG	QENDCQNMLF	170
180	190	200 PIKPGETQSG	EKPHVCDITR
RSHRHHEHLT	QHHKIQTLLQ	210	
220	230	240 TFQCNEQGKT	FNTEAMFFIH
KRVHIVQTFG	KYNEYEKACN	250	
260	270	280 NSAVIVQVIT	QVGQPTCCRK
SDFTKHQQTH	TGEKPYECVE	290	
300	310	320 CEKPSISKSD	LMLQCKMPTE
EKPACNWCE	KLESYKSSLI	330	
340	350	360 IHQRIHTGEK	PYGCNECGKT
FRRKSFLTLH	ERTHTGDKPY	370	
380	390	400 KCIECGKTFH	CKSLTLHHR

THSGEKPYQC	SECGKTESQK	410	
420	430	440 SYLTIHHRTH	TGEKPYACDH
CEEAFSHKSR	LTVHQRTHTG	450	
460	470	480 EKPYPECNECG	KPFINKSNLR
LHQRTHTG EK	PYECNECGKT	490	
500	510	520 FHRKSFLTIH	QWTHTG EKPY
ECNECGKTER	CKSELT VHQR	530	
540	550	560 THAGEKPYAC	NECGKTYSHK
SYLTVHHRTH	TGEKPYECNE	570	
580	590	600 CGKSFHCKSF	LTIHQRTHAG
KKPYECNECE	KTFINKLNLG	610	
620	630	640 IHKITHGTGER	PYECNECGKT
FRQKSNISTH	QGTHTG EKPY	650	
660	670	680 VCGKTFHRKS	FLTIHQRTHT
GKNRMDVMNV	EKL FVRNHTL	690	
700	710	720 LYIRELTPGK	SPMNV MNVEN
PFIR RQIERS	IKVFTRGRNP	730	740
750	760 MNVANVEKPC	QKSVLIVHHR	THTG EKPYEC
NECGKTFCHK	770	780	
790	800 SNLSTHQGTH	SGEKPYECDE	CRKTFYDKTV
LTIHQRTHTG	810	820	
830	840 EKPFECKE CR	KTESQKSKLF	VHHRTHTG EK
PERCNECRKT	850	860	
870	880 FSQKSGLSIH	QRTHTG EKPY	ECKECGKTFC
QKSHLSRHQQ	890	900 THIG EKSDVA	
EAGYVFPQNH	SFEP		

A cDNA and a chromosomal sequence encoding the Q9BY31 protein is available from the NCBI database as accession no. AF226994 and AC 08724, respectively.

[0056] The following genes are positive regulators of T cells as detected by Interleukin-2 production (see Table 2): ABCBI0, ACSS2, ADAM19, ADAM23, ADAMTS5, ALKBH7, ALX4, ANXA2R, AP2AI, APOBEC3C, APOBEC3D, APOL2, ARNT, ART1, ASCL4, BEX4, BTG2, BTNL2, C11orf21, C12orf80 (also called LINC02874), CBX4, CBY1, CCDC183, CCDC71L, CD2, CD28, CD6, CDKNIB, CDKN2C, CHERP, CIPC, CLIP3, CNGBI, CNR2, CREB5, CUL3, DCTN5, DEF6, DEPDC7, DYNLL2, EAPP, EEPDI, ELFN2, EMB, EMP, I EMP3, EP300, ERCC3, ESRP1, F2, FBXL13, FBXO41, FNBPI, FOSB, FOSLI, FOXO4, FOXQI, FUZ, GABRG1, GIGTLC2, GNPDA1, GPR18, GPR20, GPR21, GPR84, GRIN3A, GSDMD, GSTM1, HCST, HELZ2, HEPHL1, IL2, IL2RB, IRX4, ISM1, KLF7, KLRC4, KRT 8, LAT, LCP2, LHX6, LMNA, MAGEA9B, MAP3KI2, MERIK, MTMRI1, NDRG3, N1T1, NLRC3, NLRP2, NPLOC4, ORC1, OSBPL7, OTOP3, OTUD7A, OTUD7B, P2RY14, PAFAH11B2, PCP4, PDE3A, PHF8, PIK3API, PLA2G3, PLCG2, POLK, POU2F2, PPHL2, PRACI, PRKCB, PRKD2, RAB6A, RACI, RAC2, RIPK3, RRAS2, RYR1, SAFB2, SCN3A, SDCCA(uG8, SERPINF1, SGTA, SHOC2, SIGLECI, SIRT1, SLC16AI, SLC44A5, SLC5A5, SMC4, SPPL2B, SSU-12, SWAP70, TAF15, THEMIIS, TM4SF4, TMEM79, TNFRSF1(B, TNFSF11, TNRC6A, TPGS2, TRAF3IP2, TRIM21, TRMT5, TRPM4, TRPV5, TSPYL5, UBA52, UBL5, VAV1, WARS2, ZAP70, ZNF141, ZNF296, and ZNF701. Example 2 provides additional positive regulators T cells that were detected by Interleukin-2 production.

[0057] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases, which are incorporated by reference.

[0058] A few examples of protein sequences encoded by some of the genes detected as positive

regulators of T cells by Interleukin-2 production are provided. For example, an amino acid sequence for the protein encoded by the human ADAMTS5 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9UNA0, shown below as SEQ ID NO: 12.

TABLE-US-00012	10	20	
30	40 MLLGWASLLL	CAFRLPLAAV	GPAATPAQDK
AGQPPTAAAA	50	60	
70	80 AQPRRRQGEE	VQERAEP PGH	PHPLAQRRRS
KGLVQNIDQL	90	100	
110	120 YSGGGKVGYL	VYAGGRRELL	DLERDGSVGI
AGFVPAGGGT	130	140	
150	160 SAPWRHRSHC	FYRGTVDGSP	RSLAVEDLCG
GLDGFFAVKH	170	180	
190	200 ARYTLKPLLR	GPWAEEEKGR	VYGDGSARIL
HVYTREGESE	210	220	
230	240 EALPPRASCE	TPASTPEAHE	HAPAHSNPSG
			RAALASQLLD
	250	260	270
280			
QSALSPAGGS	GPQTTWRRRR	RSISRARQVE	LLLVADASMA
290	300	310	320 RLYGRGLQHY
LLTLASIANR	LYSHASIENH	IRLAVVKVVV	330
340	350	360 IGDKDKSLEV	SKNAATTLKN
FCKWQHQH NQ	LGDDHEEHYD	370	
380	390	400 AAILFTREDL	CGHHSCDTLG
MADVGTICSP	ERSCAVIEDD	410	
420	430	440 GLHAAFTVAH	EIGHLIGLSH
DDSKFCEETF	GSTEDKRLMS	450	
460	470	480 SILTSIDASK	PWSKCTSATI
TEFLDDGHGN	CLLDLPRKQI	490	
500	510	520 LGPEELPGQT	YDATQQCNLT
FGPEYSVCPG	MDVCARLWCA	530	
540	550	560 VVRQGQMVCL	TKKLPAVEGT
PCGKGRICLQ	GKCVDKTKKK	570	
580	590	600 YYSTSSHGNW	GSWGSWGQCS
RSCGGGVQFA	YRHCNNPAPR	610	
620	630	640 NNGRYCTGKR	AIYRSCSIMP
CPPNGKSFRH	EQCEAKNGYQ	650	
660	670	680 SDAKGVKTEV	EWVPKYAGVL
PADVCKLTCR	AKGTGYVVF	690	
700	710	720 SPKVTDGTEC	RLYSNSVCVR
GKCVRTGCDG	IIGSKLQYDK	730	
740	750	760 CGVCGGDNSS	CIKIVGTENK
KSKGYTDVVR	IPEGATHIKV	770	
780	790	800 RQFKAKDQTR	FTAYLALKKK
NGEYLINGKY	MISTSETIID	810	820
830	840 INGTVMNYSG	WSHRDDELHG	MGYSATKEIL
IVQILATDPT	850	860	
870	880 KPLDVRY SFF	VPKKSTPKVN	SVTSHGSNKV
GSHTSQPQWV	890	900	
910	920 TGPWLACSRT	CDTGWHTRIV	QCQDGNRKLA
KGCPLSQRPS	930 AFKQC LLKKC		

A cDNA and a chromosomal sequence encoding the protein is available from the NCBI database as accession no. AF 42099 and AP001698, respectively.

[0059] A nucleotide sequence for human Cl2orf80 cDNA (also called LINC02874) that is a positive regulator of T cells as detected by Interleukin-2 production is available from the NCBI database as accession no. NR_164127.1, shown below as SEQ ID NO: 13.

TABLE-US-00013 1 AGAGCGAGAA GATGATGCAT GTGAGCCCTG CCCTTGGGAA
41 GCTTCCAGGT TGGGAAATGA GGAATGAGCC TGACACCCAG 81
GGCCCAGAGA GACCCAGGAC AGAGGCAGGT CAGGAGGCAG 121
ACACGCGCTG CIGGGTAATG ACGACAGCAC CAGTAATCAC 161 GGCTACTCCT
TGTTAAGTAC TTACTAAGTG CAAGACTCCA 201 AGCTAAGCAA TTAATAGACC
TTTTCTTGTT TAATCCTTAC 241 CACAATTCCA TAGGGITGAG TAGGAAGTCC
TTCTTGAAGT 281 CTAATCTCAA GAATTCATGC CATGGATTGG GCCAATGTTC 321
CACCTTTATT GGAGTCTGGG GACTTAGGTG GAAAAGGCAG 361
AACTGGCTGG TTGGAGGAGC CACCTCCCTG CAAGGGGCCA 401
GGAGGGAGAT TACGGAGGCG CCAAGCCCAG AGCTCCAACA 441
TGCACCTGCC ACAGCTCCAG GGGAGATCGG GGGCCTGCCA 481 ATTTACCCCG
CCCCATGATC TCATGGCTGT GTGCGCCAGG 521 CACTGGCTCA GGGAGCAGCA
TCTCACAGAG AAGATACTTG 561 GATGGGCCAC AGGCCAAAAC
TGGGCCAAAA CCCTGGAGAG 601 GGGGACCTGG CTCAAGGCCA
CACCACATTT CCATGTCATT 641 TTCCAGTGGC ACCAACTGAG CTGGACAGAT
GTTCTCACAG 681 AGCTACACAT GCCACAGTCA TCACTAGTTG CGAGAGTCTC
721 CAGTGTTTAT TAAGCCTATT TTGCTGGAAG TTAGTCCCT 761
GGAGGATTAG TCCCTGCCCT TAAGGAGTGC CAAAGAAGAC 801 TTTGGAATCT
AAGTCACATC TCCCCTGCTG CAATTTTTTC 841 CTCTTGATAA TCAAGGAATC
TCACTGATAA AACTCTCAAT 881 GAGGAGCAGC TCTCCCCATG AGCAAGTTCT
TGCCTTCATT 921 GCCAGGGAGG CCCAGCTAA GGTATACTG GGAAAGGAAT
961 CTCTGGGGAG TACCTGGAAG AGGTCAGCTT CTCCTCAAAG 1001
GAAGCCTCTC CAGCTGGTGT ACTGCAAAGC CTTCCAAGAC 1041
CAAGTTCCCT CTTCTGGCC TCTGGAACCA GCCTGTGTGC 1081
CAGCGCTGTG TCCCAGTAA CACACATGAG TCTCTCCCTC 1121
AGAAGCCACA GAAGGATGCA GAGAGAGACC AACCCAGGAT 1161
TGATTCTCAG CACTTACTAA TTGTGTGGAC ATAAGTCTC 1201 TGGACCTCAG
CTTTGCTATC TATAAAATGA GCACCCTTTT 1241 TATAGATATG AACACTTTAA
GAGAGTCATA AAGATTGAAG 1281 TTGATTIGTG CTGCGCCTGG
CACAGAGGTC TCCCTCAGTA 1321 AATGGCAGCC ACTGCTATTG
GGGTGCCCCA ACACCCCTTG 1361 CCCCTGCCCA CCATTAGCTT TCACTTAGGC
CCACACTGAG 1401 GGTGTGGCTG TTGTGTTGGG GGAAGGAAAA AAACCATGCC
1441 TGGGTTGCAG GGCCCCCACC ACCAACTGGT CTGCTTTTGT 1481
GGGAAGCATT TTCTGATGCC CTCGCCCTAC CCGTGGGCTG 1521
GTTGACTAAG TGTCCAGCAT CATGAGGAAA AGAGCAGGGT 1561
TCAGGGACAG CTTGGCTCAG CCCTGAAACT CATCTGGGCC 1601
TAGGTCCAGA TGAGAGGCAA GCCTGGGAGG CTTCCGTTG 1641
TACCCTTGCC TATCCTGCAG CACCCTCTAG CCTGCAGGCC 1681 GCTCCTGGGT
GGCATAGCAC CTTGGATGTC AGGTGTGGGC 1721 CTTCCCAGGC
ACATGGCATG AGGGGGCACT TCTGTGGGTT 1761 GTTGGGGGCA
GGAAGGGATG TGTCTCCAGC TGGACTTGGG 1801 CTCTCGCATT
TTGGGGCCCA GCCATGCCAG GACAGCACAC 1841 ATGGGCGCTT
AGTGCGAATC TCATGATGGA GCAGAGGAGG 1881 AGCAAAGCAA
AACCAGGGAG TCCCCAGGCC CCTGCTTCTG 1921 CCCC GCCAG
AGACAGTGGA GCAGGTGTCT CCAGCTTCTT 1961 AACCTCAACG

CACAGTAAGA AATACATTTT ACAGCAACCA 2001 ATAGACACAT ACATTTACAA
AACGA

[0060] An amino acid sequence for the protein encoded by the human CCDC183 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q5T5S1, shown below as SEQ ID NO: 14.

TABLE-US-00014

	10	20	
30	40 MRRHSETDVE	EQTQELKTIT	QLQEQCRALQ
IQGVKENMDQ	50	60	
70	80 NKATLALLRS	NIRRG AQDWA	LAKKYDQWTI
SKACGKNLPL	90	100	
110	120 RLAHCRSTME	VVREKLRKYV	EDRVNMHNLL
IHLVRRRGQK	130	140	
150	160 LESMQLELDS	LRSQPDASKE	ELRLLQIIRQ
	170	180	190
200	210	220	230
240	CSELSDMKIM		
250			
260	270	280	NRLNQKKLI
DKIHTKETSE		290	
KYRRGQMDLD	FPSNLMSTET		
300	310	320	LKLRRKETST
AEMEYQSGVT		330	
340	350	360	RELAQRNTEE
NLELQMEDCE		370	
EWRVQLKALV	KQLELEEAVL		
380	390	400	KFRQKPSSIS
EKSVEKKMTD		410	
MLKEEEEERLQ	LAHSNMTKGQ		
420	430	440	ELLTIQMGI
DNLYVRLMGI		450	460
NLPATQREV	LSNILDINSK		
470	480	LAYCEGKLTY	LADRVQMVS
TEEGDTKVRD		490	500
TLESSTLMEK		510	520
YNTRISFENR	EEDMIDTEQE	PDMDHSYVPS	
RAEIKRQAQR	530	LIEGKLKAAK	KKKK

A cDNA and a chromosomal sequence encoding the protein is available from the NCBI database as accession no. AB075864 and AL355987, respectively.

[0061] An amino acid sequence for the protein encoded by the human CIPC gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9C0C6, shown below as SEQ ID NO: 15,

TABLE-US-00015

	10	20	
30	40 MERKNPSRES	PRRLSAKVGK	GTEMKKVARQ
LGMAAAESDK	50	60	
70	80 DSGESDGSSE	CLSSAEQMES	EDMLSALGWS
REDRPRQNSK	90	100	
110	120 TAKNAFPTLS	PMVVMKNVLV	KQGSSSSQLQ
SWIVQPSFEV	130	140	
150	160 ISAQPQLLFL	HPPVPSPVSP	CHTGEKKSDS
	170	180	190
200	210	220	230
240	EPTKAGAVPS		
250			
260	270	280	QTLQPVSSSH
VAKAPSLIFA		290	300
SPASPVCASD	STLHGLESNS		

310	320 PLSPLSANYS	SPLWAAEHL C	RSPDIFSEQ R	QSKHRRFQNT
	330	340	350	360
LVVLHKSGLL	EITLKTKELI	RQNQATQVEL	DQLKEQTQLF	
370	380	390 IEATKS RAPQ	AWAKLQASLT	
PGSSNIGSDL	EAFSDHPAI			

A cDNA and a chromosomal sequence encoding the CIPC protein is available from the NCBI database as accession no. AB051524 and AC007686, respectively.

[0062] An amino acid sequence for the protein encoded by the human CUL3 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q13618, shown below as SEQ ID NO: 16.

TABLE-US-00016		10	20	
30	40 MSNLSKGTGS	RKDTKMRIRA	FPMTMDEKYV	
NSIWDLLKNA	50	60		
70	80 IQEIQRKNNS	GLSFEELYRN	AYTMVLHKHG	
EKLYTGLREV	90	100		
110	120 VTEHLINKVR	EDVINSLN NN	FLQTLNQAWN	
DHQTAMVMIR	130	140		
150	160 DILMYMDRVY	VQQNNVENVY	NEGLIIERDQ	
VVRYGCIRDH	170	180		
190	200 LRQTLLDMIA	RERKGEVVDR	GAIRNACQML	
MILGLEGRSV	210	220		
230	240 YEEDFEAPFL	EMSAEFFQME	SQKFLAENSA	
SVYIKKVEAR	250	260		
270	280 INEEIERVMH	CLDKSTEEPI	VKVVERELIS	KHMKTIVEME
	290	300	310	320
NSGLVHMLKN	GKTEDLGCMY	KLFSRVPNGL	KTMCECMSSY	
330	340	350	360 LREQGKALVS	
EEGEGKNPVD	YIQGLLDLKS	RFDRELLESF	370	
380	390	400 NNDRLFKQTI	AGDFEYFLNL	
NSRSPEYLSL	FIDDKLKKG V	410		
420	430	440 KGLTEQEVET	ILDKAMVLER	
FMQEKDVFER	YYKQHLARRI	450		
460	470	480 LINKSVSDDS	EKNMISKLKT	
ECGCQFTSKL	EGMERDMSIS	490		
500	510	520 NTIMDEF RQH	LQATGVSLGG	
VDLTVRVLTT	GYWPTQSATP	530		
540	550	560 KCNIPPAPRH	AFEIFRRFY L	
AKHSGRQLTL	QHMGGSADLN	570		
580	590	600 ATFYGPVKKE	DGSEVGVGGA	
QVTGSNTRKH	ILQVSTFQMT	610		
620	630	640 ILMLENNREK	YTFEEIQQET	
DIPERELVRA	LQSLACGKPT	650	660	
670	680 QRVLTKEPKS	KEIENGHIFT	VNDQFTSKLH	RVKIQTVAAK
	690	700	710	720
QGESDPERKE	TRQKVDDDRK	HEIEAAIVRI	MKSRKKMQHN	
730	740	750	760 VIVAEVTQQL	
KARFLPSPVV	IKKRIEGLIE	REYLARTPED	RKVYTYVA	

A cDNA and a chromosomal sequence encoding the Q13618 protein is available from the NCBI database as accession no AF064087 and AC073052, respectively.

[0063] An amino acid sequence for the protein encoded by the human EMB (Embigin) gene that is

a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q6PCB8, shown below as SEQ ID NO: 17.

TABLE-US-00017		10	20
30	40 MRALPGLLEA	RARTPRLLLL	QCLLAAARPS
SADGSAPDSP	50	60	
70	80 FTSPPLREEI	MANNFSLESH	NISLTEHSSM PVEKNITLER
	90	100	110 120
PSNVNLICQF	TISGDLNAVN	VIWKKDGEQL	ENNYLVSATG
130	140	150	160 STLYTQYRFT
IINSKQMGSY	SCFFREEKEQ	RGTFNFKVPE	170
180	190	200 LHGKNKPLIS	YVG DSTVLTC
KCQNCEPLNW	TWYSSNGSVK	210	
220	230	240 VPVGVMNKY	VINGTYANET
KLKITQLLEE	DGESYWCAL	250	
260	270	280 FQLGESEEH	ELVVLSYLV
LKPFLVIVAE	VILLVATILL	290	300
310	320 CEKYTQKKKK	HSDEGKEFEQ	IEQLKSDDSN
GIENNVPRHR KNESLGQ			

A cDNA and a chromosomal sequence encoding the protein is available from the NCBI database as accession no. AK300860 and AC035145, respectively.

[0064] An amino acid sequence for the protein encoded by the human ESRP1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q6NXG1. shown below as SEQ ID NO: 18.

TABLE-US-00018		10	20
30	40 MTASPDYLVV	LFGITAGATG	AKLGSDEKEL
ILLEWKVVDL	50	60	
70	80 ANKKVGQLHE	VIVRPDQLEL	TEDCKEETKI DVESISSASQ
	90	100	110 120
LDQALRQENQ	SVSNELNIGV	GTSFCLCTDG	QLHVRQILHP
130	140	150	160 EASKKNVLLP
ECFYSSFDLR	KEFKKCCPGS	PDIDKLDVAT	170
180	190	200 MTEYLNFEKS	SSVSRYGASQ
VEDMGNIILA	MISEPYNHRE	210	
220	230	240 SDPERVNYKF	ESGTCSKMEL
IDDNTVVRAR	GLPWQSSDQD	250	
260	270	280 IARFFKGLNI	AKGGAALCIN
AQGRRNGEAL	VRVSEEHRD	290	
300	310	320 LALQRHKHHM	GTRYIEVYKA
TGEDELKIAG	GTSNEVAQFL	330	
340	350	360 SKENQVIVRM	RGLPFTATAE
EVVAFFGQHC	PITGGKEGIL	370	380
390	400 FVTYPDGRPT	GDAFVLFACE	EYAQNALRKH
KDLLGKRYIE	410	420	
430	440 LERSTAAEVQ	QVLNRESSAP	LIPLTPPII PVLPPQFVPP
	450	460	470 480
INVRDCIRER	GIPYAATIED	ILDELGEFAT	DIRTHGVH MV
490	500	510	520 LNHQGRPSGD
AFIQMKSADR	AFMAAQKCHK	KNMKDRYVEV	530
540	550	560 FQCSAEEMNF	VLMGGTLNRN
GLSPPPCKLP	CLSPPSYTEP	570	580

590 600 APAAVIPTEA AIYQPSVILN PRALQPSTAY YPAGTQLEMN
610 620 630 640
YTAYYPSPPG SPNSLGYEPT AANLSGVPPQ PGTVVVRMQGL
650 660 670 680 AYNTGVKEIL
NFFQGYQYAT EDGLIHINDQ ARTLPKEWVC I
A cDNA and a chromosomal sequence encoding the Q6NXG1 protein is available from the NCBI database as accession no. BC067098 and AP005660, respectively.
[0065] An amino acid sequence for the protein encoded by the human FBXL13 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8NEE6, shown below as SEQ ID NO:19.
TABLE-US-00019 10 20
30 40 MTPELMIKAC SFYTGHLVKT HFCTWRDIAR
TNENVVLAEK 50 60
70 80 MNRAVTCYNE RLQKSVFHHW HSYMEDQKEK
LKNILLRIQQ 90 100
110 120 IIYCHKLTI LTKWRNTARH KSKKKKEDELI LKHELQLKKW
130 140 150 160
KNRLILKRAA AEESNFPERS SSEVELVDET LKCDISLLPE
170 180 190 200 RAILQIFFYL
SLKDVIICGQ VNHAWMLMTQ INSLWNAIDE 210
220 230 240 SSVENVIPDK YIVSTLQRWR
INVIRLNERG CLLRPKTERS 250 260
270 280 VSHCRNLQEL NVSDCPTFTD ESMRHISEGC
PGVLCLNLSN 290 300
310 320 TTITNRTMRL LPRHFHNLQN LSLAYCRRFT
DKGLQYLNLG 330 340
350 360 NGCHKLIYLD LSGCTQISVQ GERYIANSCT GIMHLTINDM
370 380 390 400
PTLTDNCVKA LVEKCSRITS LVETGAPHIS DCTFRALSAC
410 420 430 440 KLRKIRFEGN
KRVIDASFKF IDKNYPNISH IYMADCKGIT 450
460 470 480 DSSLRSLSPL KQLTVLNLAN
CVRIGDMGLK QFLDGPASMR 490
500 510 520 IRELNLSNCV RLSDASVMKL
SERCPNLNYL SLRNCEHLTA 530
540 550 560 QGIGYIVNIF SLVSIIDLSGT
DISNEGLNVL SRHKKLKELS 570
580 590 600 VSECYRITDD GIQAFCKSSL
ILEHLDVSYC SQLSDMIIKA 610 620
630 640 LAIYCINITS LSIAGCPKIT DSAMEMLSAK CHYLHILDIS
650 660 670 680
GCVLLTDQIL EDLQIGCKQL RILKMQYCIN ISKKAQQRMS
690 700 710 720 SKVQQQEYNT
NDPPRWEGYD REGNPVTELD NITSSKGALE 730 LTVKKSTYSS
EDQAA

A cDNA and a chromosomal sequence encoding the FBXL13 protein is available from the NCBI database as accession no. AY359238 and AC005250, respectively.

[0066] An amino acid sequence for the protein encoded by the human FBXO41 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8TF61, shown below as SEQ ID NO:20.

TABLE-US-00020		10	20	
30		40 MASLDLPYRC	PRCGEHKRER	SLSSLRAHLE YSHTYETLYI
	50	60	70	80
LSKINSICDG	AAAAAAAAAAAA	ASGEPLAPEP	AALLAVPGAR	
90	100	110	120 REVFE	STSFQ
GKEQAAGPSP	AAPHLLHHHH	HHAPLAHFG	130	
140	150	160 DLVPASIPCE	ELAEPGIVPA	
AAARYALREI	EIPLGELFAR	170	180	
190	200 KSVASSACST	PPPGPGGPC	PGPASASPAS	PSPADVAYEE
	210	220	230	240
GLARLKIRAL	EKLEVDRRLE	RLSEEVEQKI	AGQVGRLQAE	
250	260	270	280 LERKAAELET	
ARQESARLGR	EKEELEERAS	ELSRQVDVSV	290	
300	310	320 ELLASLKQDL	VHKEQELSRK	
QQEVVQIDQF	LKETAAREAS	330		
340	350	360 AKLRLQQFIE	ELLERADRAE	
RQLQVISSC	GSTPSASIGR	370	380	
390	400 GGGGGGAGPN	ARGPGRMREH	HVGPAVPNTY	
AVSRHGSSPS	410	420		
430	440 TGASSRVPAA	SQSSGCYDSD	SLELPRPEEG	APEDSGPGGL
	450	460	470	480
GTRAQAANGG	SERSQPPRSS	GLRRQAIQNW	QRRPRRHSTE	
490	500	510	520 GEEGDVSDVG	
SRTTESEAEG	PIDAPRPGPA	MAGPLSSCRL	530	
540	550	560 SARPEGGSGR	GRRRAERVSPS	
RSNEVISPEI	LKMRAALFCI	570	580	
590	600 FTYLDTRILL	HAAEVCRDWR	FVARHPAVWT	
RVLLENARVC	610	620		
630	640 SKFLAMLAQW	CTQAHSLTLQ	NLKPRQRGKK	
ESKEEYARST	650	660		
670	680 RGCLEAGLES	LIKAAGGNLL	ILRISHCPNI	LTDRSLWLAS
	690	700	710	720
CYCRALQAVT	YRSAIDPVGH	EVIWALGAGC	REIVSLQVAP	
730	740	750	760 LHPCQQPTRF	
SNRCLQMIGR	CWPHLRALGV	GGAGCGVQGL	770	
780	790	800 ASLARNCMRL	QVLELDHVSE	
ITQEVAAEVC	REGLKGLEML	810		
820	830	840 VLTATPVTPK	ALLHENSICR	
NLKSIVVQIG	IADYFKEPSS	850	860	
870 PEAQKLFEDM	VTKLQALRRR	PGESKILHIK	VEGGC	

A cDNA and a chromosomal sequence encoding the FBXO41 protein is available from the NCBI database as accession no. AB075820 and AC010913, respectively.

[0067] An amino acid sequence for the protein encoded by the human FOSL1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. P15407, shown below as SEQ ID NO:21.

TABLE-US-00021		10	20	
30		40 MERDEGEPPG	SSGNNGGGYGG	PAQPPAAAQA
AQQKFHLVPS		50	60	
70	80 INTMSGSQEL	QWMVQPHEL	G PSSYPRPITY	PQYSPPQPRP
	90	100	110	120

GVIRALGPPP	GVRRRPCEQI	SPEEEERRRV	RRERNKLAAA
130	140	150	160 KCRNRRRKELT
DFLQAETDKL	EDEKSGLQRE	IEELQKQKER	170
180	190	200 LELVLEAHRP	ICKIPEGAKE
GDTGSTSGTS	SPPAPCRPVP	210	220
230	240 CISLSPGPVL	EPEALHTPTL	MITPSLTPFT PSLVFTYPST
	250	260	270 PEPCASAHRK

SSSSSGDPSS DPLGSPTLLA L

A cDNA and a chromosomal sequence encoding the FOSL1 protein is available from the NCBI database as accession no. X16707 and AP006287, respectively.

[0068] An amino acid sequence for the protein encoded by the human FOXO4 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. P98177, shown below as SEQ ID NO:22.

TABLE-US-00022	10	20	
30	40 MDPGNENSAT	EAAAIIDLDP	DEEPQSRPRS CTWPLPRPEI
	50	60	70 80
ANQPSEPPEV	EPDLGEKVHT	EGRSEPILLP	SRLPEPAGGP
90	100	110	120 QPGILGAVTG
PRKGGSRRNA	WGNQSYAELI	SQAIESAPEK	130
140	150	160 RLTLAQIYEW	MVRTVPYEKD
KGDSNSSAGW	KNSIRHNISL	170	
180	190	200 HSKFIKVHNE	ATGKSSWWML
NPEGGKSGKA	PRRRAASMDS	210	
220	230	240 SSKLLRGRSK	APKKKPSVLP
APPEGATPTS	PVGHEAKWSG	250	
260	270	280 SPCSRNREEA	DMWTTFRPRS
SSNASSVSTR	LSPLRPESEV	290	300
310	320 LAEEIPASVS	SYAGGVPTL	NEGLELLDGL NLTSSHLLS
	330	340	350 360
RSGLSGESLQ	HPGVTGPLHT	YSSSLESPAE	GPLSAGEGCF
370	380	390	400 SSSQALEALL
TSDTPPPPAD	VLMTQVDPIL	SQAPTLLLLG	410
420	430	440 GIPSSSKLAT	GVGLCPKPLE
APGPSSLVPT	LSMIAPPPVM	450	460
470	480 ASAPIPKALG	TPVLIPPTEA	ASQDRMPQDL
DLDMYMENLE	490	500	CDMDNIISDL

MDEGEGLDEN FEPDP

A cDNA and a chromosomal sequence encoding the FOXO4 protein is available from the NCBI database as accession no. X93996 and AL590764, respectively.

[0069] An amino acid sequence for the protein encoded by the human FUZ gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9BT04, shown below as SEQ ID NO:23.

TABLE-US-00023	10	20	
30	40 MGEEGTGGTV	HLLCLAASSG	VPLFCRSSRG
GAPARQQLPF	50	60	
70	80 SVIGSLNGVH	MFGQNLEVQL	SSARTENTTV
VWKSFHDSIT	90	100	
110	120 LIVLSSEVGI	SELRLERLLQ	MVFGAMVILV GLEELINIRN
	130	140	150 160
VERLKKDLRA	SYCLIDSELG	DSELIGDLTQ	CVDCVIPPEG

170	180	190	200
AEAAGTTEVS	LVVSGRVVAA	TEGWWRLGTP	SLLQEALSGF
220	230	240	210
PVYLP HGSPT	VPHRLTLTL	EAVLLPWLVG	SLPPQTARDY
270	280	250	260
	LPSLELCLIC	GPSPPLSQLY	PQLLERWWQP
	290	300	310
	LLHLELKRCL	ETVEPLGDKE	320
330	340	350	360
RNFYTIVIST	HEPPEPGPPE	KTEDEVYQAQ	PSPEQRRRL
380	390	400	370
VALQLGERRL	LLLLSPQSPT	LPRACYLVLG	TEEPGTGVRL
		410	LHALTPLL

A cDNA and a chromosomal sequence encoding the FUZ protein is available from the NCBI database as accession no. AK026341 and AC006942, respectively.

[0070] An amino acid sequence for the protein encoded by the human IRX4 gene is available from the UniPROT database as accession no. P78413, shown below as SEQ ID NO:23.

TABLE-US-00024	10	20	
30	40	MSYPQFGYPY	SSAPQELMAT
	50	60	70
	80		
ASAQAPVYCP	VYESRLLATA	RHEINSAAAL	GVYGGPYGGS
90	100	110	120
SEASAFYSLN	SEDSKDGSGS	AHGGLAPAAA	QGYGNYVTYG
140	150	160	130
DSGTRRKNAT	RETTSTLKAW	AYYPYEPALG	QYPYDRYGTM
180	190	170	
ITKMTLTQVS	TWFANARRRI	200	LQEHRKNPYP
220	230	210	TKGEKIMLAI
YAEGEEEEEGG	EEEAREEPLK	240	KKENKMTWPP
260	270	250	RNKCADKRP
DDEDPLEAEP	PACELKPPFH	280	SSKNAEPVGK
310	320	290	300
	SLDGGLERP	AAPDGPVKEA	SGALRMSLAA
GGGAALDEDL	330	340	
350	360	ERARSCLRSA	AAGPEPLPGA
LGFVPAGASA	370	380	EGGPQVCEAK
390	400	GLEAKPRIWS	LAHTATAAAA
PSCMLKRQGP	410	420	AATSLSQTEF
430	440	AAPAAVSSAP	ATSPSVALPH
	450	460	470
	480		
GVFHDPILRH	STINQAWATA	KGALLDPGPL	GRSLGAGANV
490	500	510	LTAPLARAFA
GAARELLALP	KAGGKPECA	PAVPQDAPAA	

A cDNA and a chromosomal sequence encoding the IRX4 protein is available from the NCBI database as accession no. AF124733 and AB690778, respectively.

[0071] An amino acid sequence for the protein encoded by the human ISM1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. B1AKI, shown below as SEQ ID NO:24.

TABLE-US-00025	10	20
30	40	MVRLAAELLL
DGPDAAAGNA	50	60
70	80	SQAQLQNNLN
HLDHQA AHQP	90	100

110	120 FPRPRERQET	GHPSLQRDEP	RSFLLDIPNF	PDLSKADING
	130	140	150	160
QNPNIQVTIE	VVDGPDSEAD	KDQHPENKPS	WSVPSPDWRA	
170	180	190	200 WWQRSLSLAR	
ANSGDQDYKY	DSTSDDSNFL	NPPRGWDHTA	210	
220	230	240 PGHRTFETKD	QPEYDSTDGE	
GDWSLWSVCS	VTCGNGNQKR	250		
260	270	280 TRSCGYACTA	TESRTCDRPN	
CPGIEDTERT	AATEVSLLAG	290	300	
310	320 SEEFNATKLF	EVDTDSCERW	MSCKSEFLKK	
YMHKVMNDLP	330	340		
350	360 SCPCSYPTVE	AYSTADIEDR	IKRKDERWKD	ASGPKEKLEI
	370	380	390	400
YKPTARYCIR	SMLSLESTIL	AAQHCCYGDN	MQLITRGKGA	
410	420	430	440 GTPNLISTEF	
SAELHYKVDV	LPWIICKGDW	SRYNEARPPN	450	
460 NGQKCTESPS	DEDYIKQFQE	AREY		

A cDNA and a chromosomal sequence encoding the ISM1 protein is available from the NCBI database as accession no. BC017997 and AL050320, respectively.

[0072] An amino acid sequence for the protein encoded by the human MTMR-11 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. A4FU01, shown below as SEQ ID NO:25.

TABLE-US-00026	10	20		
30	40 MWWGGRGQSF	NIAPQKEEPE	MGSVQENRMP	
EPRSRQPSSC	50	60		
70	80 LASRCLPGEQ	ILAWAPGVRK	GLEPELSGTL	ICTNERVTFQ
	90	100	110	120
PCGWQWNQDT	PLNSEYDFAL	VNIGRLEAVS	GLSRVQLLRP	
130	140	150	160 GSLHKFIPEE	
ILIHGRDERL	LRVGFEAGGL	EPQAFQVIMA	170	
180	190	200 IVQARAQSNQ	AQQYSGITLS	
KAGQGSGSRK	PPIPLMETAE	210		
220	230	240 DWETERKKQA	ARGWRVSTVN	
EREDVATSLP	RYFWVPNRIL	250	260	
270	280 DSEVRRAFGH	FHQGRGPRLS	WHHPGGSDDL	
RCGGFYTASD	290	300		
310	320 PNKEDIRAVE	LMLQAGHSDV	VLVDTMDELP	
SLADVQLAHL	330	340		
350	360 RIRALCLPDS	SVAEDKWLSA	LEGTRWLDYV	
RACLRKASDI	370	380		
390	400 SVIVISRVRS	VILQERGDRD	LNGLLSSLVQ	LISAPEARL
	410	420	430	440
FGFQSLVQRE	WVAAGHPELT	RIGGTGASEE	APVELLELDC	
450	460	470	480 VWQLLQQEPA	
DEEFSEFELL	ALHDSVRVPD	ILTFLRNTPW	490	
500	510	520 ERGKQSGQLN	SYTQVYTPGY	
SQPPAGNSEN	LQLSVWDWDL	530		
540	550	560 RYSNAQILQF	QNPGYDPEHC	
PDSWLPRPQP	SEMVPGPPSS	570		
580	590	600 VWLFSRGALT	PLNQLCPWRD	

SPSLAVSSR WLPRAISSE 610 620
630 640 SLADQEWGLP SHWGACPLPP GLLLPGYLGP
QIRLWRRCYL 650 660
670 680 RGRPEVQMGL SAPTISGLQD ELSHLQELLR KWTPRISPED
690 700 HSKKRDPHTI LNPTEIAGIL KGRAEGDLG

A cDNA and a chromosomal sequence encoding the MTMR11 protein is available from the NCBI database as accession no. U78556 and AL590487, respectively.

[0073] An amino acid sequence for the protein encoded by the human NDRG3 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9UGV2. shown below as SEQ ID NO:26.

TABLE-US-00027 10 20
30 40 MDELQDVQLT EIKPLLNDKN GTRNFQDEDC
QEHDIIETTHG 50 60
70 80 VVHVVTIRGLP KGNRPVILTY HDIGLNHKSC
FNAFENFEDM 90 100
110 120 QEITQHFAVC HVDAPGQQEG APSFPTGYQY
PTMDELAEML 130 140
150 160 PPVLTHLSLK SIIGIGVGAG AYILSRFALN HPELVEGLVL
170 180 190 200
INVDPCAKGW IDWAASKLSG LTINVVDIIL AHHFGQEELQ
210 220 230 240 ANLDLIQTYR
MHIAQDINQD NLQLFLNSYN GRRDLEIERP 250
260 270 280 ILGQNDNKSK TLKCSTLLVV
GDNSPAVEAV VECNSRLNPI 290 300
310 320 NTILLKMADC GGLPQVVQPG KLTEAFKYFL
QGMGYIPSAS 330 340
350 360 MTRLARSRTH STSSSLGSGE SPESRSVTSN QSDGIQESCE
370 SPDVLDRHQT MEVSC

A cDNA and a chromosomal sequence encoding the NDRG3 protein is available from the NCBI database as accession no. AB044943 and AL031662, respectively.

[0074] An amino acid sequence for the protein encoded by the human NPLOC4 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8TAT6, shown below as SEQ ID NO:27.

TABLE-US-00028 10 20
30 40 MAESIIRVQ SPDGVKRITA TKRETAATFL KKVAKKEFGFQ
50 60 70 80
NNGESVYINR NKTGEITASS NKSLNLLKIK HGDLLFLFPS
90 100 110 120 SLAGPSSEME
TSVPPGEKVE GAPNVVEDEI DQYLSKQDGK 130
140 150 160 IYRSRDPQLC RHGPLGKCVH
CVPLEPEDED YLNHLEPPVK 170
180 190 200 HMSFHAYIRK LIGGADKGKF
VALENISCKI KSGCEGHLPW 210
220 230 240 PNGICTKCQP SAITLNRQKY
RHVDNIMFEN HTVADRELDF 250
260 270 280 WRKTGNQHEG YLYGRYTEHK
DIPIGIRAEV AAIYEPPQIG 290 300
310 320 TQNSLELLED PKAEVVDEIA AKLGLRKVGW
IFTDLVSEDV 330 340
350 360 RKGIVRYSRN KDTYFLSSEE CITAGDEQNK

HPNMCRLSPD		370		380
390	400 GHFGSKEVTA		VATGGPDNQV	HFEGYQVSNQ
CMALVRDECL	410		420	
430	440 LPCKDAPELG		YAKESSEQY	VPDVIFYKDVD
KEGNEITQLA	450		460	
470	480 RPLPVEYLII	DITTTEPKDP	VYTFSISQNP	FPIENRDVLG
	490	500	510	520
ETQDEHSLAT	YLSQNTSSVE	LDTISDEHLL	LFLVINEVMP	
530	540	550		560 LQDSISILLE
AVRIRNEELA	QTKWRSEQWA	TIEQLCSTVG		570
580	590	600 GQLPGLHEYG		AVGGSTHTAT
AAMWACQHCT	FMNQPGTGHC	EMCSLPRT		

A cDNA encoding the NPLOC4 protein is available from the NCBI database as accession no. AB040932.

[0075] An amino acid sequence for the protein encoded by the human OTOP3 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q7RTS5. shown below as SEQ ID NO:28.

TABLE-US-00029	10	20		
30	40 MGRGARAAAA	QSRWGRASRA	SVSPGRTIRS	
APAVGEAQET	50	60		
70	80 EAAPEKENRV	DVGAEERAAA	TRPRQKSWLV	
RHESLLLRRD	90	100		
110	120 RQAQKAGQLF	SGLLALNVVE	LGGAFICSMI	
FNKVAVILGD	130	140		
150	160 VWILLATLKV	LSLIWLLYYV	ASTTRRPHAV	
LYQDPHAGPL	170	180		
190	200 WVRGSLVLFQ	SCTFCLNIFR	VG YDVSHIRC	KSQDLVESV
	210	220	230	240
IEMVEIGVQT	WVLWKHCKDC	VRVQTNFTRC	GLMLTLATNL	
250	260	270		280 LLWVLAVIND
SMHREIEAEL	GILMEKSTGN	ETNTCLCLNA		290
300	310	320 TACEAFRRGF		LMLYPESTEY
CLICCAVLEV	MWKNVGRHVA	330		
340	350	360 PHMGAHPATA		PFHLHGAIFG
PLLGLLVLLA	GVCVFVLFQI	370		380
390	400 EASGPAIACQ	YFTLYYAFYV	AVLPTMSLAC	LAGTAIHGLE
	410	420	430	440
ERELDTVKNP	TRSLDVVLLM	GAALGQMGIA	YESIVAIVAK	
450	460	470		480 RPHELLNRLI
LAYSLLLILQ	HIAQNLFHIE	GLHRRPLWET		490
500	510	520 VPEGLAGKQE		AEPPIRGSL
EIGQGLQRAS	LAYIHSYSHL	530		540
550	560 NWKRRALKEI	SLFLILCNIT	LWMMPAFGIH	PEFENGLEKD
	570	580		590 FYGYQIWFAI
VNFGPLPLGVF	YRMHSGVGLV	EVYLGA		

A cDNA and a chromosomal sequence encoding the OTOP3 protein is available from the NCBI database as accession no. BK000568 and AC087651, respectively.

[0076] An amino acid sequence for the protein encoded by the human OTUD7A gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q8TE49, shown below as SEQ [D NO:29.

TABLE-US-00030		10	20
30	40 MVSSVLPNPT	SAECWAALLH	DPMTLDMDAV
LSDEVIRSTGA	50	60	
70	80 EPGLARDLLE	GKNWDLTAAL	SDYEQLRQVH
TANLPHVENE	90	100	
110	120 GRGPKQPERE	PQPGHKVERP	CLQRQDDIAQ
EKRLSRGISH	130	140	
150	160 ASSAIVSLAR	SHVASECNNE	QFPLEMPIYT FQLPDLSVYS
	170	180	190 200
EDERSFIERD	LIEQATMVAL	EQAGRINWWS	TVCTSCKRLL
210	220	230	240 PLATTGDGNC
LLHAASLGMW	GFHDRDEVIR	KALYTMMRTG	250
260	270	280 AEREALKRRW	RWQQTQQNKE
EEWEREWTEL	LKLASSEPT	290	
300	310	320 HFSKNGGTGG	GVDNSEDVY
ESLEEFHVEV	LAHILRRPIV	330	340
350	360 VVADTMLRDS	GGEAFAPIPE	GGIYLPLEVP PNRCHCSPLV
	370	380	390 400
LAYDQAHESA	LVSMEQRDQQ	REQAVIPLTD	SEHKLLPLHE
410	420	430	440 AVDPGKDWEW
GKDDNDNARI	AHLILSLEAK	LNLLHSYMN	450
460	470	480 TWIRIPSETR	APLAQPESPT
ASAGEDVQSL	ADSLDSRDS	490	
500	510	520 VCSNSNSNNG	KNGKDKEKEK
QRKEKDKTRA	DSVANKLGSF	530	
540	550	560 SKILGIKLLK	NMGGLGGLVH
GKMGRANSAN	GKNGDSAERG	570	
580	590	600 KEKKAJSRKG	SKEESGASAS
TSPSEKTTPS	PTDKAAGASP	610	620
630	640 AEKGGGPRGD	AWKYSTDVKL	SLNILRAAMQ
GERKFIFAGL	650	660	
670	680 LLTSHRHQFH	EEMIGYYLTS	AQERFSAEQE
QRRRDAATAA	690	700	
710	720 AAAAAAAAAAT	AKRPPRRPET	EGVPVPERAS
PGPPTQLVLK	730	740	
750	760 LKERPSPGPA	AGRAARAAAG	GTASPGGGAR
RASASGPVPG	770	780	
790	800 RSPPAPARQS	VIHVQASGAR	DEACAPAVGA
LRPCATYPQQ	810	820	
830	840 NRSLSQSYS	PARAAALRTV	NIVESLARAV
PGALPGAAGT	850	860	
870	880 AGAAEHKSQT	YINGEGALRD	GLEFADADAP
TARNGECGR	890	900	
910	920 GGPGPVQRR	QRENCAFYGR	AETEHYCSYC
YREELRRRRE	ARGARP		

A cDNA sequence encoding the OTUD7A protein is available from the NCBI database as accession no. AJ430383.

[0077] An amino acid sequence for the protein encoded by the human PDE3A gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q14432, shown below as SEQ ID NO:30.

TABLE-US-00031	10	20	
30	40 MAVPGDAARV	RDKPVHSGVS	QAPTAGRDCH
HRADPASPRD	50	60	
70	80 SGCRGCWGD	VLQPLRSSRK	LSSALCAGSL
SELLALLVRL	90	100	
110	120 VRGEVGC	QCKEAAAAEE	EEAAPGAEGG
VEPGPRGGAP	130	140	
150	160 GGGARLSPWL	QPSALLESLL	CAFEWMGLYL
LRAGVRLPLA	170	180	
190	200 VALLAACCGG	EALVQIGLGV	GEDHLLSLPA
AGVVLSC	210	220	
230	240 ATWLVIRLRL	GVLMIALISA	VRTVSLISLE REKVAWRPYL
	250	260	270 280
AYLAGVLGIL	LARYVEQILP	QSAEAAPREH	LGSQLIAGTK
290	300	310	320 EDIPVFKRRR
RSSSVVSAEM	SGCSSKSHRR	TSLPCIPREQ	330
340	350	360 LMGHSEWDHK	RGPRGSQSSG
TSITVDIAVM	GEAHGLITDL	370	380
390	400 LADPSLPPNV	CISLRAVSNL	LSTQLTFQAI HKPRVNPVTS
	410	420	430 440
ISENYTCSDS	EESSEKDKLA	IPKRIRRS	PGLLRVSST
450	460	470	480 WTTTTSATGE
PTLEPAPVRR	DRSTSIKLQE	APSSSPDSWN	490
500	510	520 NPVMMTLTKS	RSFTSSYAIS
AANHVKAKKQ	SRPGALAKIS	530	
540	550	560 PLSSPCSSPL	QGTPASSLVS
KISAVQFPES	ADTTAKQSLG	570	580
590	600 SHRALTYTQS	APDLSPQILT	PPVICSSCGR PYSQGNPADE
	610	620	630 640
PLERSGVATR	TPSRIDDTAQ	VISDYEINNN	SDSSDIVQNE
650	660	670	680 DETECLREPL
RKASACSTYA	PETMMELDKP	ILAPEPLVMD	690
700	710	720 NLDSIMEQLN	TWNFPIEDLV
ENIGRKCGRI	LSQVSYRLFE	730	740
750	760 DMGLFEAFKI	PIREEMNYFH	ALEIGYRDIP YHNRIHATDV
	770	780	790 800
LHAVWYLTTQ	PIPGLSTVIN	DHGSTSDSDS	DSGFTHGHMG
810	820	830	840 YVESKTYNVT
DDKYGCISGN	IPALELMALY	VAAAMHDYDH	850
860	870	880 PGRTNAFLVA	TSAPQAVLYN
DRSVLENHHA	AAAWNLEMSR	890	
900	910	920 PEYNFLINLD	HVEFKHFREL
VIEAILATDL	KKHEDFVAKF	930	940
950	960 NGKVND	DWTNENDRLL	VCQMCIKLAD
INGPAKCKEL	970	980	
990	1000 HLQWTDGIVN	EFYEQGDEEA	SLGLPISPFM DR
	1010	1020	1030 1040
QESFISHIVG	PLCNSYDSAG	LMPGKWVEDS	DESGDIDDPE
1050	1060	1070	1080 EEEEEAPAPN
EEETCENNES	PKKKTEKRRK	IYCQITQHLL	1090

1100	1110	1120 QNHKMWKKVI	EEEQRLAGIE
NQSLDQTPQS	HSSEQIQAIK	1130	1140 EEEEEKGKPR
GEEIPTQKPD	Q		

A cDNA sequence encoding the PDE3A protein is available from the NCBI database as accession no. M91667.

[0078] An amino acid sequence for the protein encoded by the human POLK gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database, shown below as SEQ ID NO:31.

TABLE-US-00032	10	20	
30	40 MDSTKEKCDS	YKDDLRLRMG	INDNKAGMEG
LDKEKINKII	50	60	
70	80 MEATKGSRFY	GNELKKEKQV	NQRIENMMQQ
KAQITSQQLR	90	100	
110	120 KAQLQVDRFA	MELEQSRNLS	NTIVHIDMDA
FYAAVEMRDN	130	140	
150	160 PELKDKPIAV	GSMSMLSTSN	YHARREGVRA
AMPGFIAKRL	170	180	
190	200 CPQLIIVPPN	FDKYRAVSKE	VKEILADYDP
NEMAMSLDEA	210	220	
230	240 YLNITKHL EE	RQNWPE DKRR	YFIKMGSSVE
NDNPGKEV NK	250	260	
270	280 LSEHERSISP	LIFEESPSDV	QPPGDPFQVN FEEQNNPQIL
	290	300	310 320
QNSVVFGTSA	QEVVKEIRER	IEQKTTLTAS	AGIAPNTMLA
330	340	350	360 KVCSDKNKPN
GQYQILPNRQ	AVMDFIKDLP	IRKVSGIGKV	370
380	390	400 TEKMLKALGI	ITCTELYQQR
ALLSLLESET	SWHYFLHISL	410	420
430	440 GLGSTHLIRD	GERKSMSVER	TESEINKAEE QYSICQELCS
	450	460	470 480
ELAQDLQKER	LKGRTVTIKL	KNVNFEVKTR	ASTVSSVVST
490	500	510	520 AEEIFAIAKE
LLKTEIDADE	PHPLRLRIMG	VRISFPNEE	530
540	550	560 DRKHQQRSII	GELQAGNQAL
SATECTLEKT	DKDKFVKPLE	570	
580	590	600 MSHKKS FEDK	KRSERKWSHQ
DTEKCEAVNK	QSFQTSQPEQ	610	
620	630	640 VLKKKMENL	EISENSDDCQ
ILTCPVCFRA	QGCISLEALN	650	660
670	680 KHVDECLDGP	SISENEKMFS	CSHVSATKVN
KKENVPASSL	690	700	
710	720 CEKQDYEAHP	KIKEISSVDC	IALVDTIDNS SKAESIDALS
	730	740	750 760
NKHSKEECSS	LPSKSFNIEH	CHQNSSSTVS	LENEDVGSFR
770	780	790	800 QEYRQPYLCE
VKTGQALVCP	VCNVEQKTS D	LTLENVHVDV	810
820	830	840 CINKSFIQEL	RKDKENPVNQ
PKESSRSTGS	SSGVQKAVIR	850	860
870 TKRPGLMTKY	STSKKIKPNN	PKHTLDIFEK	

[0079] A cDNA and a chromosomal sequence encoding the POLK protein is available from the

NCBI database as accession no. AB027564 and AY273797, respectively.

[0080] An amino acid sequence for the protein encoded by the human PRACI gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q96KF2, shown below as SEQ ID NO:32.

TABLE-US-00033	10	20
30	40 MICAHFSDQG PAHLTTSKSA	FLSNKKTSTL KHELLGETRSD
	50 GSACNSGISG	GRGRKIP

A cDNA and a chromosomal sequence encoding the PRACI protein is available from the NCBI database as accession no. AF331 165 and CH471 109, respectively.

[0081] An amino acid sequence for the protein encoded by the human SERPINF1 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database, shown below as SEQ ID NO:33.

TABLE-US-00034	10	20
30	40 MQALVLLLCI	GALLGHSSCQ NPASPPEEGS PDPDSTGALV
	50	60 70 80
EEEDPFFKVP	VNKLAAAVSN	FGYDLYRVRS STSPTINVLL
90	100	110 120 SPLSVATALS
ALSLGAEQRT	ESIIHRALYY	DLISSPDIHG 130
140	150	160 TYKELLDTVT APQKNLKSAS
RIVFEKKLRI	KSSFVAPLEK	170 180
190	200 SYGTRPRVLT	GNPRLDLQEI NNWVQAQMKG
KLARSTKEIP	210	220
230	240 DEISILLGV	AHFKGQWVIK FDSRKTSLED FYLDEERTVR
	250	260 270 280
VPMMSDPKAV	LRYGLSDLS	CKIAQLPLIG SMSIIFLPL
290	300	310 320 KVTQNLTIE
ESLTSEFIHD	IDRELKTVQA	VLTVPKLKLS 330
340	350	360 YEGEVIKSLQ EMKLQSLEDS
PDFSKITGKP	IKLTQVEHRA	370 380
390	400 GFEWNEDGAG	TTPSPGLQPA HLTFPLDYHL
NQPFIFVERD	410 TDTGALLFIG	KILDPRGP

A cDNA and a chromosomal sequence encoding the SERPINF1 protein is available from the NCBI database as accession no. M76979 and U29953, respectively.

[0082] An amino acid sequence for the protein encoded by the human SSUH-12 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q9Y2M2, shown below as SEQ ID NO:34.

TABLE-US-00035	10	20
30	40 MPSPVGLIRA	LPLWPQFLA CTLRRLAGPR ESTGPSQKPP
	50	60 70 80
PLCSVPCRVP	AMTEEVAREA	LLSEVDSKCC YSSTVAGDLV
90	100	110 120 IQELKRQTLC
RYRLETESES	RISEWTFQPF	TNHSVDGPQR 130
140	150	160 GASPRLWDIK VQGPPMFQED
TRKFQVPHSS	LVKECHKCHG	170
180	190	200 RGRYKCSGCH GAGTVRCPSC
CGAKRKAKQS	RRCQLCAGSG	210
220	230	240 RRCSTCSGR GNKICATCKG
EKKLLHFIQL	VIMWKNSLEE	250
260	270	280 EVSEHRLNCP RELAKAKGE
NLFKDENSVV	YPIVDFPLRD	290

300 310 320 ISLASQRGIA EHSAAALASRA

RVLQQRQTIE LIPLTEVHYW 330 340

350 YQGKTYVYYI YGTDHQVYAV DYPERYCCGC TIV

A cDNA and a chromosomal sequence encoding the SSUH2 protein is available from the NCBI database as accession no. AB024705 and AC034187, respectively,

[0083] An amino acid sequence for the protein encoded by the human TMISF4 gene that is a positive regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no, P48230, shown below as SEQ ID NO:35.

TABLE-US-00036

10	20
30	40 MCTGGCARCL GGTLIPLAFF GFLANILLFF PGGKVIDDND
50	60
70	80
90	100
110	120 EGCGKRFAMF
130	
140	150
160	170
180	190
200	SILLVVGGIQ MVLCAIQVVN
GLLGTLCGDC QCCGCCGGDG PV	

A cDNA and a chromosomal sequence encoding the TM4SF4 protein is available from the NCBI database as accession no. U31449 and C1471052, respectively.

[0084] The following genes are positive regulators of T cells as detected by increased T cell proliferation (see Table 3): ABCB1, ASAP1, ATP10A, DEAF1, FO XK1, ITGAX, LCE6A, LCP2, LEFTY1, MYC, NAT8B, OLFM3, and PLD6. Table 7 provides additional positive regulators of T cells as detected by increased T cell proliferation.

[0085] An amino acid sequence for the protein encoded by the human ATP10A gene that is a positive regulator of T cells as detected by increased cell proliferation is available from the UniPROT database as accession no. 060312, shown below as SEQ ID NO:36.

TABLE-US-00037

10	20
30	40 MEREPAGTEE PGPPGRRRRR EGRTRTVRSN LLPPPGAEDP
50	60
70	80
90	100
110	120 EQFHRPANVY
130	
140	150
160	170
180	190
200	210
220	
230	240 DGETNLKRRQ VVRGESELVS EENPLTFTSV IECEKPNNDL
250	260
270	280
290	300
310	320 YAGHETKALL
330	
340	350
360	370
380	
390	400 SELTMIIVIQ VLIPISLYVS IEIVKACQVY FINQDMQLYD
410	420
430	440
450	460
470	480 RCTVSGVEYS
490	
500	510
520	RGSIGSHQSV RVVHRTQSTK

SHRRIGSRAE	AKRASMLSKH	530	
540	550	560 TAFSSPMEKD	ITPDPKLLEK
VSECDKSLAV	ARHQEHLLAH	570	
580	590	600 LSPELSDVED	FFIALTICNT
VVVTSPDQPR	TKVRVRFELK	610	
620	630	640 SPVKTIEDEL	RRFTPSCLTS
GCSSIGSLAA	NKSSHKIGSS	650	660
670	680 FPSTPSSDGM	LIRLEERLGQ	PTSAIASNGY SSQADNWASE
	690	700	710 720
LAQEQESERE	LRYEAESPDE	AALVYAARAY	NCVIVERLHD
730	740	750	760 QVSVELPHLG
RLTFELLHTL	GEDSVRKRMS	VVIRHPLIDE	770
780	790	800 INVYTKGADS	VVMDLLQPCS
SVDARGRHQK	KIRSKTQNYL	810	
820	830	840 NVYAAEGLRT	ICIAKRVISK
EEYACWLQSH	LEAESSLENS	850	
860	870	880 EELLFQSAIR	LETNIHLIGA
TGIEDRLQDG	VPETISKLRQ	890	900
910	920 AGLQIWVLTG	DKQETAVNIA	YACKLLDHDE
EVITLNATSQ	930	940	
950	960 EACAALLDQC	LCYVQSRGLQ	RAPEKTKGKV
SMRFSSLCPP	970	980	
990	1000 STSTASGRRP	SLVIDGRSLA	YALEKNLEDK FLFLAKQCRS
	1010	1020	1030 1040
VICCRSTPLQ	KSMVVKLVR	KLKAMTLAIG	DGANDVSMIQ
1050	1060	1070	1080 VADVGVGISG
QEGMQAVMAS	DEAVPKERYL	ERLLILHGHW	1090
1100	1110	1120 CYSRLANMVL	YFFYKNTMEV
GLLFWFQFFC	GFSASTMIDQ	1130	1140
1150	1160 WYLIFENLLF	SSLPPLVTGV	LDRDVPANVL LINPQLYKSG
	1170	1180	1190 1200
QNMEEYRPRT	FWENMADA AF	QSLVCFSIPY	LAYYDSNVDL
1210	1220	1230	1240 FTWGTPIVTI
ALLTFLHLG	IETKIWTWIN	WITCGFSVLL	1250
1260	1270	1280 FFTVALIYNA	SCATCYPPSN
PYWTMQALLG	DPVFYLTCLM	1290	1300
1310	1320 TPVAALLPRL	FERSLQGRVE	PTQLQLARQL TRKSPRRCSA
	1330	1340	1350 1360
PKETFAQGRL	PKDSGTEHSS	GRTVKTSVPL	SQPSWHTQQP
1370	1380	1390	1400 VCSLEASGEP
STVDMSMPVR	EHTLLEGLSA	PAPMSSAPGE	1410
1420	1430	1440 AVLRSPGGCP	EESKVRAAST
GRVTPLSSLF	SLPTESLLNW	1450	1460
1470	1480 ISSWSLVSRL	GSVLQFSRTE	QLADGQAGRG
LPVQPHSGRS	1490 GLQGPDRHLL	IGASSRRSQ	

An amino acid sequence for the protein encoded by the human LCE6A gene that is a positive regulator of T cells as detected by increased cell proliferation is available from the UniPROT database as accession no. A0A183, shown below as SEQ U) NO:37.

TABLE-US-00038	10	20
30	40	50 MSQKQKQSWK PPNVPKCSP

QRSNPCLAPY STPCGAPHSE GCHSSSQRPE 60
 70 80 VQKPRRRARQK LRCLSRGTTY HCKEEEECEGD
 A cDNA and a chromosomal sequence encoding the LCE6A protein is available from the NCBI database as accession no. DQ991251 and AL162596, respectively.
 [0086] An amino acid sequence for the protein encoded by the human NAT8B gene that is a positive regulator of T cells as detected by increased cell proliferation is available from the UniPROT database as accession no. Q9UHF3, shown below as SEQ ID NO:38.
 TABLE-US-00039 10 20
 30 40 50 MAPYHIRKYQ ESDRKSVVGL
 LSGGMAEHAP ATFRRLKLP RTLILLGGA 60
 70 80 90 100 LALLLVSGSW
 ILALVFSLSL LPALWFLAKK PWTRYVDIAL RTDMSDITKS
 110 120 130 140 150
 YLSECGSCFW VAEEEEKVVG TVGALPVDDP TLREKRLQLF HLSVDNEHRG
 160 170 180
 190 200 QGIAKALVRT VLQFARDQGY SEVVLDTNSNI
 QLSAMGLYQS LGFKKTGQSF 210 220
 FHVWARLVDL HTVHFIYHLP SAQAGRL

A cDNA sequence encoding the NAT8B protein is available from the NCBI database as accession no. AF185571

Negative Regulators of T Cells

[0087] The following genes are negative regulators of T cells as detected by interferon- γ production (see Table 4): ACER2, ADGRVI, AIF1L, ALPL, AMACR, AMZ1, ARHGAP30, ARHGDIB, ARHGEF11, ARLI1, ATP2A2, B3GNT5, BAC H2, BLM, BSG, BTBD2, BTLA, BTRC, CAIL, CASTOR2, CBLB, CCNT2, CCSER1, CD37, CD44, CD5, CD52, CD55, CDK6, CEACAMI, CEBPA, CEBPB, CEP164, CKAP2L, CLCN2, CLDN25, COLQ, CST5, CTNNA1, CYP24A, DDIT4L, DENND3, DGKG, DGKK, DGKZ, DSCI, EBF2, ECEL1, EIF3K, EPB41, EPS8L1, FAM35A, FAM53B, FAM83A, FKRP, FOXA3, FOXF1, FOXF2, FOXI3, FOXJ1, FOXL2, FOXL2NB, GiABRQ, GATA3, GATA4, GATA6, GCM2, GCSAM, GCSAML, GMFG, GNL3L, GRAP, GRTB2, GRIA1, GTSF1IL, HRI-2, HiYLS1, IKZF1, IKZF3, IL2RB, INPPL1, JMJD1C, KCNVI, KRIT1, LAIIBI, LAPIM5, LAT2, LAXI, LCK, LENE, LMO4, LRRC25, LRRC4B, LYN, MAB21L2, MAP4K1, MBIP, MBOA Ti, MIETITL23, MIPEP, MIPOLI, MMP2I, MSMB, MUCI, MUC21, MUC8, N4BPI, NAIF1, NDNF, NFFAIC1, NFKB2, NFKBIA, NKX2-1, NKX2-3, NMB, NR2F1, ODF4, OPRD1, ORC5, OTUD4, PASDI, PBK, PCBP2, PDLIM1, PDPN, PECAIM1, PIP5KI A, PIP5K1B, PITPNA, POGZ, POLK, POU2AF1, PSTPI1, PTPN12, PTPRC, PVRIG, RAB14, RBP7, RETREGI, RFC2, RHCE, RN.sup.F19B, RNF2, RUSC2, SELPLG, SETD1IB, SH3KBP1, SIGLEC6, SIPA1L1, SLA, SLA2, SLC26A4, SLC44A5, SLC45A1, SLC6A8, SLC6A9, SMAD9, SMLkGP, SOCS3, SOX13, SPATA31A1, SPN, SPOCK3, SPREDI, STAP1, STK35, SULLT6B1, SYTI5, TEC, TIAMI, TMEML51A, TMEML87B, TMPRSSI IE, TNNT2, TRIB2, TRLM28, TSPAN UBASH3B, UBQLN4, UBXN7, UNC119, UPP1, VPS28, WLS, ZKSCAN4, ZNF445, and ZNF474. Table 7 provides additional negative regulators of T cells as detected by interferon- γ production.

[0088] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases, which are incorporated by reference.

[0089] A few examples of protein sequences encoded by some of the genes detected as negative regulators of T cells by interferon- γ production are provided. For example, an amino acid sequence for the protein encoded by the human ATI 1L gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BQI0, shown below as SEQ ID NO:39.

TABLE-US-00040		10	20	
30	40	50	MSGELSNRFQ	GGKAFGLLKA
RQERRLA EIN	REFLCDQKYS	DEENLPEKLT		60
70	80	90		100 AFKEK YMEFD
LNNEGEIDLM	SLKRMMEKLG	VPKTHLEMKK	MISEVTGGVS	
110	120	130	140	150
DTISYRDFVN	MMLGKRS AVL	KLVM MFEGKA	NESSPKPVGP	PPERDIASLP

A cDNA and a chromosomal sequence encoding the AIF1L protein is available from the NCBI database as accession no. AL136566 and AL157938, respectively.

[0090] An amino acid sequence for the protein encoded by the human ARHGDIB gene that is a negative regulator of T cells as detected by interferon-γ production is available from the UniPROT database as accession no. P52566, shown below as SEQ ID NO:40.

TABLE-US-00041		10	20	
30	40	50	MTEKAPEPHV	EEDDDDELDS
KINYKPPPQK	SLKELQEMDK	DDESLIKYKK		60
70	80	90		100 TLLGDGPVVT
DPKAPNVVVT	RLTLVCESAP	GPITMDLTGD	LEALKKETIV	
110	120	130	140	150
LKEGSEYRVK	IHFKVNRDIV	SGLKYVQHTY	RTGVKVDKAT	FMVGSYGPRP
	160	170	180	
190	200	EEYEFLTPVE	EAPKGMLARG	TYHNKSFFTD
DDKQDHLSWE	WNLSIKKEWT	E		

A cDNA and a chromosomal sequence encoding the ARHGDIB protein is available from the NCBI database as accession no. L20688 and CH471094, respectively.

[0091] An amino acid sequence for the protein encoded by the human BLM gene that is a negative regulator of T cells as detected by interferon-γ production is available from the UniPROT database as accession no. P54132, shown below as SEQ ID NO:41.

TABLE-US-00042		10	20	
30	40	50	MAAVPQNNLQ	EQLERHSART
LNNKLSLSKP	KFSGFTFKKK	TSSDNNVSVT		60
70	80	90		100 NVSVAKTPVL
RNKDVNVTED	FSFSEPLPNT	TNQQRVKDFF	KNAPAGQETQ	
110	120	130	140	150
RGGSKSLLPD	FLQTPKEVVC	TTQNTPTVKK	SRDTALKKLE	FSSSPDSLST
	160	170	180	
190	200	INDWDDMDDF	DTSETSKSFV	TPPQSHFVRV
STAQSKKKGK	RNFFKAQLYT		210	
220	230	240		250 TNTVKTDLPP
PSSESEQIDL	TEEQKDDSEW	LSSDVICIDD	GPIAEVHINE	
260	270	280	290	300
DAQESDSLKT	HLEDERDNSE	KKKNLEEAE L	HSTEKVPCIE	FDDDDYDTDF
	310	320	330	
340	350	VPPSPEE IIS	ASSSSSKCLS	TLKDLDTS DR
LLSKPEKMSM		360	370	KEDVLSTSKD
380	390	400	QELNPETSTD	CDARQISLQQ
QLIHVMEHIC	KLIDTIPDDK	LKLLDCGNEL		410
420	430	440		450 LQQRNIRRKL
LTEVDFNKSD	ASLLGSLWRY	RPDSL DGPME	GDSCPTGNSM	
460	470	480	490	500
KELNFSHLPS	NSVSPGDCLL	TTTLGKTGFS	ATRKNL FERP	LFNTHLQKSF

510	520	530		
540	550 VSSNWAETPR	LGKKNESSYF	PGNVLTSTAV	
KDQNKHTASI	NDLERETQPS	560		
570	580	590	600 YDIDNFDIDD	
FDDDDDDWEDI	MHNLAASKSS	TAAYQPIKEG	RPIKSVSERL	
610	620	630	640	650
SSAKTDCLPV	SSTAQNINFS	ESIQNYTDKS	AQNLASRNLK	HERFQSLSFP
	660	670	680	
690	700 HTKEMMKIFH	KKFGLHNFRT	NQLEAINAAL	
LGEDCFILMP	TGGGKSLCYQ	710		
720	730	740	750 LPACVSPGVT	
VVISPLRSLI	VDQVQKLTSL	DIPATYLTGD	KTDSEATNIY	
760	770	780	790	800
LQLSKKDPII	KLLYVTPEKI	CASNRLISTL	ENLYERKLLA	RFVIDEAHCV
	810	820	830	
840	850 SQWGHDFRQD	YKRMNMLRQK	FPSVPVMALT	
ATANPRVQKD	ILTQLKILRP	860	870	
880	890	900 QVFSMSFNH	NLKYYVLPKK	
PKKVAFDCLE	WIRKHHPYDS	GIIYCLSRRE	910	
920	930	940	950 CDTMADTLQR	
DGLAALAYHA	GLSDSARDEV	QQKWINQDGC	QVICATIAFG	
960	970	980	990	1000
MGIDKPDVRF	VIHASLPKSV	EGYYQESGRA	GRDGEISHCL	LFYTYHDVTR
	1010	1020	1030	
1040	1050 LKRLIMMEKD	GNHHTRETHF	NNLYSMVHYC	
ENITECRRIQ	LLAYFGENG	1060	1070	
1080	1090	1100 NPDFCKKHPD	VSCDNCCKTK	
DYKTRDVTDD	VKSIVRFVQE	HSSSQGMRNI	1110	
1120	1130	1140	1150 KHVGPSGRFT	
MNMLVDIFLG	SKSAKIQSGI	FGKGSAYSRH	NAERLFKKLI	
1160	1170	1180	1190	1200
LDKILDEDLY	INANDQAIAY	VMLGNKAQTV	LNGNLKVDFM	ETENSSSVKK
	1210	1220	1230	
1240	1250 QKALVAKVSQ	REEMVKKCLG	ELTEVCKSLG	
KVFGVHYFNI	FNTVTLLKLA	1260	1270	
1280	1290	1300 ESLSDPEVL	LQIDGVTEDEK	
LEKYGAEVIS	VLQKYSEWTS	PAEDSSPGIS	1310	
1320	1330	1340	1350 LSSSRGPGRS	
AAEELDEEIP	VSSHYPASKT	RNERKRKKMP	ASQRSKRRKT	
1360	1370	1380	1390	1400
ASSGSKAKGG	SATCRKISSK	TKSSSIIGSS	SASHTSQATS	GANSKLGIMA
	1410 PPKPINRPFL	KPSYAFS		

A cDNA and a chromosomal sequence encoding the BLM protein is available from the NCBI database as accession no. U39817 and AY886902, respectively.

[0092] An amino acid sequence for the protein encoded by the human BSG gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q7KTJ7, shown below as SEQ ID NO:42.

TABLE-US-00043	10	20	
30	40	50 MEAKFLASAL	SFLSIFLAIY
AQSLANDLSK	ESTEFEEEST	IYYGDPVVNL	60

70			90	100	GQPF	SITCII	
PITDQIHWLK	NGEPITRHNL	RHGRDDHAYV	LSESAIEGEK				
110	120	130	140				150
HKIEAHL SVR	HALKVHEGRY	QCNRRRGSYI	LHVRDPKGVG	AGAGEPTESG			
	160	170	180				
190	200	YQTIDELTPN	SADDFFTRAW	LEQQQQQQQL			
PHQSHKLHKS	HLGYGNASLS		210				
220	230	240		250	GSQPWHPSAG		
GGGIHRVYSA	TPPDFPPRL	NLLEQTVAPP	EPPTILYNPN				
260	270	280	290				300
PTHPTASATA	TETSVLLTTA	HHHAHHQQQL	QQSQHTLNA	FQLPLPPRPN			
	310	320	330				
340	350	PGQNER YQTY	APHYVPPVVV	SGAGAGAGAD			
PGAGASGEQT	TISAATSTRA		360	370			
380	390	400	MMGGGGGVAG	AGFSAGASGP			
MLGAGGHMLM	GGQGHQVHLQ	HQTLLPVKMD					
410	420	430	440				450
KLVPNYDNAE	HQMKFYDIRS	PLVLSCNVKD	GTPGGVLIWK	KNGTAVTDVP			
	460	470	480				
490	500	SLRGRFKLIA	DENKFIIDKT	DTNDDGKYSC			
EFDGVSKEIE	VIARVVVRVP		510	520			
530	540	550	SNTAVVEGEK	MSVTCSVVG T			
KPELTWTFAN	VTLTNATDRF	ILKPDDNGVP		560			
570	580	590	600	NAILTLDNVT			
LDDRGEYKCI	GRNAANVYGG	NTTTPASDVT	TVRVKGKFAA				
610	620	630	640				650
LWPFLGICAE	VLILCIILI	YEKRRNKSEL	EESDTPQEQ	KKKRRNYD			

A cDNA and a chromosomal sequence encoding the BSG protein is available from the NCBI database as accession no, AE014134 and AAN10661.2, respectively.

[0093] An amino acid sequence for the protein encoded by the human BTBD2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9BX70, shown below as SEQ ID NO: 43.

TABLE-US-00044		10	20		
30	40	50	MAAGGSGGRA	SCPPGVGVGP	
GTGGSPGPSA	NAAATPAPGN	AAAAAAAAAA			
60	70	80	90		
100	AAAAPGPTPP	APPGPGTDAQ	AAGAERAEEA	AGPGAAALQR	EAAYNWQASK
	110	120	130		
140	150	PTVQERFAFL	FNNEVLCDVH	FLVGKGLSSQ	
RIPAHRFVLA	VGSAVFDAMF		160		
170	180	190	200	NGGMATTSTE	
IELPDVEPAA	FLALLKFLYS	DEVQIGPETV	MTTLYTAKKY		
210	220	230	240		250
AVPALEAHCV	EFLKKNLRAD	NAFMLLTQAR	LFDEPQLASL	CLENIDKNTA	
	260	270	280		
290	300	DAITAEGFTD	IDLDTLVAVL	ERDTLGIREV	
RLFNAVVRWS	EAECQRQQLQ		310		
320	330	340	350	VTPENRRKVL	
GKALGLIRFP	LMTIEEFAAG	PAQSGILVDR	EVVSLFLHFT		
360	370	380	390		400

VNPKPRVEFI DRPRCCLRGK ECSINRFQV ESRWGYSGTS DRIRFSVNKR
410 420 430
440 450 IFVVGFGLYG SIHGPTDYQV NIQIIHTDSN
TVLGQNDTGF SCDGSASTFR 460
470 480 490 500 VMFKEPVEVL
PNVNYTACAT LKGPDSHYGT KGLRKVTHES PTTGAKTCFT
510 520 FCYAAGNNNG TSVEDGQIPE VIFYT

A cDNA and a chromosomal sequence encoding the BTBD2 protein is available from the NCBI database as accession no. AF355797 and AC004678, respectively.

[0094] An amino acid sequence for the protein encoded by the human CASTOR2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. A6NHX0, shown below as SEQ ID NO:44.

TABLE-US-00045 10 20
30 40 50 MELHILEHRL QVASVAKESI
PLFTYGLIKL AFLSSKTRCK FFSLTETPED 60
70 80 90 100 YTIIVDEEGF
LELPSSSEHLS VADATWLALN VVSGGGSFSS SQPIGVTKIA
110 120 130 140 150
KSVIAPLADQ NISVFMLSTY QTDFILVRER DLPFVTHTLS SEFTILRVVN
160 170 180
190 200 GETVAAENLG ITNGFVKPKL VQRPVIHPLS
SPSNRFCVTS LDPDTLPAVA 210 220
230 240 250 TLLMDVMFYS NGVKDPMATG
DDCGHIRFFS FSLIEGYISL VMDVQTQQR 260
270 280 290 300 PSNLLFTSAS
GELWK MVRIG GQPLGFDECG IVAQISEPLA AADIPAYYIS
310 320 TFKFDHALVP EENINGVISA LKVSQAEKH

A cDNA and a chromosomal sequence encoding the CASTOR2 protein is available from the NCBI database as accession no. BC147030 and AC245150, respectively.

[0095] An amino acid sequence for the protein encoded by the human CCSER1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no, Q9C013, shown below as SEQ ID NO:45.

TABLE-US-00046 10 20
30 40 50 MGDSGSRRST LVSRLPIFRR
SINRRHDSLP SSPSSSNTVG VHSSSPSSTN 60
70 80 90 100 SSSGSTGKRR
SIFRTPSISF HHKKGSEPKQ EPTNQNLIS NGAQPGHSNM
110 120 130 140 150
QKLSLEEHIK TRGRHSVGFS SSRNKKITRS LTEDFEREKE HSTNKNVFIN
160 170 180
190 200 CLSSGKSEGD DSGFTEDQTR RSVKQSTRKL
LPKSFESSHYK FSKPVLQSQS 210
220 230 240 250 ISLVQQSEFS
LEV TQYQERE PVLVRASPSC SVDVTERAGS SLQSPLLSAD
260 270 280 290 300
LTTAQTPSEF LALTEDSVSE MDAFSKSGSM ASHCDNFGHN DSTSQMSLNS
310 320 330
340 350 AAVIKTTTEL TGTVPCAIMS PGKYRLEGQC
STESNSLPET SAANQKEVLL 360
370 380 390 400 QIAELPATSV

SHSESNLPAD	SEREENIGLQ	NGETMLGTNS	PRKLGFYEQH	
410	420	430	440	450
KAIAEHVKGI	HPISDSKIIP	TSGDHHIFNK	TSHGYEANPA	KVLASSLSPE
	460	470	480	
490	500	REGRFIERRL	RSSEGTAGS	SRMILKPKDG
NIEEVNSLRK	QRAGSSSSKM		510	
520	530	540	550	NSLDVLNNLG
SCELEDDLM	LDLEFLEEQS	LHPSVCREDS	YHSVVSCAAV	
560	570	580	590	600
VLTPMEPMIE	MKKREEPEFP	EPSKQNLCLK	LTKDVDQEAR	CSHISRMFNS
	610	620	630	
640	650	PSADWPLQGV	EENGIDSLP	FRLMLQDCTA
VKTLCLKMKR	VLQESADMSP		660	
670	680	690	700	ASSTTSLPVS
PLTEEPVPFK	DIMKDECSML	KLQLKEKDEL	ISQLQEELGK	
710	720	730	740	750
VRHLQKAFAS	RVDKSTQTEL	LCYDGLNLKR	LETVQGGREA	TYRNRIVSQN
	760	770	780	
790	800	LSTRDRKAIH	TPTEDRFRYS	AADQTSPYKN
KTCQLPSLCL	SNFLKDKELA		810	
820	830	840	850	EVIKHSRGTY
ETLTSDVTQN	LRATVGQSSL	KPTAKTEGLS	TFLEKPKDQV	
860	870	880	890	900
ATARQHSTFT	GRFGQPPRGP	ISLHMYSRKN	VFLHHNLHST	ELQTLGQQDG

A cDNA and a chromosomal sequence encoding the CCSER1 protein is available from the NCBI database as accession no. AB051467 and AC093729, respectively.

[0096] An amino acid sequence for the protein encoded by the human CLCN2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P51788, shown below as SEQ ID NO:46,

TABLE-US-00047	10	20		
30	40	50	MAAAAAEEGM	EPRALQYEQT
LMYGRYTQDL	GAFAKEEAAR	IRLGGPEPWK		
60	70	80	90	
100	GPPSSRAAPE	LLEYGRSRCA	RCRVCSVRCH	KFLVSRVGED
	110	120	130	WIFLVLLGLL
140	150	MALVSWVMDY	AIAACLQAQQ	WMSRGLNTSI
LLQYLAWVTY	PVVLITFSAG		160	
170	180	190	200	FTQILAPQAV
GSGIPEMKTI	LRGVVLKEYL	TLKTFIAKVI	GLTCALGSGM	
210	220	230	240	250
PLGKEGPFVH	IASMCAALLS	KFLSLFGGIY	ENESRNTEML	AAACAVGVGC
	260	270	280	
290	300	CFAAPIGGVL	FSIEVTSTFF	AVRNYWRGFF
AATFSAFIFR	VLAVWNRDEE		310	
320	330	340	350	TITALFKTRF
RLDFPFDLQE	LPAFAVIGIA	SGFGGALFVY	LN RKIVQVMR	
360	370	380	390	400
KQKTINRFLM	RKRLLPALV	TLLISTLTFP	PGFGQFMAGQ	LSQKETLVTL
	410	420	430	
440	450	FDNRTWVRQG	LVEELEPPST	SQAWNPPRAN

VFLTLVIFIL	MKFWMSALAT			460	470
480	490		500 TIPVPCGAFM	PVEVIGAAFG	
RLVGESMAAW	FPDGIHTDSS	TYRIVPGGYA		510	
520	530		540	550 VVGAAALAGA	
VTHTVSTAVI	VFELTGQIAH	ILPVMIAVIL	ANAVAQSLQP		
560	570		580	590	600
SLYDSIIRIK	KLPYLPGLGW	GRHQYRVRV	EDIMVRDVP	VALSCTFRDL	
	610	620		630	
640	650 RLALHRTKGR	MLALVESPE	MILLGSIERS		
QVVALGAL	SPARRRQHM		660		
670	680		690	700 ERRATQTSPL	
SDQEGPPTPE	ASVCFQVNTE	DSAFPAARGE	THKPLKPALK		
710	720		730	740	750
RGPSVTRNLG	ESPTGSAESA	GIALRSLFCG	SPPPEAASEK	LESCEKRKLK	
	760	770		780	
790	800 RVRISLASDA	DLEGEMSPEE	ILEWEEQQLD		
EPVNFSDCKI	DPAPFQLVER		810	820	
830	840		850 TSLHKTHTIF	SLLGVDHAYV	
TSIGRLIGIV	TLKELRKAIE	GSVTAQGVKV		860	
870	880		890 RPPLASFRDS	ATSSSDTETT	
EVHALWGPHS	RHGLPREGSP	SDSDDKCQ			

A cDNA and a chromosomal sequence encoding the CLCN2 protein is available from the NCBI database as accession no. S77770 and AC078797, respectively.

[0097] An amino acid sequence for the protein encoded by the human EBF2 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9 AK2, shown below as SEQ ID NO:47.

TABLE-US-00048		10		20	
30	40		50 MFGIQDTLGR	GPTLKEKSLG	
AEMDSVRSWV	RNVGVVDANV	AAQSGVALSR			
60	70		80	90	
100 AHFEKQPPSN	LRKSNNFFHFV	LALYDRQGQP	VEIERTAFVD	EVENDKEQGN	
	110	120		130	
140	150 EKTNNNGTHYK	LQLLYSNGVR	TEQDLYVRLI		
DSVTKQPIAY	EGQNKNPEMC		160		
170	180		190	200 RVLLTHEVMC	
SRCCEKKSCG	NRNETPSDPV	IIDRFFLKFF	LKCNQNCLKT		
210	220		230	240	250
AGNPRDMRRF	QVVLSTTVNV	DGHVLAVSDN	MFVHNNNSKHG	RRARRLDPSE	
	260	270		280	
290	300 ATPCIKAISP	SEGWTTGGAM	VIIIGDNFFD		
GLQVVFGTML	VWSELITPHA		310		
320	330		340	350 IRVQTTPRHI	
PGVVEVTLSY	KSKQFCKGAP	GRFIYTALNE	PTIDYGFQRL		
360	370		380	390	400
QKVIPRHPGD	PERLAKEMLL	KRAADLVEAL	YGTPHNNQDI	ILKRAADIAE	
	410	420		430	
440	450 ALYSVPRNPS	QLPALSSSPA	HSGMMGINSY		
GSQLGVSISE	STQGNNGQYI		460	470	
480	490		500 RNTSSISPRG	YSSSSTPQQS	
NYSTSSNSMN	GYSNVPMANL	GVPGSPGFLN		510	

520 530 540 550 GSPTGSPYGI
MSSSPTVGSS STSSILPFSS SVFPAVKQKS AFAPVIRPQG
560 570 SPSPACSSGN GNGFRAMTGL VVPPM
A cDNA and a chromosomal sequence encoding the EBF2 (COE2) protein is available from the NCBI database as accession no. AY700779 and AC023566, respectively,
[0098] An amino acid sequence for the protein encoded by the human FAM83A gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q86UY5, shown below as SEQ ID NO:48.

TABLE-US-00049

	10	20		
30	40	50 MSRSRHLGKI	RKRLEDVKSQ	
WVRPARADFS	DNESARLATD	ALLDGGSEAY		
60	70	80	90	
100 WRVLSQEGEV	DFLSSVEAQY	IQAQAREPPC	PPDTLGGAEA	GPKGLDSSSL
	110	120	130	
140	150 QSGTYFPVAS	EGSEPALLHS	WASAEKPYLK	
EKSSATVYFQ	TVKHNNIRD	160		
170	180	190	200 VRRCITRISQ	
VLVILMDVFT	DVEIFCDILE	AANKRGVFVC	VILDQGGVKL	
210	220	230	240	250
FQEMCDKVQI	SDSHLKNISI	RSVEGEIYCA	KSGRKFAQI	REKFIISDWR
	260	270	280	
290	300 FVLSGSYSFT	WLCGHVHRNI	LSKFTGQAVE	
LFDEEFRHLY	ASSKPVMGLK	310		
320	330	340	350 SPRLVAPVPP	
GAAPANGRLS	SSSGSASDRT	SSNPFSGRSA	GSHPGTRSVS	
360	370	380	390	400
ASSGPCSPAA	PHPPPPPRFQ	PHQGPWGAPS	PQAHLSRPH	DGPPAAVYSN
	410	420	430 LGAYRPTRLQ	
LEQLGLVPRL	TPTWRPFLQA	SPHF		

A cDNA sequence encoding the FAM83A protein is available from the NCBI database as accession no. DQ280322.

[0099] An amino acid sequence for the protein encoded by the human FOXF1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database, shown below as SEQ ID NO:49.

TABLE-US-00050

	10	20		
30	40	50 MSSAPEKQQP	PHGGGGGGGG	
GGGAAMDPAS	SGPSKAKKTN	AGIRRPEKPP	60	
70	80	90	100 YSYIALIVMA	
IQSSPTKRLT	LSEIYQFLQS	RFPFFRGSYQ	GWKNSVRHNL	
110	120	130	140	150
SLNECFIKLP	KGLGRPGKGH	YWTIDPASEF	MFEEGSFRRR	PRGFRRKCQA
	160	170	180	
190	200 LKPMYSMMNG	LGFNHLPD	TY GFQGSAGGLS	
CPPNSLALEG	GLGMMNGHLP	210		
220	230	240	250 GNVDGMALPS	
HSVPHLPSNG	GHSYMGGCGG	AAAGEYPHHD	SSVPASPLLP	
260	270	280	290	300
TGAGGVMEPH	AVYSGSAAAW	PPSASAALNS	GASYIKQQPL	SPCNPAANPL
	310	320	330	
340	350 SGSLSTHSLE	QPYLHQNSHN	APAELQGIPR	

YHSQSPSMCD RKEFVFSFNA 360 370

MASSSMHSAG GGSYYHQQVT YQDIKPCVM

A cDNA and a chromosomal sequence encoding the FOXF1 protein is available from the NCBI database as accession no. U13219 and AF085343, respectively.

An amino acid sequence for the protein encoded by the human FOXL3 gene that is a negative regulator of T cells as detected by interferon-production is available from the UniPROT database as accession no. A8MTJ6, shown below as SEQ ID NO:50.

TABLE-US-00051

10	20		
30	40	50	MALYCGDNFG VYSQPGLPPP
AATAAAPGAP	PAARAPYGLA	DYAAPPAAAA	
60	70	80	90
100	NPYLWLNGPG	VGGPPSAAAA	AAAAYLGAPP PPPPPGAAAG PFLQPPPAAG
	110	120	130
140	150	TFGCSQRPEA	QPAPAAPASP AAPAGPGELG
WLSMASREDL	MKMVRPPYSY	160	
170	180	190	200
APERKLTLSH	IYQFVADSF	FYQRSKAGWQ	NSIRHNLSLN
210	220	230	240 250
DCFKKVPRDE	DDPGKGNWT	LDPNCEKMFD	NGNFRRKRKR RSEASNGSTV
	260	270	280
290	300	AAGTSKSEEG	LSSGLGSGVG GKPEEESPST
LLRPSHSPEP	PEGTKSTASS	310	320
330	340	350	PGGPMLTSTP CLNTFFSSLS
SLSVSSSVST	QRALPGSRHL	GIQGAQLPSS	360
370	380	390	400
ASADTLQLSN	STSNSTGQRS	SYYSPPFAST	SGGQSSPFSS
410	420	PFHNFSMVNS	LIYPREGSEV

A cDNA and a chromosomal sequence encoding the *FOXL3* protein is available from the NCBI database as accession no. BN001222 and AC012671, respectively.

[0100] An amino acid sequence for the protein encoded by the human FOXL2NB gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q6ZUIU3, shown below as SEQ ID NO:51.

TABLE-US-00052

10	20		
30	40	50	MTRTPVGSAR TRPKPRKLGP
QRGKALQASS	RLSESPALVK	KRMPDACTLG	
60	70	80	90
100	RAGIGLPKMC	LHMAVRHASKA	QKTGPGILQQ RQKPPAPRAS GGPALLGKRR
	110	120	130
140	150	GCSEAGSASL	EPLSSSRAAA GCLNQVPLSP
FLAGPRNTRR	LPAPERERIE	160	170
PLRCLASKGK	LHCVY		LAATLCLEGW

A cDNA and a chromosomal sequence encoding the FOXL2NB protein is available from the NCBI database as accession no. AK125319 and AC092947, respectively.

[0101] An amino acid sequence for the protein encoded by the human HYLS1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q96M1 1, shown below as SEQ ID NO:52.

TABLE-US-00053

10	20		
30	40	50	MEELLPDGQI WANMDPEERM
LAAATAFTHI	CAGQGEDVR	REAQSIQYDP	60
70	80	90	100
			YSKASVAPGK

RPALPVQLQY	PHVESNPVSE	TVSEASQRLR	KPVMKRKVLR	
110	120	130	140	150
RKPDGEVLVT	DESIISESES	GTENDQDLWD	LRQRLMNVQF	QEDKESSFDV
	160	170	180	
190	200	SQKFNLPEHY	QGISQDQLIC	SLQREGMGSP
AYEQDLIVAS	RPKSFILPKL		210	220
230	240		250	DQLSRNRGKT
RDWDSIRLPG	EDHRKELRWG	VREQMLCRAE		260
270	280	290	PQSKPQHIYV	PNNYLVPTK
KRSALRWGVR	CDLANGVIPR	KLPFPLSPS		
A cDNA and a chromosomal sequence encoding the HYLS1 protein is available from the NCBI database as accession no, AK057477 and AP000842, respectively.				
[0102] An amino acid sequence for the protein encoded by the human LAMB1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P07942, shown below as SEQ ID NO:53,				
TABLE-US-00054		10	20	
30	40	50	MGLLQLLAFS	FLALCRARVR
AQEPEFSYGC	AEGSCYPATG	DLIGRAQKL		60
70	80	90		100
PEPYCIVSHL	QEDKKCFICN	SQDPYHETLN	PDSHLIENVV	
110	120	130	140	150
TTFAPNRLKI	WWQSENGVEN	VTIQLDLEAE	FHFTHLIMTF	KTFRPAAMLI
	160	170	180	
190	200	ERSSDFGKTW	GVYRYFAYDC	EASFPGISTG
PMKKVDDIIC	DSRYSDIEPS		210	220
230	240	250	TEGEVIFRAL	DPAFKIEDPY
SPRIQNLLKI	TNLRIKFVKL	HTLGDNLLDS		260
270	280	290		300
AVYDMVVRGN	CFCYGHASEC	APVDGFNEEV	EGMVHGHCMC	
310	320	330	340	350
RHNTKGLNCE	LCMDFYHDLP	WRPAEGRNSN	ACKKCNCNEH	SISCHFDMAV
	360	370	380	
390	400	YLATGNVSGG	VCDDCQHNTM	GRNCEQCKPF
YYQHPERDIR	DPNFCERCTC		410	
420	430	440		450
CDSYTDFTSG	LIAGQCRCKL	NVEGEHCDVC	KEGFYDLSSE	
460	470	480	490	500
DPFGCKSCAC	NPLGTIPGGN	PCDSETGHCY	CKRLVIGQHC	DQCLPEHWGL
	510	520	530	
540	550	SNDLDGCRPC	DCDLGGALNN	SCFAESGQCS
CRPHMIGRQC	NEVEPGYYFA		560	
570	580	590		600
EANIGPGVSI	VERQYIQDRI	PSWTGAGFVR	VPEGAYLEFF	
610	620	630	640	650
IDNIPYSMEY	DILIRYEPQL	PDHWEKAVIT	VQRPGRIPTS	SRCGNTIPDD
	660	670	680	
690	700	DNQVVSLSPG	SRYVVLPRPV	CFEKGTTYTV
RLELPQYTSS	DSDVESPYTL		710	720
730	740	750	IDSLVLMPYC	KSLDIFTVGG
SGDGVVTNSA	WETFQRYRCL	ENSRSVVKTP		760

770		780	790	800	MTDVCRNIIF	
SISALLHQTG	LACECDPQGS	LSSVCDPNGG	QCQCRPNVVG			
810	820	830	840			850
RTCNRCAPGT	FGFGPSGCKP	CECHLQGSVN	AFCNPVTGQC	HCFQGVYARQ		
	860	870	880			
890	900	CDRCLPGHWG	FPSCQPCQCN	GHADDCDPVT		
GECLNCQDYT	MGHNCRCLA		910			
920	930	940		950	GYYGDPHIGS	
GDHCRPCPCP	DGPDSGRQFA	RSCYQDPVTL	QLACVCDPGY			
960	970	980	990			1000
IGSRCDDCAS	GYFGNPSEVG	GSCQPCQCHN	NIDTTDPEAC	DKETGRCLKC		
	1010	1020	1030			
1040	1050	LYHTEGEHCQ	FCRFGYYGDA	LQQDCRKCVC		
NYLGTVQEH	NGSDCQCDKA		1060	1070		
1080	1090	1100	TGQCLCLPNV	IGQNCDRCAP		
NTWQLASGTG	CDPCNCNAAH	SFGPSCNEFT		1110		
1120	1130	1140		1150	GQCQCMPGFG	
GRTCSECQEL	FWGDPDVECR	ACDCDPRGIE	TPQCDQSTGQ			
1160	1170	1180	1190			1200
CVCVEGVEGP	RCDKCTRGYS	GVFPDCTPCH	QCFALWDVII	AELTNRTHRF		
	1210	1220	1230			
1240	1250	LEKAKALKIS	GVIGPYRETV	DSVERKVSEI		
KDILAQSPAA	EPLKNIGNLF		1260	1270		
1280	1290	1300	EEAEKLIKDV	TEMMAQVEVK		
LSDTTSQSNS	TAKELDSLQT	EAESLDNTVK		1310		
1320	1330	1340		1350	ELAEQLEFIK	
NSDIRGALDS	ITKYFQMSLE	AEERVNASTT	EPNSTVEQSA			
1360	1370	1380	1390			1400
LMRDRVEDVM	MERESQFKEK	QEEQARLLDE	LAGKLQSLDL	SAAAEMTCGT		
	1410	1420	1430			
1440	1450	PPGASCSETE	CGGPNCRTDE	GERKCGGPGC		
GGLVTVAHNA	WQKAMDLDQD		1460	1470		
1480	1490	1500	VLSALAEVEQ	LSKMVSEAKL		
RADEAKQSAE	DILLKTNATK	EKMDKSNEEL		1510		
1520	1530	1540		1550	RNLIKQIRNF	
LTQDSADLDS	IEAVANEVLK	MEMPSTPQQL	QNLTEDIRER			
1560	1570	1580	1590			1600
VESLSQVEVI	LQHSAADIAR	AEMLLLEEAKR	ASKSATDVKV	TADMVKEALE		
	1610	1620	1630			
1640	1650	EAEKAQVAAE	KAIKQADEDI	QGTQNLITSI		
ESETAASEET	LFNASQRISE		1660	1670		
1680	1690	1700	LERNVEELKR	KAAQNSGEAE		
YIEKVVYTVK	QSAEDVKKTL	DGELDEKYKK		1710		
1720	1730	1740		1750	VENLIAKKTE	
ESADARRKAE	MLQNEAKTLL	AQANSKLQLL	KDLERKYEDN			
1760	1770	1780	QRYLEDKAQE	LARLEGEVRS		
LIKDISQKVA	VYSTCL					

A cDNA and a chromosomal sequence encoding the LAMB1 protein is available from the NCBI database as accession no. M61916 and M61950, respectively.

[0103] An amino acid sequence for the protein encoded by the human LENEP gene that is a

negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9Y5L5, shown below as SEQ ID N0:54.

TABLE-US-00055	10	20	
30	40	50 MQPRTQPLAQ	TLPFFLGAP
RDTGLRVPVI	KMG TGWEGFQ	RTLKEVAYIL	60 LCCWCIKELL D

A cDNA and a chromosomal sequence encoding the LENEP protein is available from the NCBI database as accession no. AF268478 and AF144412, respectively.

[0104] An amino acid sequence for the protein encoded by the human LRRC4B gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9NT99, shown below as SEQ ID NO:55.

TABLE-US-00056	10	20	
30	40	50 MARARGSPCP	PLPPGRMSWP
HGALLFLWLF	SPPLGAGGGG	VAVTSAAGGG	
60	70	80	90
100 SPPATSCPVA	CSCSNQASRV	ICTRRDLAEV	PASIPVNTRY LNLQENGIQV
	110	120	130
140	150 IRTDTFKHLR	HLEILQLSKN	LVRKIEVGAF
NGLPSLNTLE	LFDNRLTTVP		160 170
180	190	200 TQAFEYLSKL	RELWLRNNPI
ESIPSYAFNR	VPSLRRLDLG	ELKRLEYISE	210
220	230	240	250 AAFEGLVNLR
YLN LGMCNLK	DIPNL TALVR	LEELELSGNR	LDLIRPGSFQ
260	270	280	290 300
GLTSLRKLWL	MHAQVATIER	NAFDDLKSLE	ELNLSHNNLM SLPHDLFTPL
	310	320	330
340	350 HRLERVHLNH	NPWHCNC DVL	WLSWWLKETV
PSNTTCCARC	HAPAGLKGRY		360
370	380	390	400 IGELDQSHFT
CYAPVIVEPP	TDLNVTEGMA	AELKCRTGTS	MTSVNWLTPN
410	420	430	440 450
GTLMTHGSYR	VRISVLHDGT	LNFTNVTVQD	TGQYTCMVTN SAGNTTASAT
	460	470	480
490	500 LNVSAVDPVA	AGGTGSGGGG	PGGSGGVGGG
SGGYTYFTTV	TVETLETQPG		510
520	530	540	550 EEALQPRGTE
KEPPGPTTDG	VWGGGRPGDA	AGPASSTTA	PAPRSSRPTE
560	570	580	590 600
KAFTVPITDV	TENALKDLDD	VMKTTKIIIG	CFVAITFMAA VMLVAFYKLR
	610	620	630
640	650 KQHQLHKHHG	PTRTVEIINV	EDELPAASAV
SVAAAAVAS	GGGVGGDSHL		660
670	680	690	700 ALPALERDHL
NHHHYVAAAF	KAHYSSNPSG	GGCGGKGPPG	LNSIHEPLL F 710
KSGSKENVQE	TQI		

A cDNA and a chromosomal sequence encoding the LRRC4B protein is available from the NCBI database as accession no. BC019687 and AC008743, respectively.

[0105] An amino acid sequence for the MAB21L2 protein encoded by the human gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9Y586, shown below as SEQ [D NO:56.

TABLE-US-00057	10	20
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30		50	MIAAQAKLVY	QLNKYYTERC	
QARKAAIAKT	IREVCKVVSD	VLKEVEVQEP		60	
70	80	90		100	RFISSLSEID
ARYEGLEVIS	PTEFEVVLVL	NQMGVFNFVD	DGSLPGCAVL		
110	120	130	140		150
KLSDGRKRSM	SLWVEFITAS	GYLSARKIRS	RFQTLVAQAV	DKCSYRDVVK	
	160	170	180		
190	200	MIADTSEVKL	RIRERYVVQI	TPAFKCTGIW	
PRSAAQWPMP	HIPWPGPNRV		210		
220	230	240		250	AEVKAEGFNL
LSKECYSLTG	KQSSAESDAW	VLQFGEAENR	LLMGGCRNKC		
260	270	280	290		300
LSVLKTLRDR	HLELPGQPLN	NYHMKTLLEY	ECEKHPRETD	WDESCLGDRD	
	310	320	330		
340	350	NGILLQLISC	LQCRRCPHYF	LPNLDLFQGK	
PHSALESAAK	QTWRLAREIL	TNPKSLDKL			

A cDNA and a chromosomal sequence encoding the MAB21L2 protein is available from the NCBI database as accession no. AF262032 and A-155219. respectively.

[0106] An amino acid sequence for the protein encoded by the human RETREGI gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q9H6L5, shown below as SEQ ID NO:57.

TABLE-US-00058		10	20	
30	40	50	MASPAPPEHA	EEGCPAPAAE
EQAPPSPPPP	QASPAERQQQ	EEEAQEAGAA		60
70	80	90		100
AGRAAAAVTW	LLGEPVLWLG	CRADELLSWK	RPLRSLLGFV	
110	120	130	140	150
AANLLFWFLA	LTPWRVYHLI	SVMILGRVIM	QIIKDMVLSR	TRGAQLWRSL
	160	170	180	
190	200	SEWEVINSK	PDERPRLSHC	IAESWMNFSI
FLQEMSLFKQ	QSPGKFCLLV		210	
220	230	240		250
GSYIPGVILS	YLLLLCAFLC	PLEKCNDIGQ	KIYSKIKSVL	
260	270	280	290	300
LKLDFGIGEY	INQKKRERSE	ADKEKSHKDD	SELDFSALCP	KISLTVAAKE
	310	320	330	
340	350	LSVSDTDVSE	VSWTDNGTFN	LSEGYTPQTD
TSDDLDRPSE	EVFSRDLSDF		360	370
380	390	400	PSLENGMGTN	DEDELSLGLP
TELKRKKEQL	DSGHRPSKET	QSAAGLTLPL		410
420	430	440		450
NLAGDVITAA	VIAAIKDQLE	GVQQALSQAA	PIPEEDTDTE	
460	470	480		490
SELDQIESEL	GLTQDQHAEA	QQNKKSSGFL	SNLLGGH	

[0107] A cDNA sequence encoding the RETREG1 protein is available from the NCBI database as accession no. AK000159.

[0108] An amino acid sequence for the protein encoded by the human SMAD9 gene that is a negative regulator of T cells as detected by interferon-production is available from the UniPROT database as accession no. 015198, shown below as SEQ ID NO:58.

TABLE-US-00059	10	20
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30	40	50	60	70	80	90	
LLGWKQGDEE	EKWAEEKAVDS	LVKKLKKKKG					
100	110	120	130	140	150	160	170
AMDELERALS	CPGQPSKCVT	IPRSLDGRLQ	VSHRKGLPHV	IYCRVWRWPD			
140	150	160	170	180	190	200	210
YRRVETPVLP	PVLVPRHSEY	ECCEFPGFSK	QKEVCINPYH				
180	190	200	210	220	230	240	250
MPHNATYPDS	FQQPPCSALP	PSPSHAFSQS	RSASLHSEPL				
220	230	240	250	260	270	280	290
GSPSEPESPY	QHSVDTPPLP	YHATEASETQ	SGQPVDATAD				
260	270	280	290	300			
RHVVLSSIPNG	DFRPVCYEEP	QHWCSVAYYE	LNNRVGETFQ	ASSRSVLIDG			
340	350	360	370	380	390	400	410
HIGKGVHLYY	VGGEVYAECV	FCLGLLSNVN	RNSTIENTRR				
370	380	390	400	410	420	430	440
NCNYQHGFHP	ATVCKIPSGC	SLKVFNNQLF	AQLLAQSVHH				
410	420	430	440	450			
GFEVVYELTK	MCTIRMSFVK	GWGAEYHRQD	VTSTPCWIEI	HLHGPLQWLD			
	460						

A cDNA and a chromosomal sequence encoding the SMAD9 protein is available from the NCBI database as accession no. D83760 and AL138706, respectively.

[0109] An amino acid sequence for the protein encoded by the human SPATA31A1 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. Q5TZJ5, shown below as SEQ ID NO:59.

TABLE-US-00060		10	20	
30	40	50	60	
STPWVLDIFL	TLVFALGFFF	LLLPLYLSYFR	LSASSLNAPS	
70	80	90	100	
KRKCPVGRRR	RPRGRMKNHS	LRAGRECPRG	LQETSDLLSQ	
110	120	130	140	150
LQSLLGPHLD	KGDFGQLSGP	DPPGEVGERA	PDGASQSSHE	PMEDAAPILS
190	200	210	220	
LSASQPPEPS	LPLEHPSPEP	HPQDLASTPS	PGPMTTSVSS	
230	240	250	260	
PKGFTAPPLR	DSTLITPSHC	DSVALPLGTV	PDPLACSPPP	
270	280	290	300	
VASVPAISGL	GGSNSHVSAS	SRWQETARTS	CAFNSSVQQD	
310	320	330	340	350
HLSRHPPETY	QMEAGSLFLL	SSDGQNAVGI	QVTETAKVNI	WEEKENVGSF
390	400	410	420	
PFWNMGGENSK	QLPGPQKLS	NSLRNLAKSL	DAEQDTTNPK	
420	430	440	450	
NYSQLFWGLP	SLHSESLVAN	AWVTDRSYTL	QSPPELFNEM	
460	470	480	490	500
SNVCPIQRET	TMSPLLFQAQ	PPSHLGPECQ	PFISSTPQFR	PTPMAQAEAQ
	510	520	530	

540	550	560	570	580	590	600	610	620	630	640	650
KVQALSLPET	QHPEWPLLRR										
570	580	590	600	610	620	630	640	650			
RVQKSQDVFS	VSTPNLPQES	LTSILPENFP	VSPELRRQLE								
610	620	630	640	650							
QHIKKWIIQH	WGNLGRIQES	LDLMQLRDES	PGTSQAKGKP	SPWQSSMSTG							
	660	670	680								
690	700	710	720	730	740	750	760	770	780	790	800
NLSRDMKSF	RKVLGVTSEE										
720	730	740	750	760	770	780	790	800			
SDSGSDLLRC	TERTHIENIL	KAHMGRNLGQ	TNEGLIPVRV								
760	770	780	790	800							
RRSWLAVNQA	LPVSNTHVKT	SNLAAPKSGK	ACVNTAQVLS	FLEPCTQQGL							
	810	820	830								
840	850	860	870								
VSSLSTQLA	GPSSATCESG										
880	890	900	910	920	930	940	950	960	970	980	990
KQVLTKASDH	MPESLLASSP	AWKQFQRAPR									
920	930	940	950	960	970	980	990	1000			
LKPPPAGQEG	RWPSKPLTYS	LTGSTQQSRS	LGAQSSKAGE								
960	970	980	990	1000							
TREAVPQCRV	PLETCMLANL	QATSEDMHGF	EAPGTSKSSL	HPRVSVSQDP							
	1010	1020	1030								
1040	1050	1060	1070								
VVLLPDGQAS	VVPHASENLV										
1080	1090	1100	1110	1120	1130	1140	1150	1160	1170	1180	1190
ELHDLMAARR	SKLVHEEPRN	PNCQGSCKNQ									
1120	1130	1140	1150	1160	1170	1180	1190	1200			
EKSARKPNLEK	HEERLEGLRT	PQLTPVRKTE	DTHQDEGVQL								
1160	1170	1180	1190	1200							
LPSKKQPPSV	SHFGGNIKQF	FQWIFSKKKS	KPAPVTAESQ	KTVKNRSCVY							
	1210	1220	1230								
1240	1250	1260	1270								
KVNQHKQKFQ	APVCGFPCNH										
1280	1290	1300	1310	1320	1330	1340					
ATLKSQGCNP	RDRQIRNQQP	LKSVRCNNEQ									
1320	1330	1340									
LQHWPKTSGA	SSHHHHCPRH	CLLWEGI									

A chromosomal sequence encoding the SPATA31A1 protein is available from the NCBI database as accession no. BX005214.

[0110] An amino acid sequence for the protein encoded by the human ZNF445 gene that is a negative regulator of T cells as detected by interferon- γ production is available from the UniPROT database as accession no. P59923, shown below as SEQ ID NO:60.

TABLE-US-00061	10	20			
30	40	50	MPPGRWHAAY	PAQAQSSRER	
GRLQTVKKEE	EDESYPVQA	ARPQTLNRPG			
60	70	80	90		
100	110	120	130		
QELFRQLFRQ	LRYHESSGPL	ETLSRLRELC	RWWLRPDVLS	KAQILELLVL	
	110	120	130		
140	150				
EQFLSILPGE	LRVWVQLHNP	ESGEEAVALL			

EELQRDL DGT	SWRDPGPAQS		160	
170	180	190	200	PDVHWMGTGA
LRSAQIWSLA	SPLRSSSALG	DHLEPPYEIE	ARDFLAGQSD	
210	220	230	240	250
TPAAQMPALF	PREGCPGDQV	TPTRSLTAQL	QETMTFKDVE	VTFSQDEWGW
	260	270	280	
290	300	LDSAQRNLYR	DVMLENYRNM	ASLVGPFTKP
ALISWLEARE	PWGLNMQAAQ		310	
320	330	340	350	PKGPNVAAPT
GDDLQSKTNK	FILNQEPLEE	AETLAVSSGC	PATSVSEGIG	
360	370	380	390	400
LRESFQQKSR	QKDQCENPIQ	VRVKKEETNF	SHRTGKDSEV	SGSNSLDLKH
	410	420	430	
440	450	VTYLRVSGRK	ESLKHGCGKH	FRMSSHHDYDY
KKYGKGLRHM	IGGFSLHQRI		460	
470	480	490	500	HSGLKGNKKD
VCGKDFSLSS	HHQRGQSLHT	VGVSFKCSDC	GRTFSHSSHL	
510	520	530	540	550
AYHQLRLHTQE	KAFKCRVCGK	AFRWSSNCAR	HEKIHTGVKP	YKCDLCEKAF
	560	570	580	
590	600	RRLSAYRLHR	ETHAKKKFLE	LNQYRAALTY
SSGFDHHLGD	QSGEKLFDCS		610	
620	630	640	650	QCRKSFHCKS
YVLEHQRIHT	QEKPYKCTKC	RKTFRWRSNF	TRHMLRHEEE	
660	670	680	690	700
KFYKQDECRE	GFRQSPDCSQ	PQGAPAVEKT	FLCQQCGKTF	TRKKTLDVDHQ
	710	720	730	
740	750	RIHTGEKPYQ	CSDCGKDFAY	RSFIVHKKK
HAMKRKPEGG	PSFSQDTV FQ		760	
770	780	790	800	VPQSSH SKEE
PYKCSQCGKA	FRNHSFLLIH	QRVHTGEKPY	KCRECGKA FR	
810	820	830	840	850
WSSNLYRHQR	IHSLQKQYDC	HESEKTPNVE	PKILTGEKRF	WCQECGKTFT
	860	870	880	
890	900	RKRTL LDHKG	IHSGEKRYKC	NLCGKSYDRN
YRLVNHQRIH	STERPFKCQW		910	
920	930	940	950	CGKEFIGRHT
LSSHQRKHTR	AAQAERSPPA	RSSSQDTKLR	LQKLKPSEEM	
960	970	980	990	1000
PLEDCKEACS	QSSRLTGLQD	ISIGKKCHKC	SICGKTFNKS	SQLISHKRFH
	1010	1020	1030	TRERPFKCSK
CGKTFRWSSN	LARHMKNHIR	D		

A cDNA encoding the ZNF445 protein is available from the NCBI database as accession 45 no. AY262260.

[0111] The following genes are negative regulators of T cells as detected by Interleukin-2 production (see Table 5): ABI3BP, AEBP1, AHR, ANTXR2, ARHGAP15, ARHGAP27, ARHGDIB, ARID3A, ARL4D, B4GALNT3, BICD1, C10orf82, C17orf75, C19orf35, C1 RL, C2orf69, C6orf132, C9orf84, CABP1, CBLB, CCSER1, CD34, CD4, CD5, CD52, CEACAM11, CEACAM7, CEBPB, CES3, CGB3, COL11A1, COL4A3, COLQ, CPEB3, CRELD2, CST9L, DDX55, DLG4, DOK1, EBF3, EIF3K, EN2, EOMES, EPB41, ETSI, F5, FAM96A, FH1, FOXA3,

FOX13, FOX13, FOX13, FUS, FUT4, GCSAM, GCSAML, GDAP1L1, GPD2, (GMIP, GNL3L, GOLPH3, CRAP, GRB2, HAUS7, HERCI, HLA-DQB2, I-1SD17B11, IKZF1, IKZF3, INPPL1, INTS1, ITIH2, ITPKA, 1T1 KB, ITPKC, JDP2, JKAMP, JMJDIC, KIAA1024, KIF15, KIF5A, KNTC1, LAT2, LAX1, LGR5, LLIME, LMBRD2, LOC401052, LONP2, LRCH3, LRRC23, LRRC25, LRRC52, LYN, LYPDI, MAATS1, MAB21L2, MAGEB17, MAP4K1, MEF2C, METTL9, MICU1, MRPL17, MUC1, NAI1F1, NCF2, NDNF NDLUFB1, NH-P2, NKX2-6, NLGN4Y, NNL, NPIPB9, NR4AA, NR4A3, NRCAM, NRPI, NRSN2, NSUN7, OLFML1, OM1P, OPRD1, ORIK1, OR2BI1, OSBPL11, OTOG, OTUD4, PATL2, PAX5, PFKL, PHF2, PILBF1, P11P5KIA, PIP5KIB, P1TPNC1, PLCL1, PLEKHM2, PPARG, PP1C, PSRC1, PSTPIP1, PTPN12, PTPN22, PTPN6, PTPRC, PVRIG, RBP4, RPL13A, S100A2, SALL4, SAMD8, SENP6, SETDIB, SEZ6L, SFT2DI, SH3TC1, SIGIRR, SIT1, SLA, SLA2, SLC20A2, SLC39A2, SLC6A8, SMAGP, SNRNP48, SOCS2, SORBS1, SOX13, SPN, SPRED1, SPRFD2, SRPK1, STAP1, STK38L, SYPL1, TCF12, TEX35, TFCP2L1, TMEM14C, TME 1223, TMEIEM262, TNTNT2, TPRA1, TREM6-TRIM34, TSPAN1, UBASH3B, UBE2W, UBR4, UBXN7, UCPI, UTMCL, ULLKI, U7PK3B, VPS28, VSTM5, XKR9, YLPM1, ZDHCC7, EB1, ZEB2, ZNF445, ZNF70, and ZNF831. Table 7 provides additional negative regulators of T cells as detected by interleukin-2 production

[0112] Sequences and other information relating to these genes, and their encoded proteins, is available, for example from the NCBI and UniPROT databases,

[0113] A few examples of protein sequences encoded by some of the genes detected as negative regulators of T cells by Interleukin-2 production are provided. For example, an amino acid sequence for the protein encoded by the human ABI3BP gene that is a negative regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. Q727G0, shown below as SEQ ID NO:61.

TABLE-US-00062		10	20	
30	40	50	MRGGKCNMLS	SLGCLLLCGS
ITLALGNAQK	LPKGKRPNLK	VHINTTSDSI		60
70	80	90		100 LLKFLRPSPN
VKLEGLLLGY	GSNVSPNQYF	PLPAEGKFTE	AIVDAEPKYL	
110	120	130	140	150
IVVRPAPPPS	QKKSCSGKTR	SRKPLQLVVG	TLTPSSVFLS	WGFLINPHHD
	160	170	180	
190	200	WTLPSHCPND	RFYTIRYREK	DKEKKWIFQI
CPATETIVEN	LKPNTVYEF		210	220
230	240	250	VKDNVEGGIW	SKIFNHKTVV
GSKKVNGKIQ	STYDQDHTVP	AYVPRKLIPI		260
270	280	290		300 TIIKQVIQNV
THKDSAKSPE	KAPLGGVILV	HLIIPGLNET	TVKLPASLMF	
310	320	330	340	350
EISDALKTQL	AKNETLALPA	ESKTPEVEKI	SARPTTVTPE	TVPRSTKPTT
	360	370	380	
390	400	SSALDVSETT	LASSEKPVIV	PTAKISED
VLQPQTATYD	VFSSPTTSDE		410	420
430	440	450	PEISDSYTAT	SDRILDSIPP
KTSRTLEQPR	ATLAPSETPF	VPQKLEIFTS		460
470	480	490		500 PEMQPTTPAP
QQTTSIPSTP	KRRRPRPKPPR	TKPERTTSAG	TITPKISKSP	
510	520	530	540	550
EPTWTTPAPG	KTQFISLKPK	IPLSPEVTHT	KPAPKQTPRA	PPKPKTSRP
	560	570	580	

590 TPAPKDVLLP HKPYPEVSQS 610
 620 630 640 650 EPAPLETRGI
 PFIPMISPSP SQEELQTTLE ETDQSTQEPF TTKIPRTTEL
 660 670 680 690 700
 AKTTQAPHRF YTTVRPRTSD KPHIRPGVKQ APRPSGADRN VSVDSTHPTK
 710 720 730
 740 750 KPGTRRPPLP PRPTHPRRK LPPNNVTGKP
 GSAGIISGP ITTPPLRSTP 760 770
 780 790 800 RPTGTPLERI ETDIKQPTVP
 ASGELENIT DFSSSPTRET DPLGKPRFKG 810
 820 830 840 850 PHVRYIQKPD
 NSPCSITDSV KRFPKEEATE GNATSPQNP PTNLTVVTV
 860 870 880 890 900
 GCPSFVILDW EKPLNDTVTE YEVISRENGS FSGKNKSIQM TNQTFSTVEN
 910 920 930
 940 950 LKPNTSYEFQ VKPKNPLGEG PVSNTVAFST
 ESADPRVSEP VSAGRDAIWT 960
 970 980 990 1000 ERPFNSDSYS
 ECKGKQYVKR TWYKKFVG VQ LCNSLRYKIY LSDSLTGKFY
 1010 1020 1030 1040 1050
 NIGDQRGHGE DHCQFVDSFL DGRTGQQLTS DQLPIKEGYF RAVRQEPVQF
 1060 1070 GEIGGHTQIN YVQWYECGTT IPGKW

A cDNA and a chromosomal sequence encoding the ABI3BP protein is available from the NCBI database as accession no. AB056106 and CH471052, respectively.

[0114] An amino acid sequence for the protein encoded by the human GCSAML gene that is a negative regulator of T cells as detected by Interleukin-2 production is available from the UniPROT database as accession no. 043741, shown below as SEQ ID NO:62.

TABLE-US-00063 10 20
 30 40 50 MGNTTSDRVS GERHGAKAAR
 SEGAGGHAPG KEHKIMVGST DDPSVFSLPD 60
 70 80 90 100 SKLPGDKEFV
 SWQQDLEDSV KPTQQARPTV IRWSEGGKEV FISGSFNNWS
 110 120 130 140 150
 TKIPLIKSHN DFVAILDLPE GEHQYKFFVD GQWVHDPSEP VVTSQLGTIN
 160 170 180
 190 200 NLIHVKKSDF EVFDALKLDS MESSETSCRD
 LSSSPGPYG QEMYAFRSEE 210 220
 230 240 250 RFKSPILPP HLLQVILNKD
 TNISCDPALL PEPNHVMLNH LYALSIKDSV 260 270
 MVLSATHRYK KKYVTTLTYK PI

A cDNA and a chromosomal sequence encoding the GCSAML protein is available from the NCBI database as accession no. AJ224538 and AL356378, respectively. PGP-83, DNA The following genes are negative regulators of T cells as detected by reduced cellular proliferation (see Table 6): ABCB1, ASAP1, ATP10A, DEAF1, FOXK1, ITGAX, LCE6A, LCP2, LEFTY1, MYC, NAT8B, OLFM3, PLD6, PREP, SULT1A1, SULT1A4, A-INAK, AR-IODLB, B3GNT5, CASZ1, CD27, CEBPB, CRIBP, FL11, FOSL2, HLX, MAP4K1, MUC21, MXI, NDRG1, NEUROD2, SLC2A1, SLC43A3, SMAGP, SOX13, SP140, TP11, and TTC39C. Table 7 provides additional negative regulators of T cells as detected by reduced cellular proliferation.

[0115] An amino acid sequence for the protein encoded by the human SULT1A4 gene that is a

negative regulator of T cells as detected by reduced cellular proliferation is available from the UniPROT database as accession no. PODMIN0, shown below as SEQ ID NO:63.

TABLE-US-00064

		10		20	
30	40		50	MELIQDTSRP	PLEYVKGVP
IKYFAEALGP	LQSFQARPDD	LLINTYPKSG		60	
70	80		90		100
IYQGGDLEKC	NRAPIYVRVP	FLEVNDPGE	SGLETLKDTP		
110	120		130	140	150
PPRLIKSHLP	LALLPQTLLD	QKVKVVYVAR	NPKDVAVSYY	HFHRMEKAHP	
	160	170	180		
190	200	EPGTWDSFLE	KFMAGEVSYG	SWYQHVQEW	
ELSRTHPVLY	LFYEDMKENP		210		
220	230		240		250
VGRSLPEETM	DFMVQHTSFK	EMKKNPMTNY	TTVPQELMDH		KREIQKILEF
260	270		280		290
AGDWKTTFTV	AQNERFDADY	AEKMAGCSLS	FRSEL		SISPFMRKGM

A chromosomal sequence encoding the SULT1A4 protein is available from the NCBI database as accession no. AC106782.

[0116] An amino acid sequence for the protein encoded by the human SLC43A3 gene that is a negative regulator of T cells as detected by reduced cellular proliferation is available from the UniPROT database as accession no. Q8NB15, shown below as SEQ ID NO:64.

TABLE-US-00065

		10		20	
30	40		50	MAGQGLPLHV	ATLLTGLLEC
LGFAGVLFGW	PSLVFVFKNE	DYFKDLCGPD			
60	70		80		90
100	AGPIGNATGQ	ADCKAQDERF	SLIFTLGSEFM	NNFMTFPTGY	IFDRFKTTVA
	110		120		130
140		150	RLIAIFFYTT	ATLIIAFTSA	GSVLLFLAM
LITNLQIGNL		160		170	PMLTIGGILF
180		190		200	FGQHRSTIIT
AVFLIIKLLY	EKGISLRASF	IFISVCSTWH		210	LYNGAFDSSS
220		230		240	
GHIPYPLPPN	YSYGLCPGNG	TTKEEKETAE	HENRELQSKE		250
260		270		280	
FLSAKEETPG	AGQKQELRSF	WSYAFSRRFA	WHLVWLSVIQ	LWHYLFITL	300
	310		320		330
340		350	NSLLTNMAGG	DMARVSTYTN	AFAFTQFGVL
CAPWNGLLMD	RLKQKYQKEA		360		
370		380		390	
ALCSTVPSLA	LTSLCLLGFA	LCASVPILPL	QYLTFILQVI		400
410		420		430	
SRSFLYGSNA	AFLTAFPSE	HFGKLFGLVM	ALSAVVSLQ	FPIFTLIKGS	450
	460		470		480
490		500	LQNDPFYVNV	MFMLAILLTF	FHPFLVYREC
RTWKESPSAI	A				

A cDNA and a chromosomal sequence encoding the SLC43A3 protein is available from the NCBI database as accession no. AB028927 and AP000781 Any of these genes or the proteins encoded by these genes that are described herein can regulate T cells.

[0117] The sequences provided herein are exemplary. Isoforms and variants of these sequences and of any of regulators listed in Tables 1-7 or FIGS. 1-4 can also be used in the methods and

compositions described herein.

[0118] For example, isoforms and variants of the proteins and nucleic acids can be used in the methods and compositions described herein when they are substantially identical to the genes or the encoded proteins listed in Tables 1-7 or FIGS. 1-4. The terms “substantially identity” indicates that a polypeptide or nucleic acid comprises a sequence with between 55-100% sequence identity to a reference sequence, for example with at least 55% sequence identity, preferably 60%, preferably 70%, preferably 80%, preferably at least 90%, preferably at least 95%, preferably at least 96%, preferably at least 97% sequence, preferably at least 98%, preferably at least 99% identity to a reference sequence over a specified comparison window. Optimal alignment may be ascertained or conducted using the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48:443-53 (1970).

[0119] An indication that two polypeptide sequences are substantially identical is that both polypeptides have the same function acting as a regulator of T cells or T cell activity. The polypeptide that is substantially identical to a regulator sequence and may not have exactly the same level of activity as the regulator. Instead, the substantially 10 identical polypeptide may exhibit greater or lesser levels of regulator activity than the those listed in Tables 1-7 or FIG. 1-4, or any of the sequences recited herein. For example, the substantially identical polypeptide or nucleic acid may have at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90%, or at least about 95%, or at least about 97%, or at least about 98%, or at least about 100%, or at least about 105%, or at least about 110%, or at least about 120%, or at least about 130%, or at least about 140%, or at least about 150%, or at least about 200% of the activity of a regulator described herein a when measured by similar assay procedures.

[0120] Alternatively, substantial identity is present when second polypeptide is immunologically reactive with antibodies raised against the first polypeptide (e.g., a polypeptide with encoded by any of the genes listed in Tables 1-7 or FIGS. 1-4). Thus, a polypeptide is substantially identical to a first polypeptide, for example, where the two polypeptides differ only by a conservative substitution. In addition, a polypeptide can be substantially identical to a first polypeptide when they differ by a non-conservative change if the epitope that the antibody recognizes is substantially identical. Polypeptides that are “substantially similar” share sequences as noted above except that some residue positions, which are not identical, may differ by conservative amino acid changes.

Expression Systems

[0121] Nucleic acid segments encoding one or more regulator proteins, or nucleic acid segments that are inhibitory nucleic acids or such regulators, can be inserted into or employed with any suitable expression system. Nucleic acids segments encoding one or more agents that can modulate a regulator protein expression or activity can be inserted into or employed with any suitable expression system. A therapeutically effective quantity of one or more regulator proteins or modulators of such regulator proteins can be generated from such expression systems. A therapeutically effective of one or more inhibitory nucleic acids can also be generated from such expression systems.

[0122] Recombinant expression of nucleic acids (or inhibitory nucleic acids) is usefully accomplished using a vector, such as a plasmid. The vector can include a promoter operably linked to nucleic acid segment encoding one or more regulator/modulator proteins. In another example, a vector can include a promoter operably linked to nucleic acid segment that encodes a regulator/modulator inhibitory nucleic acid.

[0123] The vector can also include other elements required for transcription and translation. As used herein, vector refers to any carrier containing exogenous DNA. Thus, vectors are agents that transport the exogenous nucleic acid into a cell without degradation and include a promoter yielding expression of the nucleic acid in the cells into which it is delivered. Vectors include but are not limited to plasmids, viral nucleic acids, viruses, phage nucleic acids, phages, cosmids, and

artificial chromosomes. A variety of prokaryotic and eukaryotic expression vectors suitable for carrying, encoding and/or expressing regulator/modulator. A variety of prokaryotic and eukaryotic expression vectors suitable for carrying, encoding and/or expressing regulator/modulator inhibitory nucleic acids can be employed. Such expression vectors include, for example, pET, pET3d, pCR2.1, pBAD, pUC, and yeast vectors. The vectors can be used, for example, in a variety of in vivo and in vitro situations.

[0124] The expression cassette, expression vector, and sequences in the cassette or vector can be heterologous. As used herein, the term “heterologous” when used in reference to an expression cassette, expression vector, regulatory sequence, promoter, or nucleic acid refers to an expression cassette, expression vector, regulatory sequence, or nucleic acid that has been manipulated in some way. For example, a heterologous promoter can be a promoter that is not naturally linked to a nucleic acid of interest, or that has been introduced into cells by cell transformation procedures. A heterologous nucleic acid or promoter also includes a nucleic acid or promoter that is native to an organism but that has been altered in some way (e.g., placed in a different chromosomal location, mutated, added in multiple copies, linked to a non-native promoter or enhancer sequence, etc.). Heterologous nucleic acids may comprise sequences that comprise cDNA forms; the cDNA sequences may be expressed in either a sense (to produce mRNA) or anti-sense orientation (to produce an anti-sense RNA transcript that is complementary to the mRNA transcript).

Heterologous coding regions can be distinguished from endogenous coding regions, for example, when the heterologous coding regions are joined to nucleotide sequences comprising regulatory elements such as promoters that are not found naturally associated with the coding region, or when the heterologous coding regions are associated with portions of a chromosome not found in nature (e.g., genes expressed in loci where the protein encoded by the coding region is not normally expressed). Similarly, heterologous promoters can be promoters that are linked to a coding region to which they are not linked in nature.

[0125] Viral vectors that can be employed include those relating to lentivirus, adenovirus, adeno-associated virus, herpes virus, vaccinia virus, polio virus, AIDS virus, neuronal trophic virus, Sindbis and other viruses. Also useful are any viral families which share the properties of these viruses which make them suitable for use as vectors. Retroviral vectors that can be employed include those described in by Verma, T. M., *Retroviral vectors for gene transfer*. In *Microbiology-1985*, American Society for Microbiology, pp. 229-232, Washington, (1985). For example, such retroviral vectors can include Murine Moloney Leukemia virus, MMLV, and other retroviruses that express desirable properties. Typically, viral vectors contain, nonstructural early genes, structural late genes, an RNA polymerase III transcript, inverted terminal repeats necessary for replication and encapsidation, and promoters to control the transcription and replication of the viral genome. When engineered as vectors, viruses typically have one or more of the early genes removed and a gene or gene/promoter cassette is inserted into the viral genome in place of the removed viral nucleic acid.

[0126] A variety of regulatory elements can be included in the expression cassettes and/or expression vectors, including promoters, enhancers, translational initiation sequences, transcription termination sequences and other elements. A “promoter” is generally a sequence or sequences of DNA that function when in a relatively fixed location in regard to the transcription start site. For example, the promoter can be upstream of the nucleic acid segment encoding a regulator protein. In another example, the promoter can be upstream of an inhibitory nucleic acid segment of a modulating agent for one or more regulators.

[0127] A “promoter” contains core elements required for basic interaction of RNA polymerase and transcription factors and can contain upstream elements and response elements. “Enhancer” generally refers to a sequence of DNA that functions at no fixed distance from the transcription start site and can be either 5′ or 3′ to the transcription unit. Furthermore, enhancers can be within an intron as well as within the coding sequence itself. They are usually between 10 and 300 bp in

length, and they function in cis. Enhancers function to increase transcription from nearby promoters. Enhancers, like promoters, also often contain response elements that mediate the regulation of transcription. Enhancers often determine the regulation of expression.

[0128] Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human or nucleated cells) can also contain sequences for the termination of transcription, which can affect mRNA expression. These regions are transcribed as polyadenylated segments in the untranslated portion of the mRNA encoding tissue factor protein. The 3' untranslated regions also include transcription termination sites. It is preferred that the transcription unit also contains a polyadenylation region. One benefit of this region is that it increases the likelihood that the transcribed unit will be processed and transported like mRNA. The identification and use of polyadenylation signals in expression constructs is well established. It is preferred that homologous polyadenylation signals be used in the transgene constructs.

[0129] The expression of regulator/modulator proteins or inhibitory nucleic acid molecules therefor from an expression cassette or expression vector can be controlled by any promoter capable of expression in prokaryotic cells or eukaryotic cells. Examples of prokaryotic promoters that can be used include, but are not limited to, SP6, T7, T5, tac, bla, frp, gal, lac, or maltose promoters. Examples of eukaryotic promoters that can be used include, but are not limited to, constitutive promoters, e.g., viral promoters such as CMV, SV40 and RSV promoters, as well as regulatable promoters, e.g., an inducible or repressible promoter such as the tet promoter, the hsp70 promoter and a synthetic promoter regulated by CRE. Vectors for bacterial expression include pGEX-5X-3, and for eukaryotic expression include pCIneo-CMV.

[0130] The expression cassette or vector can include nucleic acid sequence encoding a marker product. This marker product is used to determine if the gene has been delivered to the cell and once delivered is being expressed. Marker genes can include the *E. coli* lacZ gene which encodes β -galactosidase, and green fluorescent protein. In some embodiments the marker can be a selectable marker. When such selectable markers are successfully transferred into a host cell, the transformed host cell can survive if placed under selective pressure. There are two widely used distinct categories of selective regimes. The first category is based on a cell's metabolism and the use of a mutant cell line which lacks the ability to grow independent of a supplemented media. The second category is dominant selection which refers to a selection scheme used in any cell type and does not require the use of a mutant cell line. These schemes typically use a drug to arrest growth of a host cell. Those cells which have a novel gene would express a protein conveying drug resistance and would survive the selection. Examples of such dominant selection use the drugs neomycin (Southern P. and Berg, P. J. Molec. Appl. Genet, 1: 327 (1982)), mycophenolic acid, (Mulligan, R. C. and Berg, P. Science 209: 1422 (1980)) or hygromycin, (Sugden, B. et al., Mol. Cell Biol. 5: 410-413 (1985)).

[0131] Gene transfer can be obtained using direct transfer of genetic material, in but not limited to, plasmids, viral vectors, viral nucleic acids, phage nucleic acids, phages, cosmids, and artificial chromosomes, or via transfer of genetic material in cells or carriers such as cationic liposomes. Such methods are well known in the art and readily adaptable for use in the method described herein. Transfer vectors can be any nucleotide construction used to deliver genes into cells (e.g., a plasmid), or as part of a general strategy to deliver genes, e.g., as part of recombinant retrovirus or adenovirus (Rain et al. Cancer Res. 53:83-88, (1993)). Appropriate means for transfection, including viral vectors, chemical transfectants, or physico-mechanical methods such as electroporation and direct diffusion of DNA, are described by, for example, Wolff, J. A., et al., Science, 247, 1465-1468, (1990); and Wolff, J. A. Nature, 352, 815-818, (1991).

[0132] For example, the nucleic acid molecules, expression cassette and/or vectors encoding regulator/modulator proteins or encoding inhibitory nucleic acid molecules therefor can be introduced to a cell by any method including, but not limited to, calcium-mediated transformation, electroporation, microinjection, lipofection, particle bombardment and the like. The cells can be

expanded in culture and then administered to a subject, e.g., a mammal such as a human. The amount or number of cells administered can vary but amounts in the range of about 10^{sup.6} to about 10^{sup.9} cells can be used. The cells are generally delivered in a physiological solution such as saline or buffered saline. The cells can also be delivered in a vehicle such as a population of liposomes, exosomes or microvesicles.

[0133] In some cases, the transgenic cell can produce exosomes or microvesicles that contain nucleic acid molecules, expression cassettes and/or vectors encoding one or more regulator/modulator. In some cases, the transgenic cell can produce exosomes or microvesicles that contain inhibitory nucleic acid molecules that can target regulator/modulator nucleic acids, one or more nucleic acids for regulator, or a combination thereof. Microvesicles can mediate the secretion of a wide variety of proteins, lipids, mRNAs, and micro RNAs, interact with neighboring cells, and can thereby transmit signals, proteins, lipids, and nucleic acids from cell to cell (see, e.g., Shen et al., J Biol Chem, 286(16): 14383-14395 (2011); Hu et al., Frontiers in Genetics 3 (April 2012); Pegtel et al., Proc. Nat'l Acad Sci 107(14): 6328-6333 (2010); WO/201 3/084000; each of which is incorporated herein by reference in its entirety. Cells producing such microvesicles can be used to express the one or more regulator/modulator protein and/or inhibitory nucleic acids for one or more regulator/modulators, or a combination thereof. Transgenic vectors or cells with a heterologous expression cassette or expression vector can express one or more regulator, can optionally also express one or more regulator inhibitory nucleic acids, or a combination thereof. Any of these vectors or cells can be administered to a subject. Exosomes produced by transgenic cells can be used to administer regulator/modulator proteins, regulator/modulator nucleic acids, regulator/modulator inhibitory nucleic acids, or a combination thereof to a. subject or to tumor and cancer cells in the subject.

[0134] Methods and compositions that include inhibitors of one or regulators such as inhibitory nucleic acids, antibodies, or any combination thereof.

CRISPR Modifications

[0135] In some cases, clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated (Cas) systems can be used to create one or more modifications in genomic regulator genes. Such CRISPR modifications can reduce or activate the expression or functioning of the regulator gene products. CRISPR/Cas systems are useful, for example, for RNA-programmable genome editing (see e.g., Marraffini and Sontheimer. Nature Reviews Genetics 11: 181-190 (2010); Sorek et al. Nature Reviews Microbiology 2008 6: 181-6; Karginov and Hannon. Mol Cell 2010 1:7-19; Hale et al. Mol Cell 2010:45:292-302; Jinek et al. Science 2012 337:815-820; Bikard and Marraffini Curr Opin Immunol 2012 24:15-20; Bikard et al. Cell Host & Microbe 2012 12: 177-186; all of which are incorporated by reference herein in their entireties).

[0136] A CRISPR guide RNA can be used that can target a Cas enzyme to the desired location in the genome, where it can cleave the genomic DNA for generation of a genomic modification. This technique is described, for example, by Mali et al. Science 2013 339:823-6; which is incorporated by reference herein in its entirety. Kits for the design and use of CRISPR-mediated genome editing are commercially available, e.g. the PRECISION X CAS9 SMART NUCLEASE™ System (Cat No. CAS900A-1) from System Biosciences, Mountain View, CA.

[0137] In some cases, transcriptional activators can be linked to defective Cas9 or to one or more guide RNAs to target the transcriptional activator. Such transcriptional activators include protein domains or whole proteins that assist in the recruitment of co-factors and RNA Polymerase to increase transcription of one or more of the regulator gene(s) listed in Tables 1-7 or FIGS. 1-4.

[0138] In some cases, a cre-lox recombination system of bacteriophage P1, described by Abremski et al. 1983. Cell 32:1301 (1983), Sternberg et al., Cold Spring Harbor Symposia on Quantitative Biology, Vol. XLV 297 (1981) and others, can be used to promote recombination and alteration of the regulator genomic site(s). The cre-lox system utilizes the cre recombinase isolated from bacteriophage P1 in conjunction with the DNA sequences that the recombinase recognizes (termed

lox sites). This recombination system has been effective for achieving recombination in plant cells (see, e.g., U.S. Pat. No. 5,658,772), animal cells (U.S. Pat. Nos. 4,959,317; 5,801,030), and in viral vectors (Hardy et al., J. Virology 71:1842 (1997)).

[0139] The genomic mutations so incorporated can alter one or more amino acids in the encoded regulator gene products. For example, genomic sites modified so that the encoded regulator protein is more prone to degradation, is less stable so that the half-life of such protein(s) is reduced, or so that the regulator has improved expression or functioning. In another example, genomic sites can be modified so that at least one amino acid of a regulator polypeptide is deleted or mutated to alter its activity. For example, a conserved amino acid or a conserved domain can be modified to improve or reduce of the activity of the regulator polypeptide. For example, a conserved amino acid or several amino acids in a conserved domain of the regulator polypeptide can be replaced with one or more amino acids having physical and/or chemical properties that are different from the conserved amino acid(s). For example, to change the physical and/or chemical properties of the conserved amino acid(s), the conserved amino acid(s) can be deleted or replaced by amino acid(s) of another class, where the classes are identified in the following table.

TABLE-US-00066 Classification Genetically Encoded Hydrophobic A, G, F, I, L, M, P, V, W
Aromatic F, Y, W Apolar M, G, P Aliphatic A, V, L, I Hydrophilic C, D, E, H, K, N, Q, R, S, T, Y
Acidic D, E Basic H, K, R Polar Q, N, S, T, Y Cysteine-Like C

[0140] The guide RNAs and nuclease can be introduced via one or more vehicles such as by one or more expression vectors (e.g., viral vectors), virus like particles, ribonucleoproteins (RNPs), via nanoparticles, liposomes, or a combination thereof. The vehicles can include components or agents that can target particular cell types (e.g., antibodies that recognize cell-surface markers), facilitate cell penetration, reduce degradation, or a combination thereof.

Inhibitory Nucleic Acids

[0141] The expression of one or more regulators/modulators can be inhibited, for example by use of an inhibitory nucleic acid that specifically recognizes a nucleic acid that encodes the regulator or modulator.

[0142] An inhibitory nucleic acid can have at least one segment that will hybridize to a regulator nucleic acid or modulator under intracellular or stringent conditions. The inhibitory nucleic acid can reduce expression of a regulator/modulator nucleic acid. A nucleic acid may hybridize to a genomic DNA, a messenger RNA, or a combination thereof. An inhibitory nucleic acid may be incorporated into a plasmid vector or viral DNA. It may be single stranded or double stranded, circular or linear.

[0143] An inhibitory nucleic acid is a polymer of ribose nucleotides or deoxyribose nucleotides having more than 13 nucleotides in length. An inhibitory nucleic acid may include naturally occurring nucleotides; synthetic, modified, or pseudo-nucleotides such as phosphorothiolates; as well as nucleotides having a detectable label such as ^{32}P , biotin or digoxigenin. An inhibitory nucleic acid can reduce the expression and/or activity of a regulator/modulator nucleic acid. Such an inhibitory nucleic acid may be completely complementary to a segment of an endogenous regulator/modulator nucleic acid (e.g., an RNA). Alternatively, some variability is permitted in the inhibitory nucleic acid sequences relative to regulator/modulator sequences. An inhibitory nucleic acid can hybridize to a regulator/modulator nucleic acid under intracellular conditions or under stringent hybridization conditions and is sufficiently complementary to inhibit expression of the endogenous regulator/modulator nucleic acid. Intracellular conditions refer to conditions such as temperature, pH and salt concentrations typically found inside a cell, e.g. an animal or mammalian cell. One example of such an animal or mammalian cell is a myeloid progenitor cell, Another example of such an animal or mammalian cell is a more differentiated cell derived from a myeloid progenitor cell. Generally, stringent hybridization conditions are selected to be about 5° C. lower than the thermal melting point ($T_{\text{sub.m}}$) for the specific sequence at a defined ionic strength and pH. However, stringent conditions encompass temperatures in the range of about 1° C. to about 20°

C. lower than the thermal melting point of the selected sequence, depending upon the desired degree of stringency as otherwise qualified herein. Inhibitory oligonucleotides that comprise, for example, 2, 3, 4, or 5 or more stretches of contiguous nucleotides that are precisely complementary to a regulator/modulator coding sequence, each separated by a stretch of contiguous nucleotides that are not complementary to adjacent coding sequences, can inhibit the function of one or more nucleic acids for any of the regulators or modulators described herein. In general, each stretch of contiguous nucleotides is at least 4, 5, 6, 7, or 8 or more nucleotides in length. Non-complementary intervening sequences may be 1, 2, 3, or 4 nucleotides in length. One skilled in the art can easily use the calculated melting point of an inhibitory nucleic acid hybridized to a sense nucleic acid to estimate the degree of mismatching that will be tolerated for inhibiting expression of a particular target nucleic acid. Inhibitory nucleic acids of the invention include, for example, a short hairpin RNA, a small interfering RNA, a ribozyme or an antisense nucleic acid molecule.

[0144] The inhibitory nucleic acid molecule may be single or double stranded (e.g. a small interfering RNA (siRNA)) and may function in an enzyme-dependent manner or by steric blocking. Inhibitory nucleic acid molecules that function in an enzyme-dependent manner include forms dependent on RNase H activity to degrade target mRNA. These include single-stranded DNA, RNA, and phosphorothioate molecules, as well as the double-stranded RNAi/siRNA system that involves target mRNA recognition through sense-antisense strand pairing followed by degradation of the target mRNA by the RNA-induced silencing complex. Steric blocking inhibitory nucleic acids, which are RNase-H independent, interfere with gene expression or other mRNA-dependent cellular processes by binding to a target mRNA and getting in the way of other processes. Steric blocking inhibitory nucleic acids include 2'-O alkyl (usually in chimeras with RNase-H dependent antisense), peptide nucleic acid (PNA), locked nucleic acid (LNA) and morpholino antisense.

[0145] Small interfering RNAs, for example, may be used to specifically reduce translation of regulator/Modulator such that translation of the encoded regulator/modulator polypeptide is reduced. SiRNAs mediate post-transcriptional gene silencing in a sequence-specific manner. See, for example, website at invitrogen.com/life/technology/home/Products-and-Services/Applications/rnai.html. Once incorporated into an RNA-Induced silencing complex, siRNA mediate cleavage of the homologous endogenous mRNA transcript by guiding the complex to the homologous mRNA transcript, which is then cleaved by the complex. The siRNA may be homologous and/or complementary to any region of the regulator/modulator transcript and/or any of the transcripts of the regulators/modulators. The region of homology may be 30 nucleotides or less in length, preferable less than 25 nucleotides, and more preferably about 21 to 23 nucleotides in length. SiRNA is typically double stranded and may have two-nucleotide 3' overhangs, for example, 3' overhanging UU dinucleotides. Methods for designing siRNAs are known to those skilled in the art. See, for example, Elbashir et al. *Nature* 411: 494-498 (2001); Harborth et al. *Antisense Nucleic Acid Drug Dev.* 13: 83-106 (2003).

[0146] The pSuppressorNeo vector for expressing hairpin siRNA, commercially available from IMGENEX (San Diego, California), can be used to generate siRNA for inhibiting expression of regulators/modulators. The construction of the siRNA expression plasmid involves the selection of the target region of the mRNA, which can be a trial-and-error process. However, Elbashir et al. have provided guidelines that appear to work ~80% of the time. Elbashir, S. M., et al., *Analysis of gene function in somatic mammalian cells using small interfering RNAs. Methods*, 2002. 26(2): p. 199-213. Accordingly, for synthesis of synthetic siRNA, a target region may be selected preferably 50 to 100 nucleotides downstream of the start codon. The 5' and 3' untranslated regions and regions close to the start codon should be avoided as these may be richer in regulatory protein binding sites. As siRNA can begin with AA, have 3' UU overhangs for both the sense and antisense siRNA strands, and have an approximate 50% G/C content. An example of a sequence for a synthetic siRNA is 5'-AA(N 9)UU, where N is any nucleotide in the mRNA sequence and should be approximately 50% G-C content. The selected sequence(s) can be compared to others in the human

genome database to minimize homology to other known coding sequences (e.g., by Blast search, for example, through the NCBI website).

[0147] SiRNAs may be chemically synthesized, created by in vitro transcription, or expressed from an siRNA expression vector or a PCR expression cassette. See, e.g., website at invitrogen.com/site/us/en/home/Products-and-Services/Applications/rnai.html. When an siRNA is expressed from an expression vector or a PCR expression cassette, the insert encoding the siRNA may be expressed as an RNA transcript that folds into an siRNA hairpin. Thus, the RNA transcript may include a sense siRNA sequence that is linked to its reverse complementary antisense siRNA sequence by a spacer sequence that forms the loop of the hairpin as well as a string of U's at the 3' end. The loop of the hairpin may be of any appropriate lengths, for example, 3 to 30 nucleotides in length, preferably, 3 to 23 nucleotides in length, and may be of various nucleotide sequences including, AUG, CCC, U, UCG, CCACC, CTCGAG, AAGCUU, CCACACC and UUCAAGAGA (SEQ ID NO:61). SiRNAs also may be produced in vivo by cleavage of double-stranded RNA introduced directly or via a transgene or virus. Amplification by an RNA-dependent RNA polymerase may occur in some organisms.

[0148] An inhibitory nucleic acid such as a short hairpin RNA siRNA or an antisense oligonucleotide may be prepared using methods such as by expression from an expression vector or expression cassette that includes the sequence of the inhibitory nucleic acid. Alternatively, it may be prepared by chemical synthesis using naturally occurring nucleotides, modified nucleotides or any combinations thereof. In some embodiments, the inhibitory nucleic acids are made from modified nucleotides or non-phosphodiester bonds, for example, that are designed to increase biological stability of the inhibitory nucleic acid or to increase intracellular stability of the duplex formed between the inhibitory nucleic acid and the target regulators/modulators nucleic acids.

[0149] An inhibitory nucleic acid may be prepared using available methods, for example, by expression from an expression vector encoding a complementarity sequence of the regulator/modulator nucleic acids described herein. Alternatively, it may be prepared by chemical synthesis using naturally occurring nucleotides, modified nucleotides or any mixture of combination thereof. In some embodiments, the nucleic acids of the regulators/modulators described herein are made from modified nucleotides or non-phosphodiester bonds, for example, that are designed to increase biological stability of the nucleic acids or to increase intracellular stability of the duplex formed between the inhibitory nucleic acids and other (e.g., endogenous) nucleic acids.

[0150] For example, the regulator/modulator nucleic acids can be peptide nucleic acids that have peptide bonds rather than phosphodiester bonds.

[0151] Naturally occurring nucleotides that can be employed in the regulator/modulator nucleic acids include the ribose or deoxyribose nucleotides adenosine, guanine, cytosine, thymine and uracil. Examples of modified nucleotides that can be employed in the regulator/modulator nucleic acids include 5-fluorouracil, 5-bromouracil, 5-chlorouracil, 5-iodouracil, hypoxanthine, xanthine, 4-acetylcytosine, 5-(carboxyhydroxymethyl) uracil, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluracil, dihydrouracil, beta-D-galactosylqueosine, inosine, N6-isopentenyladenine, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-methylguanine, 3-methylcytosine, 5-methylcytosine, N6-adenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5'-methoxycarboxymethyluracil, 5-methoxyuracil, 2-methylthio-N6-isopentenyladenine, uracil-5-oxoacetic acid, wybutosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, uracil-5-oxoacetic acid methylester, uracil-5-oxoacetic acid, 5-methyl-2-thiouracil, 3-(3-amino-3-N-2-carboxypropyl) uracil, (acp3)w, and 2,6-diaminopurine.

[0152] Thus, inhibitory nucleic acids of the regulators/modulators described herein may include modified nucleotides, as well as natural nucleotides such as combinations of ribose and deoxyribose nucleotides. The inhibitory nucleic acids and may be of same length as wild type

regulators/modulators described herein. The inhibitory nucleic acids of the regulators/modulators described herein can also be longer and include other useful sequences. In some embodiments, the inhibitory nucleic acids of the regulators/modulators described herein are somewhat shorter. For example, inhibitory nucleic acids of the regulators/modulators described herein can include a segment that has a nucleic acid sequence that can be missing up to 5 nucleotides, or missing up to 10 nucleotides, or missing up to 20 nucleotides, or missing up to 30 nucleotides, or missing up to 50 nucleotides, or missing up to 100 nucleotides from the 5' or 3' end.

Antibodies

[0153] Antibodies can be used as inhibitors or activators of any of the regulators/modulators described herein. For example, in some cases, antibody preparations can target one or more of the regulators or modulators described herein to block interactions by the regulators/modulators described herein or to reduce the activities of the regulators/modulators. In other cases, for example, antibodies can activate one or more of the regulator or modulators described herein that are cell surface receptors. One example of such activation is Varlilumab (a CD27 activating antibody) currently in clinical trial and that has been shown to increase anti-tumor T cell function Ansell et al. (2020) Blood Adv. 4(9): 1917-1926.

[0154] Antibodies can be raised against various epitopes of the regulators/modulator described herein. Some antibodies for regulators/modulators described herein may also be available commercially. However, the antibodies contemplated for treatment pursuant to the methods and compositions described herein are preferably human or humanized antibodies and are highly specific for their targets.

[0155] In one aspect, the present disclosure relates to use of isolated antibodies that bind specifically to regulators/modulators described herein. Such antibodies may be monoclonal antibodies. Such antibodies may also be humanized or fully human monoclonal antibodies. The antibodies can exhibit one or more desirable functional properties, such as high affinity binding to one or more regulators/modulators described herein, or the ability to inhibit functioning of any of the regulators/modulators described herein.

[0156] Methods and compositions described herein can include antibodies that bind any of the regulators/modulators described herein, or a combination of antibodies where each antibody type can separately bind one of the regulators/modulators described herein.

[0157] The term “antibody” as referred to herein includes whole antibodies and any antigen binding fragment (i.e., “antigen-binding portion”) or single chains thereof. An “antibody” refers to a glycoprotein comprising at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, or an antigen binding portion thereof. Each heavy chain is comprised of a heavy chain variable region (abbreviated herein as V.sub.H) and a heavy chain constant region. The heavy chain constant region is comprised of three domains, C.sub.H1, C.sub.H2 and C.sub.H3. Each light chain is comprised of a light chain variable region (abbreviated herein as V.sub.L) and a light chain constant region. The light chain constant region is comprised of one domain, C.sub.L. The V.sub.H and V.sub.L regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDR), interspersed with regions that are more conserved, termed framework regions (FR). Each V.sub.H and V.sub.L is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The variable regions of the heavy and light chains contain a binding domain that interacts with an antigen. The constant regions of the antibodies may mediate the binding of the immunoglobulin to host tissues or factors, including various cells of the immune system (e.g., effector cells) and the first component (C1q) of the classical complement system.

[0158] The term “antigen-binding portion” of an antibody (or simply “antibody portion”), as used herein, refers to one or more fragments of an antibody that retain the ability to specifically bind to an antigen (e.g., a peptide or domain of any of the regulators/modulators described herein). It has been shown that the antigen-binding function of an antibody can be performed by fragments of a

full-length antibody.

[0159] Examples of binding fragments encompassed within the term “antigen-binding portion” of an antibody include (i) a Fab fragment, a monovalent fragment consisting of the V_H, V_L, C_{H1} and C_L domains; (ii) a F(ab')₂ fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the V_H and C_{H1} domains; (iv) a Fv fragment consisting of the V_L and V_H domains of a single arm of an antibody, (v) a dAb fragment (Ward et al., (1989) Nature 341:544-546), which consists of a V_H domain; and (vi) an isolated complementarity determining region (CDR). Furthermore, although the two domains of the Fv fragment, V_L, and V_H, are coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the V_L, and V_H, regions pair to form monovalent molecules (known as single chain Fv (scFv); see e.g., Bird et al. (1988) Science 242:423-426; and Huston et al. (1988) Proc. Natl. Acad. Sci. USA 85:5879-5883). Such single chain antibodies are also intended to be encompassed within the term “antigen-binding portion” of an antibody. These antibody fragments are obtained using conventional techniques known to those with skill in the art, and the fragments are screened for utility in the same manner as are intact antibodies.

[0160] An “isolated antibody,” as used herein, is intended to refer to an antibody that is substantially free of other antibodies having different antigenic specificities (e.g., an isolated antibody that specifically binds any of the regulators/modulators described herein is substantially free of antibodies that specifically bind antigens other than any of the regulators/modulators described herein). An isolated antibody that specifically binds regulators/modulators described herein may, however, have cross-reactivity to other antigens, such as isoforms or related forms of the regulators/modulators proteins from other species. Moreover, an isolated antibody may be substantially free of other cellular material and/or chemicals.

[0161] The terms “monoclonal antibody” or “monoclonal antibody composition” as used herein refer to a preparation of antibody molecules of single molecular composition. A monoclonal antibody composition displays a single binding specificity and affinity for a particular epitope.

[0162] The term “human antibody,” as used herein, is intended to include antibodies having variable regions in which both the framework and CDR regions are derived from human germline immunoglobulin sequences. Furthermore, if the antibody contains a constant region, the constant region also is derived from human germline immunoglobulin sequences. The human antibodies of the invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo). However, the term “human antibody,” as used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

[0163] The term “human monoclonal antibody” refers to antibodies displaying a single binding specificity which have variable regions in which both the framework and CDR regions are derived from human germline immunoglobulin sequences. In one embodiment, the human monoclonal antibodies are produced by a hybridoma which includes a B cell obtained from a transgenic nonhuman animal, e.g., a transgenic mouse, having a genome comprising a human heavy chain transgene and a light chain transgene fused to an immortalized cell.

[0164] The term “recombinant human antibody,” as used herein, includes all human antibodies that are prepared, expressed, created or isolated by recombinant means, such as (a) antibodies isolated from an animal (e.g., a mouse) that is transgenic or transchromosomal for human immunoglobulin genes or a hybridoma prepared therefrom (described further below), (b) antibodies isolated from a host cell transformed to express the human antibody, e.g., from a transfectoma, (c) antibodies isolated from a recombinant, combinatorial human antibody library, and (d) antibodies prepared, 10 expressed, created or isolated by any other means that involve splicing of human immunoglobulin

gene sequences to other DNA sequences. Such recombinant human antibodies have variable regions in which the framework and CDR regions are derived from human germline immunoglobulin sequences. In certain embodiments, however, such recombinant human antibodies can be subjected to in vitro mutagenesis (or, when an animal transgenic for human Ig sequences is used, in vivo somatic mutagenesis) and thus the amino acid sequences of the V.sub.H and V.sub.L regions of the recombinant antibodies are sequences that, while derived from and related to human germline V.sub.L and V.sub.H sequences, may not naturally exist within the human antibody germline repertoire in vivo.

[0165] As used herein, “isotype” refers to the antibody class (e.g., IgM or IgG1) that is encoded by the heavy chain constant region genes.

[0166] The phrases “an antibody recognizing an antigen” and “an antibody specific for an antigen” are used interchangeably herein with the term “an antibody which binds specifically to an antigen.”

[0167] The term “human antibody derivatives” refers to any modified form of the human antibody, e.g., a conjugate of the antibody and another agent or antibody.

[0168] The term “humanized antibody” is intended to refer to antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences. Additional framework region modifications may be made within the human framework sequences.

[0169] The term “chimeric antibody” is intended to refer to antibodies in which the variable region sequences are derived from one species and the constant region sequences are derived from another species, such as an antibody in which the variable region sequences are derived from a mouse antibody and the constant region sequences are derived from a human antibody.

[0170] As used herein, an antibody that “specifically binds to a human regulator/modulator protein described herein” is intended to refer to an antibody that binds to the human regulator/modulator protein described herein with a K.sub.D of 1×10^{-7} M or less, more preferably 5×10^{-8} M or less, more preferably 1×10^{-8} M or less, more preferably 5×10^{-9} M or less, even more preferably between 1×10^{-8} M and 1×10^{-10} M or less.

[0171] The term “K.sub.assoc” or “K.sub.a,” as used herein, is intended to refer to the association rate of a particular antibody-antigen interaction, whereas the term “K.sub.dis” or “K.sub.d,” as used herein, is intended to refer to the dissociation rate of a particular antibody-antigen interaction. The term “K.sub.D,” as used herein, is intended to refer to the dissociation constant, which is obtained from the ratio of K.sub.d to K.sub.a (i.e., $K_{sub.d}/K_{sub.a}$) and is expressed as a molar concentration (M). K.sub.D values for antibodies can be determined using methods well established in the art. A preferred method for determining the K.sub.D of an antibody is by using surface plasmon resonance, preferably using a biosensor system such as a Biacore™ system.

[0172] The antibodies of the invention are characterized by particular functional features or properties of the antibodies. For example, the antibodies bind specifically to a human regulator/modulator described herein. Preferably, an antibody of the invention binds to a regulator/modulator described herein with high affinity, for example with a K.sub.D of 1×10^{-7} M or less. The antibodies can exhibit one or more of the following characteristics:

[0173] (a) binds to a human regulators/modulators described herein with a K_r of 1×10^{-7} M or less;

[0174] (b) inhibits the function or activity of a human regulators/modulators described herein;

[0175] (c) inhibits cancer (e.g., metastatic cancer); or [0176] (d) a combination thereof.

[0177] Assays to evaluate the binding ability of the antibodies toward a human regulators/modulators described herein can be used, including for example, ELISAs, Western blots and RIAs. The binding kinetics (e.g., binding affinity) of the antibodies also can be assessed by standard assays known in the art, such as by Biacore™ analysis.

[0178] Given that each of the subject antibodies can bind to a human regulator/modulator described herein, the V.sub.L and V.sub.H sequences can be “mixed and matched” to create other binding molecules that bind to a human regulators/modulators described herein. The binding properties of

such “mixed and matched” antibodies can be tested using the binding assays described above and assessed in assays described in the examples. When V.sub.L and V.sub.H chains are mixed and matched, a V.sub.H sequence from a particular V.sub.H/V.sub.L pairing can be replaced with a structurally similar V.sub.H sequence. Likewise, preferably a V.sub.L sequence from a particular V.sub.H/V.sub.L pairing is replaced with a structurally similar V.sub.L sequence.

[0179] Accordingly, in one aspect, the invention provides an isolated monoclonal antibody, or antigen binding portion thereof comprising: [0180] (a) a heavy chain variable region comprising an amino acid sequence; and [0181] (b) a light chain variable region comprising an amino acid sequence; wherein the antibody specifically binds a human regulators/modulators described herein.

[0182] In some cases, the CDR3 domain, independently from the CDR1 and/or CDR2 domain(s), alone can determine the binding specificity of an antibody for a cognate antigen and that multiple antibodies can predictably be generated having the same binding specificity based on a common CDR3 sequence. See, for example, Klimka et al., *British J. Cancer* 83(2):252-260 (2000) (describing the production of a humanized anti-CD30 antibody using only the heavy chain variable domain CDR3 of murine anti-CD30 antibody Ki-4), Beiboer et al., *J. Mol. Biol.* 296:833-849 (2000) (describing recombinant epithelial glycoprotein-2 (EGP-2) antibodies using only the heavy chain CDR3 sequence of the parental murine MOC-31 anti-EGP-2 antibody); Rader et al., *Proc. Natl. Acad. Sci. U.S.A.* 95:8910-8915 (1998) (describing a panel of humanized anti-integrin alphavbetaw antibodies using a heavy and light chain variable CDR3 domain. Hence, in some cases a mixed and matched antibody or a humanized antibody contains a CDR3 antigen binding domain that is specific for regulators/modulators described herein.

Assays for Drug Development

[0183] Methods are also described herein for evaluating whether test agents can modulate the expression or activity of any of the regulators/modulators described herein. T cells, cancer cells, and combinations thereof can be evaluated for susceptibility to treatment with candidate compounds.

[0184] specifically, the methods can include assay steps for identifying a candidate test agent that selectively modulates the proliferation, functioning, or visibility of T cells or cancer cells, or for increasing or decreasing the levels or functioning of regulators described herein. For example, if the proliferation, cytokine production, activity, or viability of T cells is increased or decreased in the presence of one or more of the regulators described herein but the proliferation, cytokine production, activity, or the proliferation, activity, or viability of the T cells in the T cell-regulator assay mixture changes in the presence of a test agent then that test agent has utility for modulating the regulator of the T cells. Such a test agent is referred to as a modulator.

[0185] An assay can include determining whether a test agent can specifically cause decreased or increased numbers of T cells or whether a compound can specifically cause decreased or increased functioning of T cells. If the test agent does cause altered T cell numbers or T cell functioning, then the test agent can be selected/identified for further study, such as for its suitability as a therapeutic agent to treat a cancer or an immune condition or disease. For example, the test agent identified by the selection methods featured in the invention can be further examined for their ability to target a tumor, target an immune cell, or to treat cancer by, for example, administering the test agent (modulator) to an animal model.

[0186] The cells that are evaluated can include cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 cells, CD8 T cells, metastatic cells, benign cell samples, cell lines (including as cancer cell lines), or a combination thereof. The cells that are evaluated can also include cells from a patient with cancer (including a patient with metastatic cancer), or cells from a known cancer type or cancer cell line, or cells exhibiting an overproduction of any of the regulators described herein. A test agent that can modulate the production or activity of any of these cell types can be administered to an animal, including a patient.

[0187] For example one method can include (a) obtaining a cell sample from a patient; (b)

measuring the amount or concentration of T cells/regulators/modulators in a known number or weight of cells from the sample to generate a reference value; (c) mixing the known number or weight of cells from the sample with a test agent to generate a test assay; to generate a test assay T cell/regulator/modulator value; (e) optionally repeating steps (c) and (d) with separate samples; and (f) selecting a test agent with a lower or a test assay T cell/regulator/modulator value than the reference value. The method can further include administering a test agent to an animal model, for example, to further evaluate the toxicity and/or efficacy of the test agent. In some cases, the method can further include administering the test agent to the patient from whom the cell or tissue sample was obtained.

[0188] Test agent or modulators (e.g., top hits identified by any method described herein) can be used in a cell-based assay using T cells or cells that express any of the regulators described herein as a readout of the efficacy of the test agents or modulators.

[0189] Assay methods are also described herein for identifying and assessing the potency of agents that may modulate T cells any of the regulators listed in tables 1-7 of FIGS. 1-4.

[0190] For example, T cells can release cytokines, such as Interferon γ or Interleukin-2, T cells or T cells expressing any of the modulators described herein can be contacted with a test agent and the release of cytokines by the T-cells can be measured. Such a test agent-related level of cytokines can be compared to the level observed for T cells not contacted with a test agent..

[0191] Useful test regulators, modulators, and test agents can be administered to a. test animal or a patient.

[0192] "Treatment" or "treating" refers to both therapeutic treatment and to prophylactic or preventative measures. Those in need of treatment include those already with the disorder as well as those prone to have the disorder, or those in whom the disorder is to be prevented.

[0193] "Subject" for purposes of administration of a regulator, modulator, test agent or composition described herein refers to administration to any animal classified as a mammal or bird, including humans, domestic animals, farm animals, zoo animals, experimental animals, pet animals, such as dogs, horses, cats, cows, etc. The experimental animals can include mice, rats, guinea pigs, goats, dogs, monkeys, or a combination thereof. In some cases, the subject is human.

[0194] As used herein, the term "cancer" includes solid animal tumors as well as hematological malignancies. The terms "tumor cell(s)" and "cancer cell(s)" are used interchangeably herein.

[0195] "Solid animal tumors" include cancers of the head and neck, lung, mesothelioma, mediastinum, lung, esophagus, stomach, pancreas, hepatobiliary system, small intestine, colon, colorectal, rectum, anus, kidney, urethra, bladder, prostate, urethra, penis, testis, gynecological organs, ovaries, breast, endocrine system, skin central nervous system; sarcomas of the soft tissue and bone; and melanoma of cutaneous and intraocular origin. In addition, a metastatic cancer at any stage of progression can be treated, such as micrometastatic tumors, megametastatic tumors, and recurrent cancers.

[0196] In some cases, a hematological cancer or hematological malignancy can be treated. The term "hematological malignancies" includes adult or childhood leukemia and lymphomas, Hodgkin's disease, lymphomas of lymphocytic and cutaneous origin, acute and chronic leukemia, plasma cell neoplasm, and cancers associated with AIDS.

[0197] The inventive methods and compositions can also be used to treat leukemias, lymph nodes, thymus tissues, tonsils, spleen, cancer of the breast, cancer of the lung, cancer of the adrenal cortex, cancer of the cervix, cancer of the endometrium, cancer of the esophagus, cancer of the head and neck, cancer of the liver, cancer of the pancreas, cancer of the prostate, cancer of the thyrnus, carcinoid tumors, chronic lymphocytic leukemia, Ewing's sarcoma, gestational trophoblastic tumors, hepatoblastoma, multiple myeloma, non-small cell lung cancer, retinoblastoma, or tumors in the ovaries. A cancer at any stage of progression can be treated or detected, such as primary, metastatic, and recurrent cancers. In some cases, metastatic cancers are treated but primary cancers are not treated. Information regarding numerous types of cancer can be found, e.g., from the

American Cancer Society (cancer.org), or from, e.g., Wilson et al. (1991) Harrison's Principles of Internal Medicine, 12th Edition, McGraw-Hill, Inc.

[0198] In some embodiments, the cancer and/or tumors to be treated are hematological malignancies, or those of lymphoid origin such as cancers or tumors of lymph nodes, thymus tissues, tonsils, spleen, and cells related thereto. In some embodiments, the cancer and/or tumors to be treated are those that have been resistant to T cell therapies.

[0199] Treatment of, or treating, metastatic cancer can include the reduction in cancer cell migration or the reduction in establishment of at least one metastatic tumor. The treatment also includes alleviation or diminishment of more than one symptom of metastatic cancer such as coughing, shortness of breath, hemoptysis, lymphadenopathy, enlarged liver, nausea, jaundice, bone pain, bone fractures, headaches, seizures, systemic pain and combinations thereof. The treatment may cure the cancer, e.g., it may prevent metastatic cancer, it may substantially eliminate metastatic tumor formation and growth, and/or it may arrest or inhibit the migration of metastatic cancer cells.

[0200] Anti-cancer activity can reduce the progression of a variety of cancers (e.g., breast, lung, pancreatic, or prostate cancer) using methods available to one of skill in the art. Anti-cancer activity, for example, can be determined by identifying the lethal dose (LD₅₀) or the 50% effective dose (ED₅₀) or the dose that inhibits growth at 50% (GI₅₀) of an agent of the present invention that prevents the migration of cancer cells. In one aspect, anti-cancer activity is the amount of the agent that reduces 50%, 60%, 70%, 80%, 90%, 95%, 97%, 98%, 99% or 100% of cancer cell migration, for example, when measured by detecting expression of a cancer cell marker at sites proximal or distal from a primary tumor site, or when assessed using available methods for detecting metastases.

[0201] In another example, agents that increase or decrease regulator/modulator expression or function can be administered to sensitize tumor cells to immune therapies. Hence, by administering an agent that increases or regulates/modulates expression or function, tumor cells can become more sensitive to the immune system and to various immune therapies.

Compositions

[0202] The invention also relates to compositions containing one or more active agents such as any of the regulators described herein, modulators described herein, or combinations thereof. Such active agents can be a polypeptide, a nucleic acid encoding a polypeptide (e.g., within an expression cassette or expression vector), a modified cell, an inhibitory nucleic acid, a small molecule, a compound identified by a method described herein, or a combination thereof. The compositions can be pharmaceutical compositions.

[0203] In some embodiments, the compositions can include a pharmaceutically acceptable carrier. By “pharmaceutically acceptable” it is meant that a carrier, diluent, excipient, and/or salt is compatible with the other ingredients of the formulation, and not deleterious to the recipient thereof.

[0204] The composition can be formulated in any convenient form. In some embodiments, the compositions can include a protein or polypeptide encoded by any of the genes listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include at least one nucleic acid or expression cassette encoding a polypeptide listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include at least one nucleic acid or expression cassette that includes a nucleic acid segment complementary to a gene listed in Table 1-2 (e.g., an inhibitory nucleic acid). In other embodiments, the compositions can include at least one nucleic acid or expression cassette that includes a nucleic acid segment encoding a Cas nuclease and at least one guide RNA that can target a regulator or modulator described herein. In other embodiments, the compositions can include at least one antibody that binds at least one protein encoded by at least one gene listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include at least one small molecule that binds, that activates, or that inhibits at least one gene listed in Tables 1-7 or FIG. 1-4, or at least one

small molecule that binds, that activates, or that inhibits at least one protein encoded by at least one gene listed in Tables 1-7 or FIGS. 1-4. In other embodiments, the compositions can include cells with at least one modified genomic regulator or modulator genetic site, cells that express one or more of the regulators described herein, cells that express a cas nuclease and at least one guide RNA that can target at least one regulator or modulator gene, cells that express one or more inhibitory nucleic acids, or a combination thereof. The cells can be immune cells. In some cases, the cells can be one or more types of lymphoid cells, myeloid cells, cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof.

[0205] The amount or number of cells administered can vary but amounts in the range of about 10^{sup.6} to about 10^{sup.9} cells can be used. The cells are generally delivered in a physiological solution such as saline or buffered saline. The cells can also be delivered in a vehicle such as within a population of liposomes, exosomes or microvesicles.

[0206] In some embodiments, the active agents of the invention (e.g., polypeptide, a nucleic acid encoding a polypeptide (e.g., within an expression cassette or expression vector), an antibody, an inhibitory nucleic acid, a small molecule, a compound identified by a method described herein, modified cells, or a combination thereof), are administered in a “therapeutically effective amount.” Such a therapeutically effective amount is an amount sufficient to obtain the desired physiological effect, such a reduction of at least one symptom of disease.

[0207] The disease can be cancer or an immune disease or condition. For example, active agents can reduce the symptoms of disease by 5%, or 10%, or 15%, or 20%, or 25%, or 30%, or 35%, or 40%, or 45%, or 50%, or 55%, or 60%, or 65%, or 70%, or 80% or 90%, 95%, or 97%, or 99%, or any numerical percentage between 5% and 100%. For example, symptoms of cancer can also include tumor cachexia, tumor-induced pain conditions, tumor-induced fatigue, tumor growth, and metastatic spread. Hence, the active agents may also reduce tumor cachexia, tumor-induced pain conditions, tumor-induced fatigue, tumor growth, or a combination thereof by 5%, or 10%, or 15%, or 20%, or 25%, or 30%, or 35%, or 40%, or 45%, or 50%, or 55%, or 60%, or 65%, or 70%, or 80%, or 90%, 95%, or 97%, or 99%, or any numerical percentage between 5% and 100%.

[0208] To achieve the desired effect(s), the active agents may be administered as single or divided dosages. For example, active agents can be administered in dosages of at least about 0.01 mg/kg to about 500 to 750 mg/kg, of at least about 0.01 mg/kg to about 300 to 500 mg/kg, at least about 0.1 mg/kg to about 100 to 300 mg/kg or at least about 1 mg/kg to about 50 to 100 mg/kg of body weight, although other dosages may provide beneficial results. The amount administered will vary depending on various factors including, but not limited to, the type of small molecules, compounds, peptides, or nucleic acid chosen for administration, the disease, the weight, the physical condition, the health, and the age of the mammal. Such factors can be readily determined by the clinician employing animal models or other test systems that are available in the art.

[0209] Administration of the active agents in accordance with the present invention may be in a single dose, in multiple doses, in a continuous or intermittent manner, depending, for example, upon the recipient's physiological condition, whether the purpose of the administration is therapeutic or prophylactic, and other factors known to skilled practitioners. The administration of the active agents and compositions of the invention may be essentially continuous over a preselected period of time or may be in a series of spaced doses. Both local and systemic administration is contemplated.

[0210] To prepare the composition, small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, and other agents are synthesized or otherwise obtained, purified as necessary or desired. These small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, and other agents can be suspended in a pharmaceutically acceptable carrier and/or lyophilized or otherwise stabilized. The

small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, other agents, and combinations thereof can be adjusted to an appropriate concentration, and optionally combined with other agents. The absolute weight of a given small molecule, compound, polypeptide, nucleic acid, ribonucleoprotein complex, and/or other agents included in a unit dose can vary widely. For example, about 0.01 to about 2 g, or about 0.1 to about 500 mg, of at least one molecule, compound, polypeptide, nucleic acid, ribonucleoprotein complexes, and/or other agent, or a plurality of molecules, compounds, polypeptides, nucleic acids, ribonucleoprotein complexes, and/or other agents can be administered. Alternatively, the unit dosage can vary from about 0.01 g to about 50 g, from about 0.01 g to about 35 g, from about 1 g to about 25 g, from about 0.5 g to about 12 g, from about 0.5 g to about 8 g, from about 0.5 g to about 4 g, or from about 0.5 g to about 2 g.

[0211] Daily doses of the active agents of the invention can vary as well. Such daily doses can range, for example, from about 0.1 g/day to about 50 g/day, from about 0.1 g/day to about 25 g/day, from about 0.1 g/day to about 12 g/day, from about 0.5 g/day to about 8 g/day, from about 0.5 g/day to about 4 g/day, and from about 0.5 g/day to about 2 g/day.

[0212] It will be appreciated that the amount of active agent for use in treatment will vary not only with the particular carrier selected but also with the route of administration, the nature of the cancer condition being treated and the age and condition of the patient. Ultimately the attendant health care provider can determine proper dosage. In addition, a pharmaceutical composition can be formulated as a single unit dosage form.

[0213] Thus, one or more suitable unit dosage forms comprising the active agent(s) can be administered by a variety of routes including parenteral (including subcutaneous, intravenous, intramuscular and intraperitoneal), oral, rectal, dermal, transdermal, intrathoracic, intrapulmonary and intranasal (respiratory) routes. The active agent(s) may also be formulated for sustained release (for example, using microencapsulation, see WO 94/07529, and U.S. Pat. No. 4,962,091). The formulations may, where appropriate, be conveniently presented in discrete unit dosage forms and may be prepared by any of the methods well known to the pharmaceutical arts. Such methods may include the step of mixing the active agent with liquid carriers, solid matrices, semi-solid carriers, finely divided solid carriers or combinations thereof, and then, if necessary, introducing or shaping the product into the desired delivery system. For example, the active agent(s) can be linked to a convenient carrier such as a nanoparticle, albumin, polyalkylene glycol, or be supplied in prodrug form. The active agent(s), and combinations thereof can be combined with a carrier and/or encapsulated in a vesicle such as a liposome.

[0214] The compositions of the invention may be prepared in many forms that include aqueous solutions, suspensions, tablets, hard or soft gelatin capsules, and liposomes and other slow-release formulations, such as shaped polymeric gels. Administration of inhibitors can also involve parenteral or local administration of the in an aqueous solution or sustained release vehicle.

[0215] Thus, while the active agent(s) and/or other agents can sometimes be administered in an oral dosage form, that oral dosage form can be formulated so as to protect the small molecules, compounds, polypeptides, nucleic acids, expression cassettes, ribonucleoprotein complexes, and combinations thereof from degradation or breakdown before the small molecules, compounds, polypeptides, nucleic acids encoding such polypeptides, expression cassettes, ribonucleoprotein complexes, and combinations thereof provide therapeutic utility. For example, in some cases the small molecules, compounds, polypeptides, nucleic acids encoding such polypeptide, expression cassettes, ribonucleoprotein complexes, and/or other agents can be formulated for release into the intestine after passing through the stomach. Such formulations are described, for example, in U.S. Pat. No. 6,306,434 and in the references contained therein.

[0216] Liquid pharmaceutical compositions may be in the form of, for example, aqueous or oily suspensions, solutions, emulsions, syrups or elixirs, dry powders for constitution with water or other suitable vehicle before use. Such liquid pharmaceutical compositions may contain

conventional additives such as suspending agents, emulsifying agents, non-aqueous vehicles (which may include edible oils), or preservatives. The pharmaceutical compositions may take such forms as suspensions, solutions, or emulsions in oily or aqueous vehicles, and may contain formulatory agents such as suspending, stabilizing and/or dispersing agents. Suitable carriers include saline solution, encapsulating agents (e.g., liposomes), and other materials. The active agent(s) and/or other agents can be formulated in dry form (e.g., in freeze-dried form), in the presence or absence of a carrier.

[0217] If a carrier is desired, the carrier can be included in the pharmaceutical formulation, or can be separately packaged in a separate container, for addition to the inhibitor that is packaged in dry form, in suspension or in soluble concentrated form in a convenient liquid.

[0218] An active agent(s) and/or other agents can be formulated for parenteral administration (e.g., by injection, for example, bolus injection or continuous infusion) and may be presented in unit dosage form in ampoules, prefilled syringes, small volume infusion containers or multi-dose containers with an added preservative.

[0219] The compositions can also contain other ingredients such as active agents, anti-viral agents, antibacterial agents, antimicrobial agents and/or preservatives. Examples of additional therapeutic agents that may be used include, but are not limited to: alkylating agents, such as nitrogen mustards, alkyl sulfonates, nitrosoureas, ethylenimines, and triazenes; antimetabolites, such as folate antagonists, purine analogues, and pyrimidine analogues; antibiotics, such as anthracyclines, bleomycins, mitomycin, dactinomycin, and plicamycin; enzymes, such as L-asparaginase; farnesyl-protein transferase inhibitors; hormonal agents, such as glucocorticoids, estrogens/antiestrogens, androgens/antiandrogens, progestins, and luteinizing hormone-releasing hormone antagonists, octreotide acetate; microtubule-disruptor agents, such as ecteinascidins or their analogs and derivatives; microtubule-stabilizing agents such as paclitaxel (Taxol®), docetaxel (Taxotere®), and epothilones A-F or their analogs or derivatives; plant-derived products, such as *vinca* alkaloids, epipodophyllotoxins, taxanes; and topoisomerase inhibitors; prenyl-protein transferase inhibitors, and miscellaneous agents such as, hydroxyurea, procarbazine, mitotane, hexamethylmelamine, platinum coordination complexes such as cisplatin and carboplatin; and other agents used as anti-cancer and cytotoxic agents such as biological response modifiers, growth factors; immune modulators, and monoclonal antibodies. The compositions can also be used in conjunction with radiation therapy.

[0220] The present description is further illustrated by the following examples, which should not be construed as limiting in any way. The contents of all cited references (including literature references, issued patents, published patent applications as cited throughout this application) are hereby expressly incorporated by reference.

Example 1: CRISPRa Screening Primary Human T cells to Identify Genetic Regulators

[0221] This Example describes use of CRISPRa for screening of primary human T cells to identify genetic regulators of therapeutically relevant T cell phenotypes.

[0222] T cells were isolated from two separate donors. The two populations of T cells were transduced with a dCas9-VP64-expressing lentivirus (CRISPRa), or KRAB-dCas9-expressing lentivirus (CRISPRi) and T cells that stably expressed dCas9 were selected with mCherry. The dCas9-VP64 or KRAB-dCas9 expressing T cell populations were then transfected with two genome-wide sgRNA libraries each to initiate CRISPR activation or interference of the T cells' genomes. For CRISPR activation Calabrese Sets A & B were used (see website at addgene.org/pooled-library/broadgpp-human-crispra-calabrese-p65hsf/). For CRISPR interference, Dolcetto Sets A & B were used (see, addgene.org/pooled-library/broadgpp-human-crispri-dolcetto/).

[0223] The T cell populations were stimulated with Immunocult0' CD3/CD28/CD2 T cell activator (Stemcell Technologies, Vancouver, Canada), and the stimulated CRISPRa/i edited T cells from the two donors were sorted using fluorescent activated cell sorting (FACS) for the following markers:

IL-2 cytokine production, IFN- γ production, and CellTraceM Violet for cell proliferation. Sorted cells were subjected to genomic DNA extraction, and sgRNAs were PCR amplified, followed by next-generation-sequencing, to determine sgRNA frequencies in each population. Data was analyzed using MaGeck version 0.5.9.2 Li et al. Genome Biol 15:544 (2014).

[0224] These screens identified 1074 unique genes with significant responses to those phenotypes (FDR<-0.01), including both known and novel genes in T cell function.

[0225] Table 1 below lists positive regulators of T cell functions as detected by IFN- γ production.

TABLE-US-00067 TABLE 1 Positive Regulators of T Cell Functions As detected by Interferon- γ

Production	Positive	Regulator	Interferon- γ	Gene	Production
APOBEC3C	IFNG				
APOBEC3D	IFNG	APOL2	IFNG	ASB12	IFNG
				BACE2	IFNG
				BCL9	IFNG
				BICDL2	IFNG
C15orf52	IFNG	C1orf94	IFNG	CD2	IFNG
				CD247	IFNG
				CD28	IFNG
				CNGB1	IFNG
				CTSK	IFNG
DEAF1	IFNG	DEF6	IFNG	DEPDC7	IFNG
				DKK2	IFNG
				EMP1	IFNG
				EOMES	IFNG
				EP300	
				FLT3	IFNG
				FOSL1	IFNG
				FOXQ1	IFNG
				GIN53	IFNG
				GLMN	IFNG
				GNA11	IFNG
				HELZ2	IFNG
				HRASLS5	IFNG
				IFNG	IFNG
				IL1R1	IFNG
				IL9R	IFNG
				KLHDC3	IFNG
				KLRC4	IFNG
				LAT	IFNG
				LCP2	IFNG
				LDB2	IFNG
				LTBR	IFNG
				MVB12A	IFNG
				NBPF6	IFNG
				NIT1	IFNG
				NLRC3	IFNG
				ORC1	IFNG
				OTUD7A	IFNG
				OTUD7B	IFNG
				PIK3AP1	IFNG
				PLCG2	IFNG
				PRDM1	IFNG
				PRKD2	IFNG
				PROCA1	IFNG
				RELA	IFNG
				RNF217	IFNG
				SAFB2	IFNG
				SLC16A1	IFNG
				SLC5A10	IFNG
				SLC7A3	IFNG
				SPPL2B	IFNG
				TAGAP	IFNG
				TBX21	IFNG
				TMEM150B	IFNG
				TMIGD2	IFNG
				TNFRSF12A	IFNG
				TNFRSF14	IFNG
				TNFRSF1A	IFNG
				TNFRSF1B	IFNG
				TNFRSF8	IFNG
				TNFRSF9	IFNG
				TOR1A	IFNG
				TPGS2	IFNG
				TRADD	IFNG
				TRAF3IP2	IFNG
				TRIM21	IFNG
				VAV1	IFNG
				WT1	IFNG
				ZNF630	IFNG
				ZNF717	IFNG

[0226] Table 2 below lists positive regulators of T cell functions as detected by Interleukin-2 production.

TABLE-US-00068 TABLE 2 Positive Regulators of T Cells as Detected by Interleukin-2

Production	Positive	Regulator	Positive	IL2	Gene	Production
ABCB10	IL2	ACSS2	IL2	ADAM19		
IL2	ADAM23	IL2	ADAMTS5	IL2	ALKBH7	IL2
				ALX4	IL2	ANXA2R
				IL2	AP2A1	IL2
				APOBEC3C	IL2	APOBEC3D
				IL2	APOL2	IL2
				ARNT	IL2	ART1
				IL2	ASCL4	IL2
				BEX4	IL2	BTG2
				IL2	BTNL2	IL2
				C11orf21	IL2	C12orf80
				IL2	CBX4	IL2
				CBY1	IL2	CCDC183
				IL2	CCDC71L	IL2
				CD2	IL2	CD28
				IL2	CD6	IL2
				CDKN1B	IL2	CDKN2C
				IL2	CHERP	IL2
				CIPC	IL2	CLIP3
				IL2	CNGB1	IL2
				CNR2	IL2	CREB5
				IL2	CUL3	IL2
				DCTN5	IL2	DEF6
				IL2	DEPDC7	IL2
				DYNLL2		
				IL2	EAPP	IL2
				EEP1	IL2	ELFN2
				IL2	EMB	IL2
				EMP1	IL2	EMP3
				IL2	EP300	IL2
				ERCC3	IL2	
				ESRP1	IL2	F2
				IL2	FBXL13	IL2
				FBXO41	IL2	FBNP1L
				IL2	FOSB	IL2
				FOSL1	IL2	FOXO4
				IL2	FOXQ1	IL2
				FUZ	IL2	GABRG1
				IL2	GGTLC2	IL2
				GNPDA1	IL2	GPR18
				IL2	GPR20	IL2
				GPR21		
				IL2	GPR84	IL2
				GRIN3A	IL2	GSDMD
				IL2	GSTM1	IL2
				HCST	IL2	HELZ2
				IL2	HEPHL1	IL2
				IL2	IL2RB	IL2
				IRX4	IL2	ISM1
				IL2	KLF7	IL2
				KLRC4	IL2	KRT18
				IL2	LAT	IL2
				LCP2	IL2	LHX6
				IL2	LMNA	IL2
				MAGEA9B	IL2	MAP3K12
				IL2	MERTK	IL2
				MTMR11	IL2	NDRG3
				IL2	NIT1	
				NLRC3	IL2	NLRP2
				IL2	NPLOC4	IL2
				ORC1	IL2	OSBPL7
				IL2	OTOP3	IL2
				OTUD7A	IL2	
				OTUD7B	IL2	P2RY14
				IL2	PAFAH1B2	IL2
				PCP4	IL2	PDE3A
				IL2	PHF8	IL2
				PIK3AP1	IL2	
				PLA2G3	IL2	PLCG2
				IL2	POLK	IL2
				POU2F2	IL2	PPIL2
				IL2	PRAC1	IL2
				PRKCB	IL2	PRKD2
				IL2	RAB6A	IL2
				RAC1	IL2	RAC2
				IL2	RIPK3	IL2
				RRAS2	IL2	RYS1
				IL2	SAFB2	IL2
				SCN3A	IL2	
				SDCCAG8	IL2	SERPINF1
				IL2	SGTA	IL2
				SHOC2	IL2	SIGLEC1
				IL2	SIRT1	IL2
				SLC16A1	IL2	
				SLC44A5	IL2	SLC5A5
				IL2	SMC4	IL2
				SPPL2B	IL2	SSUH2
				IL2	SWAP70	IL2
				TAF15	IL2	
				THEMIS	IL2	TM4SF4
				IL2	TMEM79	IL2
				TNFRSF10B	IL2	TNFSF11
				IL2	TNRC6A	IL2
				TPGS2		
				IL2	TRAF3IP2	IL2
				TRIM21	IL2	TRMT5
				IL2	TRPM4	IL2
				TRPV5	IL2	TSPYL5
				IL2	UBA52	IL2
				UBL5	IL2	VAV1
				IL2	WARS2	IL2
				ZAP70	IL2	ZNF141
				IL2	ZNF296	IL2
				ZNF701	IL2	

[0227] Table 3 below lists positive regulators of T cell functions as detected by T cell proliferation.

TABLE-US-00069 TABLE 3 Positive Regulators of T Cells as Detected Cell Proliferation Positive

Regulator	Increased	Cell	Gene	Proliferation
ABCB1	Proliferation	ASAP1	Proliferation	ATP10A
Proliferation	DEAF1	Proliferation	FOKK1	Proliferation
ITGAX	Proliferation	LCE6A	Proliferation	LCP2
Proliferation	LEFTY1	Proliferation	MYC	Proliferation
NAT8B	Proliferation	OLFM3		

[0228] Table 4 below lists negative regulators of T cell functions as detected by reduced IFN- γ production.

TABLE-US-00070 TABLE 4 Negative Regulators of T Cell Functions As detected by Less Interferon- γ Production

Negative Regulator	Interferon- γ Genes	Production
ACER2	IFNG	
ADGRV1	IFNG	
AIF1L	IFNG	
ALPL	IFNG	
AMACR	IFNG	
AMZ1	IFNG	
ARHGAP30	IFNG	
ARHGDIB	IFNG	
ARHGEF11	IFNG	
ARL11	IFNG	
ATP2A2	IFNG	
B3GNT5	IFNG	
BACH2	IFNG	
BLM	IFNG	
BSG	IFNG	
BTBD2	IFNG	
BTLA	IFNG	
BTRC	IFNG	
CA11	IFNG	
CASTOR2	IFNG	
CBLB	IFNG	
CCNT2	IFNG	
CCSER1	IFNG	
CD37	IFNG	
CD44	IFNG	
CD5	IFNG	
CD52	IFNG	
CD55	IFNG	
CDK6	IFNG	
CEACAM1	IFNG	
CEBPA	IFNG	
CEBPB	IFNG	
CEP164	IFNG	
CKAP2L	IFNG	
CLCN2	IFNG	
CLDN25	IFNG	
COLQ	IFNG	
CST5	IFNG	
CTNNA1	IFNG	
CYP24A1	IFNG	
DDIT4L	IFNG	
DENND3	IFNG	
DGKG	IFNG	
DGKK	IFNG	
DGKZ	IFNG	
DSC1	IFNG	
EBF2	IFNG	
ECEL1	IFNG	
EIF3K	IFNG	
EPB41	IFNG	
EPS8L1	IFNG	
FAM35A	IFNG	
FAM53B	IFNG	
FAM83A	IFNG	
FKRP	IFNG	
FOXA3	IFNG	
FOXF1	IFNG	
FOXF2	IFNG	
FOXI3	IFNG	
FOXJ1	IFNG	
FOXL2	IFNG	
FOXL2NB	IFNG	
GABRQ	IFNG	
GATA3	IFNG	
GATA4	IFNG	
GATA6	IFNG	
GCM2	IFNG	
GCSAM	IFNG	
GCSAML	IFNG	
GMFG	IFNG	
GNL3L	IFNG	
GRAP	IFNG	
GRB2	IFNG	
GRIA1	IFNG	
GTSF1L	IFNG	
HRH2	IFNG	
HYLS1	IFNG	
IKZF1	IFNG	
IKZF3	IFNG	
IL2RB	IFNG	
INPPL1	IFNG	
JMJD1C	IFNG	
KCNV1	IFNG	
KRIT1	IFNG	
LAMB1	IFNG	
LAPTM5	IFNG	
LAT2	IFNG	
LAX1	IFNG	
LCK	IFNG	
LENEP	IFNG	
LMO4	IFNG	
LRRC25	IFNG	
LRRC4B	IFNG	
LYN	IFNG	
MAB21L2	IFNG	
MAP4K1	IFNG	
MBIP	IFNG	
MBOAT1	IFNG	
METTL23	IFNG	
MIPEP	IFNG	
MIPOL1	IFNG	
MMP21	IFNG	
MSMB	IFNG	
MUC1	IFNG	
MUC21	IFNG	
MUC8	IFNG	
N4BP1	IFNG	
NAIF1	IFNG	
NDNF	IFNG	
NEATC1	IFNG	
NFKB2	IFNG	
NFKBIA	IFNG	
NKX2-1	IFNG	
NKX2-3	IFNG	
NMB	IFNG	
NR2F1	IFNG	
ODF4	IFNG	
OPRD1	IFNG	
ORC5	IFNG	
OTUD4	IFNG	
PASD1	IFNG	
PBK	IFNG	
PCBP2	IFNG	
PDLIM1	IFNG	
PDPN	IFNG	
PECAM1	IFNG	
PIP5K1A	IFNG	
PIP5K1B	IFNG	
PITPNA	IFNG	
POGZ	IFNG	
POLK	IFNG	
POU2AF1	IFNG	
PSTPIP1	IFNG	
PTPN12	IFNG	
PTPRC	IFNG	
PVRIG	IFNG	
RAB14	IFNG	
RBP7	IFNG	
RETREG1	IFNG	
RFC2	IFNG	
RHCE	IFNG	
RNF19B	IFNG	
RNF2	IFNG	
RUSC2	IFNG	
SELPLG	IFNG	
SETD1B	IFNG	
SH3KBP1	IFNG	
SIGLEC6	IFNG	
SIPA1L1	IFNG	
SLA	IFNG	
SLA2	IFNG	
SLC26A4	IFNG	
SLC44A5	IFNG	
SLC45A1	IFNG	
SLC6A8	IFNG	
SLC6A9	IFNG	
SMAD9	IFNG	
SMAGP	IFNG	
SOCS3	IFNG	
SOX13	IFNG	
SPATA31A1	IFNG	
SPN	IFNG	
SPOCK3	IFNG	
SPRED1	IFNG	
STAP1	IFNG	
STK35	IFNG	
SULT6B1	IFNG	
SYT15	IFNG	
TEC	IFNG	
TIAM1	IFNG	
TMEM151A	IFNG	
TMEM87B	IFNG	
TMPRSS11E	IFNG	
TNNT2	IFNG	
TRIB2	IFNG	
TRIM28	IFNG	
TSPAN1	IFNG	
UBASH3B	IFNG	
UBQLN4	IFNG	
UBXN7	IFNG	
UNC119	IFNG	
UPP1	IFNG	
VPS28	IFNG	
WLS	IFNG	
ZKSCAN4	IFNG	
ZNF445	IFNG	
ZNF474	IFNG	

[0229] Table 5 below lists negative regulators of T cell functions as detected by reduced Interleukin-2 production.

TABLE-US-00071 TABLE 5 Negative Regulators of T Cell Functions As detected by Less Interleukin-2 Production

Negative Regulator	Negative IL2 Gene	Production
ABI3BP	IL2	
AEBP1	IL2	
AHR	IL2	
ANTXR2	IL2	
ARHGAP15	IL2	
ARHGAP27	IL2	
ARHGDIB	IL2	
ARID3A	IL2	
ARL4D	IL2	
B4GALNT3	IL2	
BICD1	IL2	
C10orf82	IL2	
C17orf75	IL2	
C19orf35	IL2	
C1RL	IL2	
C2orf69	IL2	
C6orf132	IL2	
C9orf84	IL2	
CABP1	IL2	
CBLB	IL2	
CCSER1	IL2	
CD34	IL2	
CD4	IL2	
CD5	IL2	
CD52	IL2	
CEACAM1	IL2	
CEACAM7	IL2	
CEBPB	IL2	
CES3	IL2	
CGB3	IL2	
COL11A1	IL2	
COL4A3	IL2	
COLQ	IL2	
CPEB3	IL2	
CRELD2	IL2	
CST9L	IL2	
DDX55	IL2	
DLG4	IL2	
DOK1	IL2	
EBF3	IL2	
EIF3K	IL2	
EN2	IL2	
EOMES	IL2	
EPB41	IL2	
ETS1	IL2	
F5	IL2	
FAM96A	IL2	
FHL1	IL2	
FOXA3	IL2	
FOXE1	IL2	
FOXI3	IL2	
FOXL2NB	IL2	
FUS	IL2	
FUT4	IL2	
GCSAM	IL2	
GCSAML	IL2	
GDAP1L1	IL2	
GDPD2	IL2	
GMIP	IL2	
GNL3L	IL2	
GOLPH3	IL2	
GRAP	IL2	
GRB2	IL2	
HAUS7	IL2	
HERC1	IL2	
HLA-DQB2	IL2	
HSD17B11	IL2	
IKZF1	IL2	
IKZF3	IL2	
INPPL1	IL2	
INTS10	IL2	
ITIH2	IL2	
ITPKA	IL2	
ITPKB	IL2	
ITPKC	IL2	
JDP2	IL2	
JKAMP	IL2	
JMJD1C	IL2	
KIAA1024	IL2	
KIF15	IL2	
KIF5A	IL2	
KNTC1	IL2	
LAT2	IL2	
LAX1	IL2	
LGR5	IL2	
LIME1	IL2	

LMBRD2 IL2 LOC401052 IL2 LONP2 IL2 LRCH3 IL2 LRRC23 IL2 LRRC25 IL2 LRRC52 IL2 LYN IL2 LYPD1 IL2 MAATS1 IL2 MAB21L2 IL2 MAGEB17 IL2 MAP4K1 IL2 MEF2C IL2 METTL9 IL2 MICU1 IL2 MRPL17 IL2 MUC1 IL2 NAIF1 IL2 NCF2 IL2 NDNF IL2 NDUFB1 IL2 NHP2 IL2 NKX2-6 IL2 NLGN4Y IL2 NNT IL2 NP1PB9 IL2 NR4A1 IL2 NR4A3 IL2 NRCAM IL2 NRP1 IL2 NRSN2 IL2 NSUN7 IL2 OLFML1 IL2 OMP IL2 OPRD1 IL2 OR1K1 IL2 OR2B11 IL2 OSBPL11 IL2 OTOG IL2 OTUD4 IL2 PATL2 IL2 PAX5 IL2 PFKL IL2 PHF2 IL2 PIBF1 IL2 PIP5K1A IL2 PIP5K1B IL2 PITPNC1 IL2 PLCL1 IL2 PLEKHM2 IL2 PPARG IL2 PPIC IL2 PSRC1 IL2 PSTPIP1 IL2 PTPN12 IL2 PTPN22 IL2 PTPN6 IL2 PTPRC IL2 PVRIG IL2 RBP4 IL2 RPL13A IL2 S100A2 IL2 SALL4 IL2 SAMD8 IL2 SENP6 IL2 SETD1B IL2 SEZ6L IL2 SFT2D1 IL2 SH3TC1 IL2 SIGIRR IL2 SIT1 IL2 SLA IL2 SLA2 IL2 SLC20A2 IL2 SLC39A2 IL2 SLC6A8 IL2 SMAGP IL2 SNRNP48 IL2 SOCS2 IL2 SORBS1 IL2 SOX13 IL2 SPN IL2 SPRED1 IL2 SPRED2 IL2 SRPK1 IL2 STAP1 IL2 STK38L IL2 SYPL1 IL2 TCF12 IL2 TEX35 IL2 TFCEP2L1 IL2 TMEM14C IL2 TMEM223 IL2 TMEM262 IL2 TNNT2 IL2 TPRA1 IL2 TRIM6- IL2 TRIM34 TSPAN1 IL2 UBASH3B IL2 UBE2W IL2 UBR4 IL2 UBXN7 IL2 UCP1 IL2 UIMC1 IL2 ULK1 IL2 UPK3B IL2 VPS28 IL2 VSTM5 IL2 XKR9 IL2 YLPM1 IL2 ZDHHC7 IL2 ZEB1 IL2 ZEB2 IL2 ZNF445 IL2 ZNF70 IL2 ZNF831 IL2

[0230] Table 6 below lists negative regulators of T cell functions as detected by reduced cell proliferation.

TABLE-US-00072 TABLE 6 Negative Regulators of T Cell Functions As detected by Less Cell Proliferation Negative Decreased Regulator Cell Gene Proliferation ABCB1 Proliferation ASAP1 Proliferation ATP10A Proliferation DEAF1 Proliferation FOXK1 Proliferation ITGAX Proliferation LCE6A Proliferation LCP2 Proliferation LEFTY1 Proliferation MYC Proliferation NAT8B Proliferation OLFM3 Proliferation PLD6 Proliferation PREP Proliferation SULT1A1 Proliferation SULT1A4 Proliferation AHNAK Proliferation ARHGDIB Proliferation B3GNT5 Proliferation CASZ1 Proliferation CD27 Proliferation CEBPB Proliferation CRHBP Proliferation FLI1 Proliferation FOSL2 Proliferation HLX Proliferation MAP4K1 Proliferation MUC21 Proliferation MXI1 Proliferation NDRG1 Proliferation NEUROD2 Proliferation SLC2A1 Proliferation SLC43A3 Proliferation SMAGP Proliferation SOX13 Proliferation SP140 Proliferation TPI1 Proliferation TTC39C Proliferation

[0231] These regulators and agents that modulate these regulators can be used as T cell related immunotherapies for cancer or autoimmune diseases

Example 2: CRISPRi Identification of Genes that Regulate T Cells

[0232] This Example describes use of CRISPRi for screening of primary human T cells to identify genetic regulators of therapeutically relevant T cell phenotypes.

[0233] The two populations of T cells were transduced with a KRAB-dCas9-expressing lentivirus (CRISPRi) and T cells that stably expressed dCas9 were selected with mCherry. The KRAB-dCas9 expressing T cell populations were then transfected with two genome-wide sgRNA libraries each to initiate CRISPR interference of the T cells' genomes. For CRISPR interference, Dolcetto Sets A & B were used (see, addgene.org/pooled-library/broadgpp-human-crispri-dolcetto/).

[0234] The T cell populations were stimulated with Immunocult™ CD3/CD28/CD2 T cell activator (Stemncell Technologies, Vancouver, Canada), and the stimulated CRISPRi edited T cells from the two donors were sorted using fluorescent activated cell sorting (FACS) for the following markers: IL-2 cytokine production, IFN-γ production, and CellTrace™ Violet for cell proliferation. Sorted cells were subjected to genomic DNA extraction, and sgRNAs were PCR amplified, followed by next-generation-sequencing, to determine sgRNA frequencies in each population, Data was analyzed using MaGeck version 0.5.9.2 Li et al. Genome Biol 15:544 (2014). Table 7 lists the genes that modulated T cell functions.

TABLE-US-00073 TABLE 7 Genes from CRISPRi Screen that Modulate T Cells Gene Screen Positive or Negative Regulator ANKRD17 IFNG positive AQP3 IFNG positive ARID4B IFNG positive ATP6V1C1 IFNG positive ATPAF1 IFNG positive ATXN7 IFNG positive BCAT2 IFNG

positive BCL10 IFNG positive CASD1 IFNG positive CBBF IFNG positive CD2 IFNG positive CD247 IFNG positive CD28 IFNG positive CD3D IFNG positive CD3E IFNG positive CD3G IFNG positive CD4 IFNG positive CHUK IFNG positive CNP IFNG positive COG3 IFNG positive CREBBP IFNG positive CUL1 IFNG positive DDA1 IFNG positive DDX60L IFNG positive DEF6 IFNG positive DHDDS IFNG positive DHX29 IFNG positive DPP9-AS1 IFNG positive ELOF1 IFNG positive ERC1 IFNG positive ETNK1 IFNG positive EXOC4 IFNG positive FAM133B IFNG positive FGFR1OP IFNG positive FITM2 IFNG positive FLVCR2 IFNG positive FNDC4 IFNG positive GOSR1 IFNG positive GPX7 IFNG positive GRAP2 IFNG positive HARS IFNG positive HNRNPL IFNG positive HOXD13 IFNG positive IFNG IFNG positive IFNGR1 IFNG positive IFNGR2 IFNG positive IKBKB IFNG positive IKBKG IFNG positive IL21R IFNG positive INPP1 IFNG positive ITK IFNG positive JAK1 IFNG positive JUN IFNG positive KAT7 IFNG positive KCNIP3 IFNG positive KIAA1109 IFNG positive KIDINS220 IFNG positive LAT IFNG positive LCK IFNG positive LCP2 IFNG positive LIMS2 IFNG positive LOC101927322 IFNG positive LRIG1 IFNG positive MALT1 IFNG positive MAP3K7 IFNG positive MBD2 IFNG positive MEAF6 IFNG positive MEN1 IFNG positive MMP24 IFNG positive MOB4 IFNG positive MYLIP IFNG positive NDFIP2 IFNG positive NSD2 IFNG positive NSFL1C IFNG positive NYNRIN IFNG positive OSBP IFNG positive PCYT2 IFNG positive PGBD5 IFNG positive PI4KB IFNG positive PLCG1 IFNG positive PRDM1 IFNG positive PRKAR1A IFNG positive PRKD2 IFNG positive PRRC2B IFNG positive PTPRC IFNG positive RAC2 IFNG positive RAET1L IFNG positive RBCK1 IFNG positive RDX IFNG positive RHOA IFNG positive RHOG IFNG positive ROPN1B IFNG positive RRAS2 IFNG positive RTP2 IFNG positive SAE1 IFNG positive SCRIB IFNG positive SEC61A1 IFNG positive SEC62 IFNG positive SEH1L IFNG positive SEL1L IFNG positive SH2D1A IFNG positive SHOC2 IFNG positive SLC38A6 IFNG positive SLC3A2 IFNG positive SPCS2 IFNG positive SPTLC2 IFNG positive SPTSSA IFNG positive SRD5A2 IFNG positive SRP19 IFNG positive SRP68 IFNG positive SRP72 IFNG positive SRPRB IFNG positive SSB IFNG positive STAT3 IFNG positive SUGT1 IFNG positive SULT2B1 IFNG positive SUPT5H IFNG positive SYT15 IFNG positive TADA1 IFNG positive TADA2B IFNG positive TAF11 IFNG positive TAF13 IFNG positive TAF2 IFNG positive TAF6L IFNG positive TARS IFNG positive TBX21 IFNG positive TLN1 IFNG positive TMX1 IFNG positive TNFRSF1A IFNG positive TRAF6 IFNG positive TRIM21 IFNG positive TXK IFNG positive UBA2 IFNG positive VAV1 IFNG positive VPS29 IFNG positive VPS35 IFNG positive VPS37C IFNG positive VPS41 IFNG positive WAS IFNG positive XPO6 IFNG positive ZAP70 IFNG positive ARHGAP15 IFNG negative BRD9 IFNG negative BRIP1 IFNG negative CAD IFNG negative CBLB IFNG negative CBLL1 IFNG negative CD5 IFNG negative CDK12 IFNG negative CHERP IFNG negative CPSF2 IFNG negative CPSF6 IFNG negative CSTF3 IFNG negative CTDSPL2 IFNG negative DGKZ IFNG negative E2F1 IFNG negative EIF3B IFNG negative EIF3D IFNG negative EIF3K IFNG negative EIF4E2 IFNG negative GCN1 IFNG negative GIGYF2 IFNG negative GNAI2 IFNG negative HIF1AN IFNG negative IKBKE IFNG negative LARGE2 IFNG negative LAT2 IFNG negative MAP4K1 IFNG negative MCM2 IFNG negative METAP2 IFNG negative METTL3 IFNG negative MTF1 IFNG negative MYB IFNG negative NFKB2 IFNG negative NIT1 IFNG negative NMT1 IFNG negative NRF1 IFNG negative NUDC IFNG negative PCBP2 IFNG negative PDGFRA IFNG negative PITPNB IFNG negative PNISR IFNG negative PPP1R8 IFNG negative PRMT1 IFNG negative PSMD13 IFNG negative PSMD4 IFNG negative PTMA IFNG negative RAB4A IFNG negative RBPJ IFNG negative RIOK2 IFNG negative RNF20 IFNG negative RNF40 IFNG negative RPL19 IFNG negative RPL26 IFNG negative RPL35A IFNG negative RPL38 IFNG negative RPL6 IFNG negative RPS13 IFNG negative RPS17 IFNG negative RPS8 IFNG negative SCRIN3 IFNG negative SF3A1 IFNG negative SLA2 IFNG negative SLAMF6 IFNG negative SMC3 IFNG negative SP1 IFNG negative SPN IFNG negative SYMPK IFNG negative THOC3 IFNG negative TONSL IFNG negative TSC1 IFNG negative U2AF2 IFNG negative UBASH3A IFNG negative

UNCX IFNG negative USP5 IFNG negative ZC3H18 IFNG negative BCL10 IL2 positive CASD1
IL2 positive CD2 IL2 positive CD247 IL2 positive CD28 IL2 positive CD3D IL2 positive CD3E
IL2 positive CD3G IL2 positive CHD7 IL2 positive DEF6 IL2 positive DNTTIP1 IL2 positive
ELOF1 IL2 positive GRAP2 IL2 positive IFNGR2 IL2 positive IL2 IL2 positive ITK IL2 positive
KIDINS220 IL2 positive LAT IL2 positive LCP2 IL2 positive NDFIP2 IL2 positive NDUFB1 IL2
positive NYNRIN IL2 positive PGBD5 IL2 positive PLCG1 IL2 positive PRKD2 IL2 positive
RAC2 IL2 positive RHOG IL2 positive RPN2 IL2 positive SCRIB IL2 positive SHOC2 IL2
positive SIN3B IL2 positive SPRYD3 IL2 positive SRP19 IL2 positive SRP68 IL2 positive SRP72
IL2 positive SULT2B1 IL2 positive TAF11 IL2 positive TAF13 IL2 positive TAF2 IL2 positive
TAF8 IL2 positive TLN1 IL2 positive TRIM21 IL2 positive VAV1 IL2 positive VPS29 IL2
positive VPS35 IL2 positive WAS IL2 positive ZAP70 IL2 positive ACTL6A IL2 negative ADSS
IL2 negative ANLN IL2 negative ARHGAP15 IL2 negative ARID2 IL2 negative ATP1A1 IL2
negative AUNIP IL2 negative BECN1 IL2 negative BMS1 IL2 negative BOP1 IL2 negative
C21orf62 IL2 negative CAD IL2 negative CBFB IL2 negative CBLB IL2 negative CD5 IL2
negative CDC23 IL2 negative CDK12 IL2 negative CENPE IL2 negative CENPI IL2 negative
CEP192 IL2 negative CHAF1B IL2 negative CHERP IL2 negative CHMP3 IL2 negative CHMP5
IL2 negative CHMP6 IL2 negative CNOT1 IL2 negative CPSF4 IL2 negative CPSF6 IL2 negative
CTDSPL2 IL2 negative CTPS1 IL2 negative DDX47 IL2 negative DGKZ IL2 negative DHODH
IL2 negative DLST IL2 negative DNTTIP2 IL2 negative DPY19L3 IL2 negative E2F1 IL2
negative EDC4 IL2 negative EFTUD2 IL2 negative EIF3B IL2 negative EIF3D IL2 negative
EIF3E IL2 negative EIF3K IL2 negative EIFSA IL2 negative EP400 IL2 negative ESF1 IL2
negative FADD IL2 negative FAM49B IL2 negative FAM60A IL2 negative FAU IL2 negative
GCN1 IL2 negative GIGYF2 IL2 negative GINS3 IL2 negative HGS IL2 negative IL2RA IL2
negative IL2RB IL2 negative IL2RG IL2 negative ILF2 IL2 negative INTS3 IL2 negative JPH1
IL2 negative KANSL3 IL2 negative KAT5 IL2 negative KLF2 IL2 negative LAMTOR2 IL2
negative LOC401052 IL2 negative MAD2L1BP IL2 negative MAK16 IL2 negative MAP4K1 IL2
negative MAU2 IL2 negative MCM2 IL2 negative MCM3AP IL2 negative MEMO1 IL2 negative
METAP2 IL2 negative MMP16 IL2 negative MRPL22 IL2 negative MST1L IL2 negative
MYCBP2 IL2 negative NARFL IL2 negative NEPRO IL2 negative NFKB2 IL2 negative NMT1
IL2 negative NRF1 IL2 negative NUDC IL2 negative OTUB1 IL2 negative PCBP1 IL2 negative
PCBP2 IL2 negative PDGFRA IL2 negative PFAS IL2 negative PITPNB IL2 negative PNISR IL2
negative POLE IL2 negative POLR1B IL2 negative PPAN IL2 negative PPIH IL2 negative
PPP1R8 IL2 negative PRMT1 IL2 negative PRPF4B IL2 negative PRR12 IL2 negative PSMD2
IL2 negative PTPN23 IL2 negative PUM1 IL2 negative RAB4A IL2 negative RASA2 IL2 negative
RBM14 IL2 negative RBM25 IL2 negative RBM42 IL2 negative RBSN IL2 negative RCL1 IL2
negative RMND5A IL2 negative RNF20 IL2 negative RNF40 IL2 negative RPL10 IL2 negative
RPL10A IL2 negative RPL13 IL2 negative RPL14 IL2 negative RPL15 IL2 negative RPL18 IL2
negative RPL19 IL2 negative RPL23A IL2 negative RPL24 IL2 negative RPL26 IL2 negative
RPL27 IL2 negative RPL34 IL2 negative RPL35 IL2 negative RPL36 IL2 negative RPL37A IL2
negative RPL38 IL2 negative RPL6 IL2 negative RPL7A IL2 negative RPL8 IL2 negative RPL9
IL2 negative RPLP1 IL2 negative RPS11 IL2 negative RPS13 IL2 negative RPS16 IL2 negative
RPS17 IL2 negative RPS20 IL2 negative RPS23 IL2 negative RPS24 IL2 negative RPS25 IL2
negative RPS3 IL2 negative RPS3A IL2 negative RPS4X IL2 negative RPS5 IL2 negative RPS7
IL2 negative RPS8 IL2 negative RUVBL2 IL2 negative SART1 IL2 negative SETD1A IL2
negative SLAMF6 IL2 negative SLBP IL2 negative SMARCE1 IL2 negative SMC1A IL2 negative
SMC3 IL2 negative SNRNP27 IL2 negative SNRNP70 IL2 negative SNRPC IL2 negative SNRPF
IL2 negative SP1 IL2 negative SRFBP1 IL2 negative SRSF1 IL2 negative STAT5B IL2 negative
SURF6 IL2 negative SYMPK IL2 negative TBL1X IL2 negative THOC3 IL2 negative TNPO3 IL2
negative TRAF2 IL2 negative TRAIIP IL2 negative TSC1 IL2 negative TSG101 IL2 negative
TUBGCP5 IL2 negative TYMS IL2 negative U2AF1 IL2 negative U2AF2 IL2 negative

UBASH3A IL2 negative UBASH3B IL2 negative UPP1 IL2 negative UTP14A IL2 negative
UTP15 IL2 negative VPS28 IL2 negative WDR45 IL2 negative WDR5 IL2 negative YEATS4 IL2
negative ZMAT2 IL2 negative AAMP Proliferation positive AARS Proliferation positive AATF
Proliferation positive AK2 Proliferation positive ALDH18A1 Proliferation positive AP2M1
Proliferation positive ATIC Proliferation positive ATP1A1 Proliferation positive ATP5O
Proliferation positive ATP6V1B2 Proliferation positive ATP6V1F Proliferation positive ATXN10
Proliferation positive BMS1 Proliferation positive BOP1 Proliferation positive BUD23
Proliferation positive C12orf60 Proliferation positive CAD Proliferation positive CARS
Proliferation positive CCDC86 Proliferation positive CCT6A Proliferation positive CD247
Proliferation positive CD3D Proliferation positive CD3E Proliferation positive CD3EAP
Proliferation positive CD3G Proliferation positive CINP Proliferation positive CLNS1A
Proliferation positive CPSF4 Proliferation positive CRCP Proliferation positive CTPS1
Proliferation positive DAD1 Proliferation positive DDX27 Proliferation positive DDX52
Proliferation positive DGCR8 Proliferation positive DHODH Proliferation positive DHX29
Proliferation positive DHX37 Proliferation positive DICER1 Proliferation positive DNAJA3
Proliferation positive DNM2 Proliferation positive DNTTIP2 Proliferation positive DPH6
Proliferation positive DROSHA Proliferation positive EIF2B2 Proliferation positive EIF2B3
Proliferation positive EIF2B4 Proliferation positive EIF5A Proliferation positive ELP4
Proliferation positive ESF1 Proliferation positive EXOSC4 Proliferation positive EXOSC5
Proliferation positive EXOSC7 Proliferation positive EXOSC9 Proliferation positive FAM149B1
Proliferation positive FARSB Proliferation positive FBL Proliferation positive FCF1 Proliferation
positive FH Proliferation positive FLVCR1 Proliferation positive FTSJ3 Proliferation positive
GFER Proliferation positive GMPS Proliferation positive GNL2 Proliferation positive GNL3L
Proliferation positive GNPAT Proliferation positive GTF3C1 Proliferation positive HARS
Proliferation positive HAUS4 Proliferation positive HCCS Proliferation positive HEATR3
Proliferation positive HSD17B10 Proliferation positive HSD17B12 Proliferation positive HSPA9
Proliferation positive IL2RG Proliferation positive IMPDH2 Proliferation positive ISG20L2
Proliferation positive KARS Proliferation positive LAGE3 Proliferation positive LAT Proliferation
positive LCP2 Proliferation positive LETM1 Proliferation positive LONP1 Proliferation positive
MARS2 Proliferation positive MDN1 Proliferation positive METTL16 Proliferation positive
MMACHC Proliferation positive MRPL16 Proliferation positive MRPL22 Proliferation positive
MRPL35 Proliferation positive MRPL36 Proliferation positive MRPL37 Proliferation positive
MRPL41 Proliferation positive MRPL42 Proliferation positive MRPL45 Proliferation positive
MRPL54 Proliferation positive MRPS11 Proliferation positive MRPS14 Proliferation positive
MRPS17 Proliferation positive MRPS18A Proliferation positive MRPS2 Proliferation positive
MRPS23 Proliferation positive MRPS33 Proliferation positive MRPS5 Proliferation positive
MRPS9 Proliferation positive MTHED1L Proliferation positive MTOR Proliferation positive
MYBBP1A Proliferation positive NAT10 Proliferation positive NCL Proliferation positive NEPRO
Proliferation positive NIFK Proliferation positive NOC2L Proliferation positive NOL10
Proliferation positive NOL6 Proliferation positive NOL8 Proliferation positive NOP14
Proliferation positive NOP2 Proliferation positive NOP56 Proliferation positive NOP58
Proliferation positive NUBP1 Proliferation positive NUFIP1 Proliferation positive ORAOV1
Proliferation positive PAM16 Proliferation positive PCYT2 Proliferation positive PDCD11
Proliferation positive PDGFRA Proliferation positive PDHA1 Proliferation positive PDSS2
Proliferation positive PELP1 Proliferation positive PGD Proliferation positive PGM3 Proliferation
positive PHB Proliferation positive PHB2 Proliferation positive PISD Proliferation positive
PITRM1 Proliferation positive PMPCA Proliferation positive PNO1 Proliferation positive PNPT1
Proliferation positive POLG2 Proliferation positive POLR1A Proliferation positive POLR1C
Proliferation positive POLR1D Proliferation positive POLR2E Proliferation positive POLR3B
Proliferation positive POLRMT Proliferation positive POP1 Proliferation positive POP4

Proliferation positive POP5 Proliferation positive POT1 Proliferation positive PPAN Proliferation positive PPAT Proliferation positive PSMG1 Proliferation positive QARS Proliferation positive RAC2 Proliferation positive RBM19 Proliferation positive RCL1 Proliferation positive RIOK1 Proliferation positive RIOK2 Proliferation positive ROMO1 Proliferation positive RPF1 Proliferation positive RPF2 Proliferation positive RPL28 Proliferation positive RPL30 Proliferation positive RPL39 Proliferation positive RPLP2 Proliferation positive RPN2 Proliferation positive RPP21 Proliferation positive RPP30 Proliferation positive RPS11 Proliferation positive RPS12 Proliferation positive RPS15 Proliferation positive RPS17 Proliferation positive RPS19BP1 Proliferation positive RPS27 Proliferation positive RPS4X Proliferation positive RPS6 Proliferation positive RPSA Proliferation positive RRP12 Proliferation positive RRP36 Proliferation positive RRP7A Proliferation positive RRP9 Proliferation positive RSL24D1 Proliferation positive SAMM50 Proliferation positive SARS Proliferation positive SDHC Proliferation positive SEH1L Proliferation positive SLC35B1 Proliferation positive SLC38A6 Proliferation positive SLC7A11 Proliferation positive SPOUT1 Proliferation positive SRFBP1 Proliferation positive SSB Proliferation positive SURF6 Proliferation positive TAF1A Proliferation positive TAF1C Proliferation positive TAF1D Proliferation positive TAF8 Proliferation positive TAMM41 Proliferation positive TARS Proliferation positive TEX10 Proliferation positive TIMM44 Proliferation positive TNKS1BP1 Proliferation positive TOMM20 Proliferation positive TOMM40 Proliferation positive TP53I13 Proliferation positive TRMT112 Proliferation positive TRMT5 Proliferation positive TRNT1 Proliferation positive TSEN2 Proliferation positive TSEN54 Proliferation positive TSR1 Proliferation positive TTI1 Proliferation positive TWISTNB Proliferation positive TWNK Proliferation positive UMPS Proliferation positive UQCR10 Proliferation positive UQCRB Proliferation positive UQCRC1 Proliferation positive UTP11 Proliferation positive UTP14A Proliferation positive UTP3 Proliferation positive UTP6 Proliferation positive VAV1 Proliferation positive VPS29 Proliferation positive VPS72 Proliferation positive WAS Proliferation positive WDR12 Proliferation positive WDR3 Proliferation positive WDR36 Proliferation positive WDR55 Proliferation positive XRCC5 Proliferation positive XRCC6 Proliferation positive YAE1D1 Proliferation positive ZAP70 Proliferation positive ZCCHC9 Proliferation positive ZNHIT3 Proliferation positive ZNHIT6 Proliferation positive ZNRD1 Proliferation positive

[0235] This screen identified quite a few of the same genes as were identified in the screen described in Example 1. The following genes were new genes identified by this CRISPRi screen: HNRNPL, IHOXD13, IFNGR1, IFNGR2, IKBKB, IKBKG, IL21R, INPPi, ITKJAKI, JUN, KA T7, KCNIP3, KIAA1 A109, KID[NS220, LIMS2, LOC101927322, LRIGI, MALTI, MAP3K7, IViBD2, MEAF6, MENI, MMP4, MOB4, MYLIP, NDFIP2, NSD2, NSFL1C, NYNRIN, OSBP~, PCY T2, PGBD5, PI4 KB, PLCG1, PRKAR1A, PRRC2B, RAETI L, RBCK1, RDX, RHOA, RIOGi, ROPNIB, RTP2, SAE1, SCRIB, SEC61AI, SEC62, SEH1L, SEL1L, SI-2D1IA, SLC38A6, SLC3A2, SPCS2, SPTLC2, SPTSSA, SRD5A2, SRP19, SRP68, SRP72, SRPRB, SSB, STAT3, SUGTI, SULT2B1, SUPTIH1, TADA1, TADA2B, TAF 1, TAF13, TAF2, TAF6L, TARS, TLN1, TMNIX1, TRAF6, TXK, UBA2, VPS29, VPS35, VPS37C, VPS41, WAS, XPO6, BRD9, BIRIPI, CAD, CBLI, CDKI2, CPSF2, CPSF6, CSTF3, CTDSPL2, E2F1, EIF3B, EJF3D, EIF4E2, GCN1, GIGYF2, iNA12, HIFIAN, IKBKE, LARGE2, MCCM2, METAP2, METTL3, M-TF1, MYB, NMT1, NRFI, NUDC, PDGFRA, PITPNB, PNISR, PPP1R8, PRMTI, PSMD13, PSMD4, PTMA, RAB4A, RBPJ, RIOK2, RNF20, RNF40, RPL19, RPL26, RPL35A, RPL38, RPL6, RPS13, RPS17, RPS8, SCRNB, SF3AI, SLAMF6, SMC3, SPI, SYMHPK, THOC3, TONSL, TSCI, U2AF2, UBASH3A, 10 UNCX, USP5, ZC3H18, BCL10, CASD1, CD3D, CD3E, CD3G, CHD7, DNTTIP1, ELOF1, GRAP2, IFNGR2, ITK, KIDINS220, NDFIP2, NYNRIN, PGBD5, PLCG1, RHOG, RPN2, SCRIB, SIN3B, SPRYD3, SRP19, SRP68, SRP72, SULT2B1, TAF11, TAF13, TAF2, TAF8, TLN1, VPS29, VPS35, WAS, ACTL6A, ADSS, ANLN, ARID2, ATP1A1, AUNIP, BECN1, BMS1, BOP1, C2Iorf62, CAD, CBFB, CDC23, CDK12, CEN.sup.PE, CENPI, CEPi192,

CHAFIB, CHMFP3, CHMP5, CHMP6, CNOT1, CPSF4, CPSF6, CTDSP2, CTPS1, DDX4.7, DHODH, DLST, DNTTIP2, DPY19L3, E2FI, EDC4, EFTUID2, EIF3B, EIF3D, EIF3E, EF5A, EP400, ESFI, FADD, FAM149B, FAM60A, FAU, GCN1, GIGYF2, HGS, IL2RA, IL2EG, ILF2, INTS3, JPH1, KANSL3, KAT5, KLF2, LAMTOR2, MAD21BP, MAK16, MAU2, MC12, MCM3AP, MEMO1, TMTPT2, MMIVP16, MRPL22, MSTIL, MYCBP2, NAREL, NEPRO, IMT, Nsup.RFI, NUDC, OTUB1, PCBPI, PDGFRA, PEAS, PITPN3, PNISR, POLE, POLRIB, PPAN, PPI-H, PPIR8, PRMT1, PRPF4B, PRRI2, PSMD2, PTPN23, PUM1, RAB4A, RASA2, RBM14, RBM25, RBM42, RBSN, RCL1, RMND5A, RNF20, RNF40, RPL10, RPL10A, RPL13, RPL14, RPL15, RPL18, RPL19, RPL23A, RPL24, RPL26, RPL27, RPL34, RPL35, RPL36, RPL37A, RPL38, RPL6, RPL7A, RPL8, RPL9, RPLP1, RPS11, RPS13, RPS16, RPS17, RPS20, RPS23, RPS24, RPS25, RPS3, RPS3A, RPS4X, RPS5, RPS7, RPS8, RUVBL2, SART1, SETD1A, SLAMF6, SLBP, SMARCE1, SMC1A, SMC3, SNRNP27, SNRNP70, SNRPC, SNRPF, SPI, SRFBP1, SRSFI, STAT5B, SURF6, SYMPK, TBLIX, THOC3, TNPO3, TRAF2, TRAP, TSCI, TSG101, 30sup.+ T1CP51sup.35, TYMS, U2AF1, U2AF2, UBASL3A, UPF1, UTP14A, UTP15, WDR45, WDR5, YEATS4, ZMAT2, AAMP, AARS, AATF, AK2, ALDH1A1, AP2M1, ATIC, ATP1A1, ATP5O, ATP6V1B2, ATP6V1F, ATXN10, BMS1, BOPI, BUD23, C12orf60, CAD, CARS, CCDC86, CCT6A, CD3D, CD3E, CD3EAP, CD3G, CINP, CLNSIA, CPSF4, CRCP, CTPS1, DAD1, DDX27, DDX52, DGCR8, DHOD1, D-1X29, DUX37, DICER1, DNAJA3, DNMT2, DNTTsup.IP2, DPH6, DROSHA, EIF2B2, EIF2B3, EIF2B4, EF5A, ELP4, ESFI, EXOSC4, EXOSC5, EXOSC7, EXOSC9, FAM149B1, FARSF, FBL, FCFI, FH, FLNVCR-1, FTSJ3, GFER, GMPS, GNL2, GNPAT, GTF3C1, HARS, HAUS4, HCCS, HEATR3, USD17B10, HSD17B12, HSPA9, IL2RG, IMPD2, ISG20L2, KARS, LAGE3, LETMI, LONP1, MARS2, MDN1, MIETTL6, MMACHC, MRPI16, MRPL22, MRPL5, MIRPL36, MRPL37, MRPL41, MIRPL42, MRPL45, MRPL54, MRPS11, MRPS14, MRPS17, MRPS18A, MRPS2, MRPS23, MRPS33, MRPS5, MRPS9, MTHFD1L, MOR, MYBBP1A, NAT10, NCL, NEPRO, NIFK, NOC2L, NOLI, NOL6, NOL8, NOP14, NOP2, NOP56, NOP58, NUBP1, NLUFIP1, ORAOV1, PAM16, PCYT2, PDCD11, PDGFRA, PDHAI, PDSS2, PELP1, PGD, PGI3, PHB, PHB2, PISD, PVTRM1, PMPCA, PNOI, PNPTI, POLG2, POLR1A, POLRIC, POLRID, POLR2E, POLR3B, POLRMT, POP1, POP4, POP5, POT1, PPAN, PPAT, PSMG1, QARS, RBM19, RCCL1, RIOK1, RJOK2, ROMO1, RPF1, RPF2, RPL28, RPL30, RPL39, RPLP2, RPN2, RPP21, RPP30, RPS11, RPS12, RPS15, RPS17, RPS19BP1, RPS27, RPS4X, RPS6, RPSA, RRP12, RRP36, RRP7A, RRP9, RSL24-D1, SAMM50, SARS, SDHC, SEH1L, SLC35B1, SLC38A6, SLC7A11, SPOUT1, SRFBP1, SSB, SURF6, TAF1A, TAF1C, TAFID, TAF8, TMM41, TARS., TEX10, TIMM44, TNKS1BP1, TOMM20, TOMM40, TP53BP1, TRMT12, TRNT1, TSEN2, TSEN54, TSRI, TT1, TWIST1, TWNK, UMPS, UQCRL10, UQCRLB, UQCRLC1, UTP11, UTP14A, UTP3, UTP6, VPS29, VPS72, WAS, WDR12, Wsup.TDR3, WDR36, WDR55, XRCC5, XRCC6, YAEID1, ZCCHC9, ZNHIT3, ZNHIT6, and ZNRD1.

[0236] These regulators and agents that modulate these regulators can be used as T cell related immunotherapies for cancer or autoimmune diseases.

Example 3: CRISPRa Screening Primary fHunan T cells to Identify Genetic Regulators

INTRODUCTION

[0237] Regulated T cell cytokine production in response to stimulation plays a role in balanced immune responses. Cytokine dysregulation can lead to autoimmunity, immunodeficiency, and immune evasion in cancer (1-4). Interleukin 2 (IL-2), secreted predominantly by CD4sup.+ cells, drives T cell expansion (5) and is therapeutically applied in autoimmunity and cancer at different doses (6). Interferon gamma (IFN-γ) is a cytokine secreted by both CD4sup.+ and CD8sup.+ cells that promotes a type I immune response against intracellular pathogens including viruses (4) and correlates with positive cancer immunotherapy responses (7-9). Much of our current understanding of the pathways leading to cytokine production in humans originates from studies in transformed T cell lines, which often are not representative of primary human cell biology (10-12).

Comprehensive understanding of pathways that control cytokine production in primary human T cells would facilitate the development of next-generation immunotherapies.

[0238] Unbiased forward genetic approaches can uncover the components of regulatory networks systematically but challenges with efficient Cas9 delivery have limited their application in primary cells. Genome-wide CRISPR knockout screens have been completed using primary mouse immune cells from Cas9-expressing transgenic mice (13-15), including a screen for regulators of innate cytokine production in dendritic cells (13). Genome-scale CRISPR studies in human primary cells have recently been accomplished using transient Cas9 electroporation to introduce gene knockouts (16, 17). However, comprehensive discovery of regulators requires both gain-of-function and loss-of-function studies. For example, CRISPRa gain-of-function screens can discover genes that may not normally be active in the tested conditions, but which can promote phenotypes of interest (18, 19). In contrast to a CRISPR knockout, CRISPRa or CRISPRi require the sustained expression of an activator-linked endonuclease-dead Cas9 (dCas9) and due to poor lentiviral delivery has been limited to small scale experiments in primary cells (20, 21). Here we developed a CRISPRa and CRISPRi screening platform in primary human T cells, which allowed for the systematic discovery of genes and pathways that can be perturbed to tune stimulation-dependent cytokine responses.

Materials and Methods

Isolation and Culture of Human IT cells

[0239] Human T cells were sourced from PBMC-enriched leukapheresis products (Leukopaks, Stemcell Technologies cat 70500.2) from healthy donors, following IRB approved informed written consent (Stemcell Technologies). Bulk T cells were isolated from Leukopaks using EasySep magnetic selection following manufacturers' recommended protocol (Stemcell Technologies cat 17951). Unless stated otherwise, bulk T cells were frozen in Bambanker Cell Freezing Medium at 5×10^7 cells/ml (Bulldog Bio cat BB01) and stored at -80°C . for short-term or in liquid nitrogen for long-term storage immediately after isolation. Unless otherwise noted, thawed T cells were cultured in X-VIVO 15 (Lonza Bioscience cat 04-418Q) supplemented with 5% FCS, 55 mM 2-mercaptoethanol, 4 mM N-acetyl L-cysteine, and 500 IU/ml of recombinant human IL-2 (Amerisource Bergen cat 10101641). Primary T cells were activated using anti-humanCD3/CD28 CTS dynabeads (Fisher Scientific cat 40203D) at a 1:1 cell-to-bead ratio at 10^6 cells/ml.

Cell line maintenance

[0240] *Lenti-X* HEK293T cells (Takara Bio cat 632180) were maintained in DMEM high glucose with GlutaMAIXTM (Fisher Scientific cat 10566024), supplemented with 10% FCS, 100 U/ml of PenStrep (Fisher Scientific cat 15140122), 1 mM sodium pyruvate (Fisher Scientific cat 11360070), 1X MEM non-essential amino acids (Fisher Scientific cat 11140050), and 10 mM HEPES solution (Sigma cat H0887-1 00ML) Cells were passaged every 2 days using Tryple Express (Fisher Scientific cat 12604013) for dissociation and maintained at <60% confluency. NALM6 cells were engineered to express NY-ESO-1 peptide in an HLA-A0201 background, recognizable with the 1G4 TCR by the Fyquem lab at UCSF and provided for TCR stimulation coculture experiments. For simplicity, these cells are referred to as NALM6. NALM6 cells were cultured in RPML (Gibco cat 21870092) supplemented with 10% FCS, 100 U/ml PenStrep (Fisher Scientific cat 15140122), 1 mM sodium pyruvate (Fisher Scientific cat 11360070), and 1X MEM non-essential amino acids (Fisher Scientific cat 11140050), 10 mM HEPES solution (Sigma cat 140887-100ML), and 2 mM L-glutamine (Lonza Bioscience cat 17-605E).

Plasmids

[0241] dCas9-VP64 originated from lentiSAMv2 (addgene 75112) and cloned into the lentiCRfSPRv2-dCas9 backbone (addgene 112233) with Gibson Assembly. The promoter was switched to SFFV and mCherry was introduced upstream of dCas9-VP64, separated by a P2A sequence resulting in the pZRI 12 plasmid. The LTR-LTR range was minimized to enhance lentiviral titer. For CRISPRi, BFP in pHR-SFFV-dCas9-BFP-KRAB (addgene 46911) was switched to mCherry with Gibson Assembly resulting in pZR0 71.

[0242] Single sgRNAs for arrayed experiments have been introduced by Golden Gate Cloning as described before (22). Briefly, DNA oligomers with Golden Gate overhangs were annealed and subsequently cloned into the non-digested target plasmid using the NEB®Golden Gate Assembly Kit (BsmBI-v2, New England Biolabs cat E1602L). sgRNAs have been cloned into pXPR_502 (addgene 96923) for CRISPRa and into CROPseq-Guide-Puro (43) (addgene 86708) for CRISPRi. All single sgRNAs used in this study are found in Table 8.

TABLE-US-00074	Target-Guide-	CRISPR #	Target	Number	system	Original	Guide	Sequence
guideRS002	EGFR	1	CRISPRa	CCACCGCTGTCCACCGCCTC	guideRS003	EGFR	2	CRISPRa
GACCCAAGGCCAGCGGCCGC	guideRS004	EGFR	3	CRISPRa	GGAGGGAGGAGAACCAGCAG	guideRS015	GARP	1
AAATTGCAGCCGGAGCGCGG	guideRS017	GARP	2	CRISPRa	TCCGGATAAACCGAGGCACG	guideRS019	GARP	3
GCGAAGCATCTTCACCACCC	guideRS005	IL1R2	1	CRISPRa	GACCCAGCACTGCAGCCTGG	guideRS006	IL1R2	2
AAACTTATGCGGCGTTTCCT	guideRS007	IL1R2	3	CRISPRa	ATCACTTTAAAACCACCTCT	guideRS001	NT-CTRL	1
CRISPRi	CGCGAGCACAGCTAAGGCCA	guideRS023	B2M	2	CRISPRi	GAGTAGCGCGAGCACAGCTA	guideRS024	B2M
GGCCGAGATGTCTCGCTCCG	guideRS025	CD4	1	CRISPRi	AACAAAGCACCCCTCCCCACT	guideRS026	CD4	2
CAAACAGGCGTATCTGTGTG	guideRS027	CD4	3	CRISPRi	CTCTGCAACCAGGAGCCCAG	guideRS028	CD45	1
CACTGTTGTCTTATCAGACG	guideRS029	CD45	2	CRISPRi	CTCGTCTGATAAGACAACAG	guideRS030	CD45	3
GTTTGTTCTTAGGGTAACAG	guideRS021	NT-Ctr	1	CRISPRi	ACTCAGCCATTTTATTAGAA	pZR073	APOBEC3C	1
CRISPRa	GAGCAGCCTGTCTTTATCGG	pZR074	APOBEC3C	2	CRISPRa	GTTCTCCGGGCCCCTCCTAC	pZR077	FOXQ1
CGCCTGGTGCGCGCCCGTTG	pZR078	FOXQ1	2	CRISPRa	GAGGCCACACTGCAGCGCGG	pZR079	IFNG	1
CRISPRa	TGGGTCTGTCTCATCGTCAA	pZR080	IFNG	2	CRISPRa	GTGGCACAGGTGGGCATAAT	pZR081	IL1R1
CRISPRa	GGGTGGAGAGTTGGGACACC	pZR082	IL1R1	2	CRISPRa	GCTCGGCTGGGCCAGTCCGC	pZR083	IL2
CRISPRa	TCCATTCAGTCAGTCTTTGG	pZR084	IL2	2	CRISPRa	GAGAGCTATCACCTAAGTGT	pZR085	IL2RB
CRISPRa	TATCTGGCCCTGGGTGCTTG	pZR086	IL2RB	2	CRISPRa	GGTGCCGCCCCCAGCGTAGG	pZR087	LAT2
CRISPRa	GACAGGCTCAGCTATGAAGA	pZR088	LAT2	2	CRISPRa	GCGGCAGTGCGGCGGATGTA	pZR089	LHX6
CRISPRa	TCCCCCTCCAGCTGCAACGG	pZR090	LHX6	2	CRISPRa	GGAGGACTACCAAGAGGGGG	pZR091	MAP4K1
CRISPRa	ACAGTCGTGCAGTGCAGCTG	pZR092	MAP4K1	2	CRISPRa	GGGGCTCTGAGAGCCTCTGA	pZR093	NT-CTRL
CRISPRa	GAGTCAACGGGGAATACCAT	pZR094	NT-CTRL	2	CRISPRa	GAACCATTAGATCAATGCGA	pZR095	OTUD7B
CRISPRa	GGGGAGCGGCGCTAAAGGCG	pZR096	OTUD7B	2	CRISPRa	GAAAACACGGGGTCACGCGC	pZR097	PIK3AP1
CRISPRa	ACCTGCACCCGCGGCCGTTG	pZR098	PIK3AP1	2	CRISPRa	GCCGAGTCCCGCAGGCGGGG	pZR099	TNFRSF1A
CRISPRa	TTGGGAGTGGTCGGATTGGT	pZR100	TNFRSF1A	2	CRISPRa	GGCACAAGGCAGCCAGATCT	pZR101	TRIM21
CRISPRa	AAAGGGTGTGTGGAGAAATG	pZR102	TRIM21	2	CRISPRa	GAGCGCGCAACCAGGACCAC	pZR103	VAV1
CRISPRa	CCAGGCCTGTGTCTGAGTGGG	pZR104	VAV1	2	CRISPRa	GAGGAGGAGCCATGGGGCGG		

[0243] The genome wide CRISPRa (Calabrese A, cat 92379 and Calabrese B, cat 92380) and CRISPRi libraries (Dolcetto A, cat 92385 and Dolcetto B, cat 92386) (22) were obtained from addgene. Forty nanograms of each library were transformed into Endura™ ElectroCompetent Cells (Lucigen cat 60242-2) following the manufacturer's instructions. After transformation, Endura cells

were grown in a shaking incubator for 16 hours at 30° C. in the presence of ampicillin. Library plasmid has been isolated using the Qiagen Plasmid Plus MaxiKit (Qiagen 12963) and sequenced for sgRNA representation as described under “Genome-wide CRISPRa and CRISPRi screens”. [0244] For cDNA mediated target overexpression, the lentiCRISPRv2 (addgene 75112) backbone was rebuilt to a lentiviral cDNA cloning plasmid with an SFFV promoter followed by BsmBT restriction sites and P2A-Puro, Transgene cDNAs were purchased from Genscript, choosing the canonical (longest) isoform for each gene, and BsmBI restriction sites were introduced by PCR. The final lentiviral transfer plasmids were assembled using the NEB®Golden Gate Assembly Kit (BsmBI-v2, New England Biolabs cat E1602L).

[0245] To clone direct-capture compatible CRISPRa-SAM plasmids for Perturb-seg, different sgRNA designs were synthesized as G-Blocks (integrated DNA technologies) and cloned into pXPR_502 (addgene 96923) by Gibson assembly, replacing its sgRNA cassette.

Lentivirus production

[0246] Unless otherwise stated, HEK293T cells were seeded in Opti-MEM™ Reduced Serum Medium (OPTI-MEM) with GlutaMAvX™ Supplement (Gibco cat 31985088) supplemented with 5% FCS, 1 mM Sodium Pyruvate (Fisher Scientific) and IX MEM non-essential amino acids (Fisher Scientific) (cOPTI-MEM) at 3.6×10^7 cells per T225 flask in 45 ml of medium overnight to achieve confluency between 85% and 95% at the time point of transfection. The following morning, HEK293T cells were transfected with second generation lentiviral packaging plasmids and transfer plasmid using Lipofectamine 3000 transfection reagent (Fisher Scientific cat 1.3000075). Briefly, 165 µl of Lipofectamine 3000 reagent was added to 5 ml of room temperature OPTI-M-TEM without supplements. Forty-two micrograms of Cas9 transfer plasmid, 30 µg of psPAX2 (addgene 12260), 13 µg of pMD2 G (addgene 12259), and 145 µl of p3000 reagent were added to 5 ml of room temperature OPTI-MEM without supplements and mixed by gentle inversion. The plasmid and Lipofectamine 3000 mixes were combined, mixed by gentle inversion, and incubated for 15 min at room temperature. Following incubation, 20 ml of medium was removed from the T225 flask and the 10 ml transfection mixture was carefully added without detaching HEK293T cells. After 6 hours, the transfection medium was replaced with 45 ml of cOPTI-MEM supplemented with IX ViralBoost (Alstern Bio cat VB100). Lentiviral supernatant was harvested 24 hours after transfection (first harvest) and replaced with 45 ml of fresh cOPTI-MEM. A second harvest was performed 48 hours after transfection. Immediately after collection, the media was centrifuged at 500g, 5 min, and 4° C. to clear cellular debris. Unless otherwise noted, *Lenti-X-Concentrator* (Takara Bio 631232) was added to the collected supernatant and lentivirus was concentrated following the manufacturer's instructions and resuspended in OPTI-MEM in 1% of the original culture volume without supplements. Lentiviral particles were subsequently aliquoted and frozen at -80C.

Flow cytometry

[0247] Aria 2, Aria 3 and Aria Fusion cell sorters (BD Biosciences) at the UCSF Parnassus Flow Core and the Gladstone Institute Flow Core were used for sorting. The Attune NxT flow cytometer (Thermo Fisher) and LSRFortessa X-20 (BD Biosciences) was used for flow cytometry. Antibodies used for flow cytometric analyses and sorting are summarized in Table 9.

TABLE-US-00075

Antigen	Name	Target	Species	Fluorochrome	Clone	Vendor
EGFR	Human BV421	EGFR.1	BD	IL1R2	Human APC 34141	Thermo
IL1R2	Human FITC 34141	Thermo	GARP	Human APC 7B11	Biolegend	IFN-gamma
Human FITC 4S.B3	Biolegend	TNF-alpha	Human APC MAb11	Biolegend	MQ1- IL-2	Human Pacific Blue 17H12
Biolegend	IFN-gamma	Human Pacific Blue B27	Biolegend	CD45	Human PE H130	Biolegend
B2M	Human APC 2M2	Biolegend	CD45	Human AF488	Biolegend	CD4
Human PE RPA-T4	Biolegend	CD4	Human BV421	38261	Biolegend	GARP
Human PE 7B11	Biolegend	CD45	Human APC H130	Biolegend	IFN-gamma	Human BV421
B27	Biolegend	MQ1- IL-2	Human APC 17H12	Biolegend	CD4	Human FITC 38261
Biolegend	IFN-gamma	Human BV605	B27	Biolegend	TNF-alpha	Human BV421
Mab11						

Biologend TNF-alpha Human BV711 Mab11 Biologend EGFR Human PE AY13 Biologend CD4 Human PE-Cy7 38261 Biologend CD22 Human PerCp-Cy5.5 HIB22 Biologend CD45RA Human BV711 HI100 Biologend CD62L Human FITC DREG-56 Biologend CD4 Human BUV395 SK3 BD CD8a Human BUV496 SK1 BD

Intracellular Cytokine Staining

[0248] Unless indicated otherwise, T cells were stimulated with ImmunoCult™ Human CD3/CD28/CD2 T Cell Activator (Stemcell Technologies cat 10990) with 6.25 µl per milliliter of culture media at 2×10^6 cells/ml. One hour after restimulation, Golgi Plug protein transport inhibitor (BD Biosciences, cat 555029) was added at a 1/1000 dilution. Nine hours after addition of Golgi Plug, T cells were stained for surface antigens prior to fixation and subsequently processed for intracellular cytokine staining following BD Cytofix/Cytoperm™ kit (BD Biosciences cat 554714) instructions.

Genome-wide CRISPRa and CRISPRi screens

[0249] One day after activation, T cells from two human blood donors were infected with 2% v/v concentrated dCas9-VP64 lentivirus. Two days after activation, T cells were split into two populations and infected with 1% v/v (MOI-0.5) Calabrese Set A (addgene 92379) or 0.8% v/v (MOI ~0.5) Calabrese Set B (addgene 92380) lentivirus. These two sets were independently cultured and processed in parallel until analysis. Three days following activation, fresh media with IL-2 (final concentration 500 TU/ml) and puromycin (final concentration 2 µg/ml) was added to bring cells to 3×10^5 cells/ml. Cells were split two days later and fresh media with IL-2 was added to bring cells to 3×10^5 cells/ml. Two days later, fresh media without IL-2 was added to bring the concentration to 10^6 /ml. Eight days after initial activation, cells were harvested, centrifuged at 500g for 5 min and resuspended at 2×10^6 cells/ml X-VIVO 15 without supplements. The following day, cells were restimulated and stained for FACS as described under “intracellular cytokine staining”. Over subsequent 2 days, cells were sorted at the Parnassus Flow Cytometry Core Facility (PFCC) into [L-21 and IL-2' CD4+ T cell and IFN-1.1 and IFN-thi CD4 T cell populations. Sorted cells were stored in EasySep Buffer (PBS with 2% FCS and 1 mM EDTA) overnight until genomic DNA isolation.

[0250] The same experimental procedure using T cells from the same donors was followed for the CRISPRi screens. T cells were infected with dCas9-mCherry-KRAB at 2% v/v and Dolcetto A (addgene 92385) and B (addgene 92386) sgRNA libraries at 10% v/v or 25% v/v unconcentrated virus, respectively (-0.5 M1 O).

[0251] Genomic DNA was extracted from fixed cells as described previously (44). Integrated sgRNA sequences were amplified as previously described (22), and sequencing libraries were subsequently agarose gel purified using NucleoSpin Gel and PCR Clean-up Mini kit (Machery Nagel cat 740609.50). Libraries were sequenced on a NextSeq500 instrument to a targeted depth of 100-fold coverage.

[0252] For the supplementary CD4' T cell set of genome wide CRISPRa screens, CD4' T cells were isolated from Leukopaks using magnetic negative selection (Stemcell Technologies, cat 17952) and subsequently stimulated as described under “Isolation and culture of human “T cells”. T cells were then cultured and infected with lentivirus as described for the primary CRISPRa screens above. For library lentivirus production, Calabrese Set A and Set B plasmid were mixed at equimolar ratios before transfection and the pooled lentiviral particles from both sets was used for transduction. CD4 flow cytometry staining on day 7 after T cell activation confirmed >98% purity. T cells were further processed and restimulated as described above. T cells were separately stained for IL-2, IFN-γ, or TNF-α for FACS. After our initial analysis, it appeared the IFN-γ screen was potentially under-sampled due to lower hit resolution than the other screens. To address this, additional fixed cells from the same experiment were stained and sorted as an additional technical replicate and then computationally merged (described below).

CRISPR screen analysis

[0253] Reads were aligned to the appropriate reference library using MAGeCK version 0.5.9.2 (45) using-trim-5 22,23,24,25,26,28,29,30 argument to remove the staggered 5' adapter. Next, raw read counts across both library sets were normalized to the total read count in each sample and each of the matching samples across two sets were merged to generate a single normalized read count table. Normalized read counts in high versus low bins were compared using mageck test with-norm-method none, paired, and control-sgrna options, pairing samples by donor and using non-targeting sgRNAs as controls, respectively. Gene hits were classified as having a median absolute log₂-fold change value greater than 0.5 and an FDR <0.05. For supplemental CD4' screens, reads were aligned to the full Calabrese A and B library in a single reference file. For the supplemental CD4.sup.T IFN-γ screen, which was sorted and sequenced as two technical replicates, normalized counts were averaged across technical replicates before analyzing with mageck test.

Gene set-enrichment analysis (GSEA)

[0254] Gene set-enrichment analysis was completed with the fgsea Bioconductor R package using default settings (46). KEGG pathways v7.4 were obtained from GSEA mSigDB <http://www.gsea-msigdb.org/gsea/downloads.jsp>. The KEGG NF-κB signaling pathway (entry hsa04064) was missing from this dataset and added manually from <https://www.genome.jp/entry/pathway+hsa04064>.

s-LDSC analysis

[0255] GWAS summary statistics were downloaded from the Price lab website (<https://alkesgroup.broadinstitute.org/sumstats> formatted and <https://alkesgroup.broadinstitute.org/UKBB/>). LD scores were created for each screen (corresponding to a set of SNPs within 100 kb of genes identified as significant hits in each screen or their corresponding matched background sets) using the 1000G Phase 3 population reference. Each annotation's heritability enrichment for a given trait was computed by adding the annotation to the baselineLD model and regressing against trait chi-squared statistics using HapMap3 SNPs with the stratified LD score regression package (47). Heritability enrichments were then meta-analyzed across immune or non-immune traits using inverse variance weighting. The sets of background genes were sampled from the set of all genes that were expressed in the control sgRNA, stimulated bulk RNA-Seq data. For each screen, the background genes were sampled to match the significant screen hits in number and based on deciles of gene expression. Immune traits used for analysis were: "Eosinophil Count", "Lymphocyte Count", "Monocyte Count", "White Count", "Autoimmune Disease All", "Allergy Eczema Diagnosed", "Asthma Diagnosed", "Celiac", "Crohn's Disease", "Inflammatory Bowel Disease", "Lupus", "Multiple Sclerosis", "Primary Biliary Cirrhosis", "Rheumatoid Arthritis", "Type I Diabetes", "Ulcerative Colitis". Non-Immune traits used were: "Heel Tscore", "Balding1", "Balding4", "Bmi", "Height", "Type 2 Diabetes", "Neuroticism", "Anorexia", "Autism", "Bipolar Disorder", "Depressive Symptoms", "Fasting Glucose", "ldl", "Ldl", "Triglycerides", "Fasting Glucose"

Arrayed CRISPRa experiments For each gene chosen to target in follow up experiments, one sgRNA was chosen from the Calabrese library used in screens. The first sgRNAs ("1") were manually chosen for consistent log₂ fold-change observed in both donors. The second sgRNA ("2") was picked from the hCRISPRa-v2 genome-wide library (48), choosing the top ranked sgRNA not present in Calabrese libraries for each gene. sgRNAs were cloned into the pXPR 502 vector as described in the plasmid section.

[0256] Primary human T cells were transduced with 2% v/v inCherry-2A-dCas9-VP64 lentivirus (pZR112) 1-day post-activation. The following day (day 2), the dCas9-VP64 transduced cells were split into 96-well flat-bottom plates, avoiding edge wells, and transduced with a different sgRNA lentivirus in each well (5% v/v). One day after sgRNA transduction, fresh medium was added with IL-2 (500 IU/ml) and 2 μg/ml puromycin (final culture concentrations). Cells were passaged 2 days later, adding fresh medium with 500 IU/ml of IL-2 and maintaining a concentration of 3×10⁵.

to 1 × 10⁶ cells/ml with 96-well plates copied as needed to maintain this concentration. On day 8, cells from copied plates were pooled and samples were counted. Cells were pelleted and resuspended at a concentration of 2 × 10⁶ cells/ml in fresh X-VIVO-15 without additives. On day 9, cells were restimulated with anti-CD3/CD28/CD2 ImmunoCult T Cell Activator (as described in “Intracellular cytokine staining”) or left resting.

RT-gcR

[0257] T cells were prepared as described under Arrayed CRISPRa experiments. Seven days post sgRNA transduction 100,000 T cells per well were pelleted at 500g, 5 mins, and 4° C. Cells were lysed and RNA was extracted using Quick-RNA 96 kit (Zymo Research), following manufacturer's protocol, skipping the option of in-well DNase treatment. DNase treatment and cDNA synthesis were subsequently completed with Maxima First Strand cDNA Synthesis Kit for RT-qPCR, with dsDNase (Thermofisher Scientific). qPCR was performed with PrimeTime PCR Master Mix (Integrated DNA technologies) and PrimeTime qPCR probe assays (Integrated DNA Technologies, list of probes used in Table 10) on an Applied Biosystems Quantstudio 5 real-time PCR system. Data was analyzed using the deltaDeltaCt method. The mean Ct values of two housekeeping genes, PPIA and (GUSB, to calculate the deltaCt, and the mean deltaCt of non-targeting controls to calculate deltaDeltaCt.

TABLE-US-00076	TABLE	10	Primer	1	Primer	2	GGA	AGT	AGA	ATG	TGC		
CTG	GAT	GTA	AGC	AGG	AAG	AGA	AGC	CAG	AG	ACC	ACA	GTT	
AGA	GAA	CCA	CTCT	TGC	TAT	TGA	CCG	ATG	CTT	CGA	CAG	TTC	
AGC	CAT	CAC	TTG	CA	ACA	AAA	AGA	AAC	GAG	ATG	AC	CAC	TGT
TTT	TCC	AAG	ACC	TCA	TTC	CTG	CTA	TGA	TTT	TCT	CCC	ACTC	
CAG	AGG	TTT	GAG	TTC	TTC	TAAA	CTC	ACC	AGG	ATG	CTC	AC	
GCG	AAG	AGA	GCC	ACT	TCT	G	GTG	TAC	TTG	CTG	ATC	AAC	TGC
CTG	GTG	TTG	CCT	CTT	GTG	ATAGT	GTC	AGT	GGT	GTT	GGC	CAG	
CTT	GGA	CAC	TGG	ATC	TC	CCT	GCA	CGG	CTA	CAT	TGA	G	AAG
CAG	ACG	GAA	AGT	GAG	G	GTG	GCT	ATG	GTT	GGA	GGT	C	GAT
CCT	CAA	GTA	CTT	TCA	GCC	A	CAC	TGC	CGA	GGA	ATG	AAG	AG
TGA	TCT	CCA	AGT	CTG	TCT	GC	GAG	TCC	TTT	CGT	TTC	CAG	CA
ACA	ACT	TCG	TGC	ACT	CCA	CAG	CCT	CTG	CCT	CAA	TGG	TCT	
GCG	TGA	ATC	CTA	GAT	TTC	TG	GCT	GAG	AAG	TTG	GAA	GTG	
GAA	GCC	GAA	CTT	CTC	ACA	GCA	CTT	AAC	AAC	CTG	CTA	CCC	CAT
GTT	TTT	GAT	CCA	GAC	CCA	GAT	G	GCC	CAT	TAT	TCA	GAG	CGA
GTACAA	GAC	TGA	GAT	GCA	CAA	GTG	GTG	GCG	GAT	TTG	ATC		
ATT	TGG	Probe	/56-FAM/CCA	CAG	ATC/ZEN/AGA	AAC	CCG	ATG	AAG				
GC/3IABKFQ/	/56-FAM/TAA	AGC	TGT/ZEN/AGC	CCG	TTG	CCT	GC/3IABKFQ/						
/56-FAM/TCG	GTA	ACT/ZEN/GAC	TTG	AAT	GTC	CAA	CGC/3IABKFQ/	/56-					
FAM/TCT	ACC	TCT/ZEN/GAC	TGT	GAT	ATT	TTT	GTG	TTT	AAA	GTC			
T/3IABKFQ/	/56-FAM/TTA	CAT	GCC/ZEN/CAA	GAA	GGC	CAC	AGA/3IABKFQ/						
/56-FAM/TGT	AAC	ACC/ZEN/CCA	GAC	CCC	TCG	AA/3IABKFQ/	/56-						
FAM/AGG	TGT	GCG/ZEN/GGC	TCA	GGA	T/3IABKFQ/	/56-FAM/CAC	TTC						
CGC/ZEN/ATC	TGC	CCG	TG/3IABKFQ/	/56-FAM/TGT	AGC	AGC/ZEN/TGA							
TCC	GAG	CCT	AGA/3IABKFQ/	/56-FAM/CGA	GGG	TGG/ZEN/AGG	CCT						
GAA	TTT	TGA/3IABKFQ/	/56-FAM/TGC	TTC	TCT/ZEN/CTC	TGT	CTT	CGG					
GTG	A/3IABKFQ/	/56-FAM/ACG	GTG	TTC/ZEN/TGT	TTC	TCC	TGG						
CA/3IABKFQ/	/56-FAM/AGA	GAG	CAG/ZEN/ACT	GGA	AGA	AAA	CAG						
TGG/3IABKFQ/	/56-FAM/CAG	ATG	TCC/ZEN/CAG	TTC	CTG	TGC							
CTT/3IABKFQ/	/5Cy5/TGC	AGG	GTT	TCA	CCA	GGA	TCC	AC/3IAbRQSp/					
/5Cy5/AAT	TCA	CGC	AGA	AGG	AAC	CAG	ACA	GT/3IAbRQSp/					
cDNA experiments													

cDNA experiments

[0258] One day after activation, T cells were transduced with the 1G4 TCR lentivirus recognizing the NY-ESO-1 antigen or non-transduced for immunocult assay. One day later, cells were transduced with the transgenes in cDNA format. Three days after initial activation, puromycin was added to obtain a final concentration of 2 μ g/ml along with fresh X-VIVO 15 media with 500 IU/ml of HL-2 and further cultured and expanded analogous to the genome wide CRISPR screens. Nine days after initial activation, T cells were centrifuged and resuspended at 2×10^6 cells/ml in X-Vivo 15 without supplements. On the same day, 1G4 TCR expression was assessed by flow cytometry following dextran staining (Immudex cat WB3247-PE) to ensure even expression across different cDNA constructs. The following day, T cells were restimulated with either 6.25 μ l per milliliter of Immunocult or NALM6 cells at an effector-target ratio of 1:2 for 1G4 TCR-transduced cells. Cells were further processed as described under “intracellular cytokine staining”. CD22 was used as a marker for NALM6 cells to discriminate them from T cells in the coculture. Overexpression of OTUD7B cDNA together with the IG4 TCR (but not alone) caused toxicity and was therefore excluded from analyses. Two donors were excluded from the 1G4 TCR assay due to poor TCR transduction.

Cytokine Luminex assay

[0259] T cells were prepared as explained under “Arrayed CRISPRa experiments.” On day 9 after activation, T cells at a concentration of 2×10^6 cells/ml were restimulated with ImmunoCult™ Human CD3/CD28/CD2 (Stemcell Technologies cat 10970) at 6.25 μ l per milliliter. Twenty-four hours after restimulation, supernatant was collected and frozen at -20° C. Following a serial pilot titration, cytokine analyses were performed at a 1/200 dilution by Eve Technologies with the Luminex xMAP technology on the Luminex 200 system (Luminex). To remove very-low-expressed cytokines for downstream analysis, any group where three of four donors had undetectable cytokines, the cytokine was removed. Additionally, the sgfLIRI-1-Donor 4 measurement for IL-1 α was removed manually, as this was an extremely high outlier.

Bulk RNA-seq sample preparation

[0260] FOXQ1 and non-targeting sgRNA control primary human T cells from four donors were transduced and expanded as described in “Arrayed CRISPRa experiments” section. On day 8, mCherry+CD4+populations were sorted and resuspended in X-VIVO-15 without additives at 2×10^6 cells/ml. On day 9, cells were restimulated with 6.25 μ l per milliliter of anti CD3/CD28/CD2 ImmunoCult™ or left unperturbed for resting (non-stimulated) condition. Twenty-four hours later, cells were lysed for RNA.

[0261] RNA was purified using Quick-RNA Microprep kit (Zymo Research) without the optional in-well DNase treatment step. Purified RNA was treated with TURBO DNase (ThermoFisher Scientific) to remove potential contaminating DNA. RNA was subsequently purified using RNA Clean & Concentrator-5 kit (Zymo Research). RNA quality control was performed using an RNA ScreenTape assay (Agilent), with all samples having an RNA integrity number >7 . RNA-seq libraries were prepared using the Illumina Stranded mRNA Prep kit, with 100 ng of input RNA. Libraries were sequenced using paired-end 72-bp reads on a NextSeq500 instrument to an average depth of 3.2×10^7 clusters per sample.

Bulk RNA-seq data analysis

[0262] Adapters were trimmed from fastq files using cutadapt version 2.10 (49) with default settings keeping a minimum read length of 20 bp. Reads were mapped to the human genome GRCh38 keeping only uniquely mapping reads using STAR version 2.7.5b (50) with the following settings “-outFilterMultimapNmax 1”. Reads overlapping genes were then counted using featureCounts version 2.0.1 (51) with the following settings “-s 2” and using the Gencode version 35 basic transcriptome annotation. The count matrix was imported into R. Only genes with at least 1 count per million (CPM) across at least four samples were kept. TMM normalized counts were used for heatmaps. Differentially expressed genes between FOXQ1 overexpression and control samples were then identified using limma version 3.44.3 (52) while controlling for any differences

between donors. Significant differentially expressed genes were defined as having an FDR-adjusted P-value <0.05.

Perturb-seq Library Design and Cloning

[0263] The CRISPRa Perturb-seq target genes were selected from the primary IL-2 and IFN- γ CRISPRa screen results. First, genes that had a significant fitness defect removed from the gene list. Next, genes were ranked by median sgRNA log.sub.2-fold change and the top ranked, not previously selected gene, was picked in the following order: (1) IL-2-positive hit, (2) IFN- γ positive hit, (3) IL-2-positive hit, (4) IFN- γ -positive hit, and (5) IL-2- or IFN- γ -positive hit (alternating each round), such that positive hits outnumbered negative hits at a 4:1 ratio. Only hits that were significant (FDR<0.05) were selected in each round. The one exception was TCF7, which was added manually as we considered it worthwhile to analyze due to its known effects on T cell function. To select sgRNAs, the top two enriched sgRNAs by log.sub.2 fold-change in the screen for which the gene was selected were used. The library was ordered as pooled single stranded oligos, PCR amplified, and cloned into the CRISPRa-SAM direct-capture design I cloning vector (pZR158).

Perturb-seq Sample Preparation and Sequencing

[0264] Bulk CD3.sup.+ primary human T cells from two donors were transduced and cultured as described in the “.sup.4Genome-wide CRISPRa and CRISPRi screens” section, except library transduction was completed at lower MOI, of 0.3. Cells in the stimulated condition were stimulated with 6.25 μ l per milliliter of anti-CD3/CD28/CD2 immunocult. Twenty-four hours later, cells from both the stimulated and non-stimulated condition were sorted for mCherryTM (marking dCas9-VP64). Sorted cells were processed to single-cell RNA-seq and sgRNA sequencing libraries by the Institute for Human Genetics (IHG) Genomics Core using Chromium Next GEM Single Cell 3' Reagent Kit v3.1 with Feature Barcoding technology for CRISPR screening, following manufacturer's protocol. Before loading the Chromium chip., sorted cells from two blood donors were normalized to 1000 cells/ μ l and mixed at a 1:1 ratio, for each condition. Twenty microliters of cell suspension was loaded into four replicate wells per condition, for a total 80,000 cells loaded per condition. Final sgRNA sequencing libraries were further purified for the correct size fragment by 4% agarose F-Gel EX Gels (ThermoFisher Scientific) and gel extracted. Libraries were sequenced over two NovaSeq S4 lanes (2 stimulated wells, two non-stimulated wells per lane), at a 2:1 molar ratio of the gene expression libraries to sgRNA libraries.

Perturb-seq Analysis Alignments and count aggregation of gene expression and sgRNA reads were completed with Cell Ranger version 6.1.1. Gene expression and sgRNA reads were aligned using cellranger count, with default settings. Gene expression reads were aligned to the “refdata-gex-GRCh38-2020-A” human transcriptome reference downloaded from 10x Genomics. sgRNA reads were aligned to the Perturb-seq library using the pattern (BC)GTTTAAGAGCTATG. Counts were aggregated with cellranger aggr with default arguments. To assign sgRNAs to cells, cellranger count output files “protospacer_calls per_cell.csv” were used, filtering out droplets with >1 sgRNA called, returning a median of 133 sgRNA UMIs in sgRNA singlets. For increased stringency, only droplets with >5 sgRNA UMIs were used in further analysis. Cell donors were genetically demultiplexed using Souporecell (53) (<https://github.com/wheaton5/souporcell>). The input for each run was the bam file and barcodes.tsv file from the cellranger count output, and the reference fasta. Donor calls across wells were harmonized using the vcf file outputs from Souporecell using a publically available python script

([https://github.com/cornihyunminkang/apigenome/blob/master/scripts/vcf-match sample-ids](https://github.com/cornihyunminkang/apigenome/blob/master/scripts/vcf-match%20sample-ids)).

[0265] Gene expression data were imported and analyzed in R with Seurat version 4.0.3 ReadIOX function (54). Cells were initially quality filtered for percent mitochondrial reads <25%, number of detected RNA features >400 and <6000, removing 4% of cells.

[0266] After filtering, we recovered a median of 401 cells per sgRNA target gene per condition (median of 127 sgRNA unique molecular indices (UMIs) per singlet), and ~2000 cells with no-

target control guides per condition. Four sgRNA targets (HELZ2, TCF7, PRDM1, and IRX4) were removed from downstream analysis due to low cell counts (<100).

[0267] Gene-expression counts were normalized and transform-ed using the Seurat SCTransform function (55), with the following variables regressed: percent mitochondrial reads, S-phase score, and G2M-phase score, performing the regression as described on the Satija Lab website (https://satijalab.org/seurat/articles/cell_cycle_vignette.html). Normalized and transformed counts were used for all downstream analysis. To call CD4' and CD8' T cells, a CD4/CD8 score for each cell using following formula was used: $\log_{\text{sub.2}}(\text{CD4}/\text{mean}(\text{CD8A}, \text{CD8B}))$, with a score ≤ -0.9 called as a CD3' cell, and >1.4 called a CD4 cell.

[0268] For both restimulated and resting conditions, UMAP reduction was performed with dimensions 1-20, and otherwise default settings of the RunUMAP Seurat function. For clustering, FindClusters was run using algorithm 3, and resolution 0.4 for restimulated, and 0.5 for resting condition. Two clusters in the restimulated condition were manually merged to form "Cluster 2: Negative Regulators". The merged clusters showed highly similar gene expression patterns, with one cluster containing the bulk of cells containing negative regulator sgRNAs, and the other cluster containing sgRNAs targeting the negative regulator, MUC1. Cluster trees shown were generated using the Seurat BuildClusterTree function with default arguments. For pseudobulk differential expression analyses the Seurat FindM/larkers function was used with the default method, Wilcoxon Rank Sum test.

[0269] To generate the T cell activation score, pseudobulk differential expression analysis was first performed on restimulated versus resting no-target control sgRNAs and log 2-fold change outputs were used as gene weights. Only genes with an absolute log-fold change >0.25 and which were detected in 10% of restimulated or resting cells were used for gene weights. For a given cell, the activation score is calculated as $\sum(\text{GE} \times \text{GW}/\text{GM})$, where GE is a gene's normalized/transformed expression count, GW is the gene's weight, and GM is the gene's mean expression in no-target control cells (to correct for differential levels of baseline expression).

Statistical analysis

[0270] All statistical analyses were performed in R version 4.0.2, unless otherwise noted. In order to deal with ties in non-parametric tests, Mann-Whitney U tests were performed using the wilcox_test function of the Coin R package (version 1.4-1), with default arguments. For q-value based multiple comparison correction, the R qvalue package (version 2.20.0) was used with default arguments.

Results

Genome-wide CRISPRa screens identify regulators of IL-2 and IFN- γ production in T cells

[0271] To enable scalable CRISPRa in primary human T cells, we developed an optimized high-titer lentiviral production protocol with a minimal dCas9-VP64 vector (pZRI12), allowing for transduction efficiencies up to 80%. A second-generation CRISPRa synergistic activation mediator (SAM) system (22, 23) induced robust increases in target expression of established surface markers. Next, we scaled up our platform to perform pooled genome wide CRISPRa screens targeting $>18,800$ protein coding genes with $>112,000$ sgRNAs (22). We used fluorescence-activated cell sorting (FACS) to separate IL-2-producing CD4-T cells and IFN- γ -producing CD8-T cells into high and low bins (FIG. 1A). Subsequent sgRNA quantification confirmed sgRNAs targeting IL-2 (IL2) and IFN- γ (IFNG) were strongly enriched in the respective cytokine high populations and non-targeting control sgRNAs were not enriched in either bin (FIG. 1B). Both CRISPRa screens were highly reproducible in two different human blood donors (FIGS. 1, C and D). Gene level statistical analysis of the IL-2 and IFN- γ CRISPRa screens revealed 444 and 471 hits, respectively, including 171 shared hits (FIG. 1E). Thus, CRISPRa screens provide a robust platform to discover gain-of-function regulators of stimulation-dependent responses in primary cells.

[0272] CRISPRa hits included components of the TCR signaling pathway and T cell transcription

factors. Activation of TBX21 (encoding T-bet), which promotes both memory CD8^{sup} T cell and CD4⁺ T helper I (Th1) cell differentiation (24-26), selectively enhanced the signature type I cytokine IFN- γ (FIG. 1E). By contrast, sgRNAs activating GATA3, which promotes type II differentiation by antagonizing T-bet (25, 27), had opposite effects (FIG. 1E). Overexpression of members of the proximal TCR signaling complex, such as VAV1, CD28, LCP2 (encoding SLP 76), and LAT (28, 29) reinforced T cell activation and were enriched in both cytokine-high bins. Conversely, negative TCR signaling regulators MAP4K1 and SLA2 were depleted in these bins (FIGS. 1, B and E) (30, 31). Thus, CRISPRa identifies critical “bottlenecks” in signals leading to cytokine production.

Complementary CRISPRa and CRISPRi screens comprehensively reveal circuits of cytokine production in T cells

[0273] CRISPRa screens were effective in identifying limiting factors in cytokine production, but they could miss necessary components that would only be identified through loss-of-function studies. We therefore performed reciprocal genome wide CRISPRi screens, adapting our optimized lentiviral protocols (FIGS. 2, A and B) Drop out of gold-standard essential genes (32) and reproducibility across two human donors confirmed the screen quality. The CRISPRi fL-2 and IFN- γ screens identified 226 and 203 gene hits, respectively, including 92 shared hits (FIGS. 2, A and B). As expected, the CRISPRi hits were biased towards genes with high mRNA expression including members of the CD8^{sup} complex, whereas CRISPRa additionally identified regulators that are expressed either at low levels or not at all in T cells under the screened conditions. (FIGS. 2, C and D). For example, PIK3AP1 and IL1RI were expressed at low levels under the screened conditions (FIG. S7A). They are potentially inducible in some T cell contexts however were detected as hits by CRISPRa but not CRISPRi.

[0274] The power of coupling activation and interference screening was exemplified further by the identification of two IFN- γ -regulating circuits. CRISPRi screens identified components of the NF- κ B pathway that are required for IFN- γ production (and to a lesser extent IL-2 production). CRISPRi detected a circuit of T cell stimulation signaling through MALTI, BCL10, TRAF6, and TAK1 (encoded by MAP3K^{sup}.7) to the inhibitor of NF- κ B complex (I κ B complex, encoded by CHUK, I κ BKIB, and I κ BKG) that promotes IFN- γ production (FIGS. 2, E and F). By contrast, CRISPRa revealed a set of positive IFN- γ regulators that included members of the tumor necrosis factor receptor superfamily (TNFRSF) and IL1R1. These regulators also signal through NF- κ B, even though they are not individually required and therefore not detected by CRISPRi (FIGS. 2, E and F). Thus, CRISPRa and CRISPRi complement each other for the comprehensive discovery of functional cytokine regulators. To gain insights into functional pathways enriched across CRISPRi and CRISPRa screens, we completed gene set enrichment analysis (GSEA) of KEGG pathways, identifying multiple immune-related pathways as enriched across screens. Furthermore, we analyzed data from numerous genome-wide association studies (GWAS) to ask if the heritability of complex immune traits was enriched in genomic regions harboring our screen hits by stratified linkage disequilibrium score (s-LDSC) regression. Both CRISPRi and CRISPRa regulators of IFN- γ and CRISPRa regulators of IL-2 were in regions enriched for immune trait heritability compared to non-immune traits or an expression matched background set. Thus, these forward genetic screens may serve as a resource to help prioritize candidate functional genes in genomic regions associated with complex immune diseases.

[0275] We next completed integrative analyses of gene hits across CRISPRa and CRISPRi screens for both cytokines. We found that a handful of genes were identified across all screens (e.g., ZAP70 as a positive regulator and CBLB as a negative regulator), representing core regulators of stimulation-responsive cytokine production in T cells. The majority of hits however were either cytokine-(IL-2 in CD4⁺ T cells or IFN- γ in CD8⁺ T 30 cells) or perturbation-(activation or interference) specific. For a few target genes including PTPRC (CD45), CRISPRa and CRISPRi both influenced cytokine production in the same direction, suggesting that for some genes

activation and interference both impair optimal levels. The striking overlap in regulators between IL-2 in CD4^{sup.}+ T cells and IFN- γ in CD8-T cells led us to perform additional genome-wide CRISPRa screens for IL-2, IFN- γ , and TNF- α in CD4⁺ T cells, allowing for direct comparisons of type 1 cytokine regulators in CD4^{sup.} T T cells. Many of the strongest positive (e.g., VAV1, CD28, and LCP2) and negative hits (e.g., MAP4K1, LAT2, and GRAP) overlapped across all CRISPRa screens, likely representing core regulators of type I cytokine production in response to stimulation/costimulation. Additionally, these screens identified hits that could potentially increase or decrease individual cytokines selectively. Thus, CRISPRi and CRISPRa hits reveal both core and context-specific regulators of cytokine production.

[0276] We used our integrated dataset combined with literature review to build a high-resolution map of tunable regulators of signal transduction pathways leading to cytokine production (FIG. 2G). This included calcium pathway signaling genes (e.g., PLCG1, PLCG2, PRKCB, PRKD2, and NFATC2), and cytokine signaling genes (e.g., STAT3, JAK1, JAK3, and SOCS3), the latter suggesting feedback circuits among cytokine signals. In particular, CRISPRa identified regulators absent from previous literature (e.g., APOBEC3A/D/C, FOXQ1, and EMIPI) (FIG. 2H), underscoring the need for gain-of-function screens for comprehensive discovery. Thus, CRISPRa and CRISPRi screens complement one another to map the tunable genetic circuits controlling T cell stimulation-responsive cytokine production.

Arrayed characterization of selected CRISPRa screen hits

[0277] We next performed arrayed CRISPRa experiments for deeper phenotypic characterization of screen hits (FIG. 3A). We selected 14 screen hits (from different screen categories) (FIG. 3B) including the established regulators VAV1, MAP4K1, and positive controls IL2 and IFNG. Notably, we included genes with relatively low expression in T cells under our experimental conditions, FOXQ1, IL1 R1, LHX6, and PIK3API. First, we validated that selected sgRNAs increased the expression of target gene mRNA. Next, we assessed IL-2, IFN γ , and TNF- α by intracellular staining in both CD4⁻ and CD8⁺ T cells. Thirteen of 14 target genes caused significant changes in the proportion of cells positive for the relevant cytokine(s), with at least one sgRNA (FIG. 3, C and D). Furthermore, we observed effects on both IL-2 and IFN- γ double- and single-positive populations. With the exception of TNFRSF1 A (and IL2 or IFNG), positive regulators did not cause spontaneous cytokine production without stimulation (FIG. 3D).

[0278] Although IL-2 was screened in CD4⁺ T cells and IFN- γ in CD8^{sup.}+ T cells, CRISPRa sgRNA effects were highly correlated across both lineages (FIG. 3F). We also assessed T cell differentiation and observed that FOXQ1 and TNFRSF1A significantly decreased the percentage of CD62L⁺-cells, indicating a shift towards effector T cell states as a potential mechanism. Thus, these studies validate the pooled CRISPRa screens and begin to characterize cytokine production and cell differentiation states promoted by activation of key target genes.

[0279] We next tested if genes identified by CRISPRa could also regulate cytokines when overexpressed as cDNA transgenes, because continuous expression of CRISPRa would present challenges in cell therapies due to Cas9 immunogenicity (33). cDNA transgene overexpression of CRISPRa hits affected cytokine production in T cells stimulated with antibodies or antigen-positive cancer cells. Thus, this strategy could potentially be used to implement CRISPRa discoveries in engineered T cell therapies.

[0280] We next assessed how individual CRISPRa perturbations reprogram cytokine production by measuring a broad panel of 48 secreted cytokines and chemokines, 32 of which were detected in control samples. After confirming that the effects on IL-2, IFN- γ , and TNF- α measurements were consistent with intracellular staining (FIG. 3F), we performed principal component analysis (PCA) and hierarchical clustering on all cytokines. We observed sgRNA categorical grouping consistent with that observed in the screens, with sgRNAs targeting genes identified as regulators of both cytokines causing broad increases or decreases in cytokine concentration (FIG. 3G). Notably, there were distinct patterns in the classes of cytokines increased by different regulators (FIG. 3H). VAV1

and FOXQ1 (a transcription factor that has not been well characterized in T cells) led to preferential increases in type 1 signature cytokines and dampened type 2 cytokines. Surprisingly, OTUDB, a positive regulator of proximal TCR signaling (34), had a distinct effect and increased type 2 cytokines. We next asked if modulations in the secretome correlated with transcriptional control of the corresponding genes. Taking FOXQ1 as an example, we performed bulk RNA-seq on FOXQ1 and control sgR.sup.NA CD4-T cells and found it correlated highly with the secretome effects. Thus, the identified regulators may not only modulate TCR stimulation and signaling, but also tune the T cell secretome towards specific signatures.

CRISPRa Perturb-seq characterizes the molecular phenotypes of cytokine regulators

[0281] To assess the global molecular signatures resulting from each CRISPRa gene induction we developed a platform to couple pooled CRISPRa perturbations with barcoded single-cell RNA sequencing (scRNA-seq) read-outs (CRISPRa Perturb-seq) (FIG. 4A). As similar CRISPRa Perturb-seq approaches have been powerful in cell lines and animal models (35-37), we incorporated a direct-capture sequence into the CRISPRa-SAM modified sgRNA scaffold to enable compatibility with droplet-based scRNA-seq methods.

[0282] We performed CRISPRa Perturb-seq characterization of regulators of stimulation responses in 56,000 primary human T cells targeting 70 hits and controls from our genome wide CRISPRa cytokine screens (FIGS. 4, A and B). First, we confirmed that sgRNAs led to significant increases in the expression of their target genes. Next, uniform manifold approximation and projection (UMAP) dimensionality reduction revealed discrete separation of the resting and restimulated cells and showed relatively even distribution of cells from two donors (FIG. 4C). Gene signatures allowed us to resolve most T cells as either CD4 or CD8' (FIG. 4D). Thus, we generated a high-quality CRISPRa Perturb-seq dataset.

[0283] Cytokine production can be tuned by reinforced TCR signaling. To identify CRISPRa gene perturbations that tune the general strength of stimulation-responsive genes, we calculated a scRNA-seq "activation" score based on a gene signature we derived from comparing resting and restimulated cells within the non-targeting control sgRNA group. Projecting activation scores on the stimulated cell UMAP revealed discrete regions of higher and lower activation scores among the restimulated cells (FIG. 4E). We next examined activation scores across CRISPRa perturbations (FIG. 4F). Strikingly, negative regulators except IKZF3 (encoding the transcription factor Aiolos) decreased activation scores, suggesting they act to broadly dampen stimulation strength. By contrast, IKZF3 reduced IFNG expression without reducing the overall activation score (FIG. 4F), indicative of a possible distinct mechanism of cytokine gene regulation. Many of the positive regulators significantly increased activation score, with VAV1 causing the strongest activation potentiation (FIG. 4F). Thus, many, but not all, hits act by tuning overall T cell activation to varying degrees.

[0284] We next asked how different perturbations affected the expression of cytokine and other effector genes in stimulated cells. We analyzed pseudobulk differential gene expression under restimulated conditions for each sgRNA target cell group, compared with no-target control cells. IFNG was differentially expressed in 29 different sgRNA targets, with only sgRNAs targeting negative regulators causing decreased expression. IL2, however, was barely detectable by scRNA-seq. Only IL2 and VAV1 sgRNAs caused its increased expression, consistent with our observations that VAV1 activation caused the greatest level of IL 2 release (FIG. 3H). Many of the negative regulators drove a stereotyped pattern of differential cytokine gene expression, whereas positive regulators generally promoted more diverse cytokine expression patterns than negative regulators. Notably, TBX21 (T-bet) modulated the expression of most detectable cytokine genes. Furthermore, unlike most perturbations, it altered cytokine expression independently of stimulation.

[0285] We next used clustering analysis to characterize CRISPRa-driven cell states in restimulated and resting T cells (FIG. 4G). For each cluster, we identified the top upregulated gene expression markers and cytokine genes, contributions of CD4'/CD8 T cells, and overrepresented sgRNAs

revealing a diverse landscape of T cell states promoted by CRISPRa (FIG. 4, H to J). Negative cytokine regulators (e.g., MAP4KI) were highly enriched in cluster 2, marked by LTB expression and low activation score. Notably, only GATA3 promoted a Th2 phenotype (cluster 3) suggesting that altered T helper differentiation was not a common mechanism among negative LFNG regulators. Thus, Perturb seq reveals cell states promoted by the overexpression of different key regulators.

[0286] We identified two 112-expressing clusters, despite poor capture of the transcript, with both of the clusters consisting primarily of CD4⁺ T cells. Cluster 13 had higher 112 expression of the two and was promoted by VAX1 and OTUD7B sgRNAs. VAV1 sgRNAs were strongly enriched in both IFNG⁻ and 1L.sup.2-expressing clusters, suggesting that VAV1-mediated potentiation of T cell stimulation may drive differentiation towards multiple distinct cytokine-producing populations.

[0287] We also identified two distinct clusters of cells expressing JFNG (clusters 1 and 12) containing both CD4⁻ and CD8'.sup.+ T cells. Cluster 1 was marked by high expression of CCL3 and CCL4 and was enriched for sgRNAs with strong activation score potentiation, such as VAV1, CD28, and FOXQ1. By contrast, Cluster 12 was enriched for sgRNAs known to activate the NF- κ B pathway, such as IL1RI, TRAF3IP2, TNFRSF1A, and TNFRSF1B. These observations suggest that potentiated stimulation/costimulation may drive T cells to an activated IFNG expressing state distinct from more specific signaling through the NF- κ B pathway. Activation of a subset of TNFRSF receptor genes (TNFRSF1A, TNFRSF1B, LTBR, and CD27), also promoted cell states (clusters 5 and 6) marked by the high expression of cell-cycle genes. LTBR and CD27 sgRNAs were almost exclusively found in cells of this cluster, whereas TNFRSF1A/B sgRNAs appeared to push cells to both proliferative and FNG-expressing states. Thus, CRISPRa Perturb seq reveals how regulators of cytokine production both tune T cell activation and program cells into different stimulation-responsive states,

Discussion

[0288] Paired CRISPRa and CRISPRi screens complement one another to decode the genetic programs regulating stimulation-responsive cytokine production in primary human T cells. CRISPRi identified required cytokine regulators, whereas CRISPRa uncovered key signaling bottlenecks in pathway function as well as regulators that are not necessarily active in ex vivo-cultured T cells. Future screens performed in various other experimental conditions have potential to identify additional regulators of T cell states and functions.

[0289] The technologies developed in this study enable screening approaches in primary human T cells and other primary cell types, such as screens for functional noncoding regions of the human genome (18, 38, 39). Furthermore, this screening framework is adaptable to other non-heritable editing applications of the CRISPR toolkit (40). continuing to expand opportunities to interrogate complex biological questions in primary cells, especially when CRISPR perturbations are coupled with single-cell analyses.

BIBLIOGRAPHY

[0290] 1. A. K. Abbas, E. Trotta, D. R Simeonov, A. Marson, J. A. Bluestone, Revisiting IL-2: Biology and therapeutic prospects. *Sci Immunol.* 3 (2018), doi:10.1126/sciimmunol.aat1482. [0291] 2. L. Ni, J. Lu, Interferon gamma in cancer immunotherapy. *Cancer Med.* 7, 4509-4516 (2018). [0292] 3. F. Castro, A. P. Cardoso, R. M. Giongalves, K. Serre, M. J. Oliveira, Interferon-Gamma at the Crossroads of Tumor Immune Surveillance or Evasion. *Front. Immunol.* 9, 847 (2018). [0293] 4. L. B. Ivashkiv, IFN γ : signalling, epigenetics and roles in immunity, metabolism, disease and cancer immunotherapy. *Nat. Rev. Immunol.* 18, 545-558 (2018). [0294] 5. T. R. Malek, The biology of interleukin-2. *Annu. Rev. Immunol.* 26, 453-479 (2008). [0295] 6. D. A. Boardman, M. K. Levings, Cancer immunotherapies repurposed for use in autoimmunity. *Nat Biomed Eng.* 3, 259-263 (2019). [0296] 7. J. Gao, L. Z. Shi, H. Zhao, J. Chen, L. Xiong, Q. He, T. Chen, J. Roszik, C. Bernatchez, S. E. Woodman, P.-L. Chen, P. Hwu, J. P. Allison, A. Futreal, J. A. Wargo, P. Sharma, Loss of IFN- γ Pathway Genes in Tumor Cells as a Mechanism of Resistance to Anti-

CTLA-4 Therapy. *Cell* 167, 397-404.e9 (2016). [0297] 8. J. M. Zaretsky, A. Garcia-Diaz, D. S. Shin, H. Escuin-Ordinas, W. Hugo, S. Hu-Lieskovan, D. Y. Torrejon, G. Abril-Rodriguez, S. Sandoval, L. Barthly, J. Saco, B. Horne Moreno, R. Mezzadra, B. Chmielowski, K. Ruchalski, 1 P. Shintakur, P. J. Sanchez, C. Puig-Saus, C. Cherry, E. Seja, X Kong, J. Pang, B. Berent-Maoz, B. Comin-Anduix, T. G. Graeber, P. C. Tumeh, T. N. M. Schumacher, R. S. Lo, A. Ribas, Mutations Associated with Acquired Resistance to PD-1 Blockade in Melanoma. *N. Engl. J. Med.* 375, 819-829 (2016). [0298] 9. M. Ayers, J. Lunceford, M. Nebozhyn, E. Murphy, A. Loboda, D. R. Kaufman, A. Albright, J. D. Cheng, S. P. Kang, V. Shankaran, S. A. Piha-Paul, J. Yearley, T. Y. Seiwert, A. Ribas, T. K. McClanahan, IFN- γ -related mRNA profile predicts clinical response to PD-1 blockade. *J. Clin. Invest.* 127, 2930-2940 (2017). [0299] 10. R. R. Bartelt, N. Cruz-Orcutt, M. Collins, J. C. D. Houtman, Comparison of T cell receptor induced proximal signaling and downstream functions in immortalized and primary T cells. *PLoS One*. 4, e5430 (2009). [0300] 11. H. Colin-York, S. Kumari, L. Barbieri, L. Cords, M. Fritzsche, Distinct actin cytoskeleton behaviour in primary and immortalised T-cells. *J. Cell Sci.* 133 (2019), doi:10.1242/jcs.232322. [0301] 12. F. Astoul, C. Edmunds, D. A. Cantrell, S. G. Ward, PI 3-K and T-cell activation: limitations of T-leukemic cell lines as signaling models. *Trends Immunol.* 22, 490-496 (2001). [0302] 13. & Parnas, M. Jovanovic, T. M. Fisenhaure, R. H. Herbst, A. Dixit, C. J. Ye, D. Przybylski, R. J. Platt, 1. Tirosh, N. F. Sanjana, O. Shalem, R. Satija, R. Raychowdhury, P. Mertins, S. A. Carr, F. Zhang, N. Hacohen, A. Regev, A Genome-wide CRISPR Screen in Primary Immune Cells to Dissect Regulatory Networks. *Cell*. 162, 675-686 (2015). [0303] 14. M. B. Dong, G. Wang, R. D. Chow, L. Ye, L. Zhu, X. Dai, J. J. Park, H. R. Kim, Y. Errami, C. D. Guzman, X. Zhou, K. Y. Chen, P. A. Renauer, Y. Du, J. Shen, S. Z. Lam, J. J. Zhou, D. R. Lannin, R. S. Herbst, S. Chen, Systematic immunotherapy Target Discovery Using Genome Scale In Vivo CRISPR Screens in CD8 T Cells. *Cell*, 178, 1189-1204.e23 (2019). [0304] 15. J. Henriksson, X. Chen, T. Comes, U. Ullah, K. B. Meyer, R. hiragaia, G. Duddy, J. Pramanik, K. Yusa., R. Lahesmaa, S. A. Teichmann, Genome-wide CRISPR_Screens in T Helper Cells Reveal Pervasive Crosstalk between Activation and Differentiation. *Cell*. 176, 882-896.ei 8 (2019). [0305] 16. E. Shifrut, J. Carnevale, V. Tobin, T. L. Roth, J. M. Woo, C. T. Biui, P. J. Li, M. E. Diolaiti, A. Ashworth, A. Marson, Genome-wide CRISPR Screens in Primary Human T Cells Reveal Key Regulators of Immune Function. *Cell*. 175, 1958-1971.e15 (2018). [0306] 17. P. Y. Ting, A. E. Parker, J. S. Lee, C. Trussell, O. Sharif, F. Luna, G. Federe, S. W. Barnes, J. R. Walker, J. Vance, M.-Y. Gao, H. E. Klock, S. Clarkson, C. Russ, L. J. Miraglia, M. P. Cooke, A. E. Boitano, P. McNamara, J. Lamb, C. Schmedt, J. L. Snead, Guide Swap enables genome-scale pooled CRISPR-Cas9 screening in human primary cells. *Nat. Methods*. 15, 941-946 (2018). [0307] 18. D. R. Sirneonov, B. G. Gowen, M. Boontanart, T. L. Roth, J. D. Gagnon, M. R. Mumbach, A. T. Satpathy, Y. Lee, N. L. Bray, A. Y. Chan, D. S. Lituiev, M. L. Nguyen, R. E. Gate, M. Subramaniam, Z. Li, J. M. Woo, T. Mitros, (G. J. Ray, G. L. Curie, N. Naddaf, J. S. Chu, H. Ma, E. Boyer, F. Van Gool, H. Huang, R. Liu, V. R. Tobin, K. Schumann, M. J. Daly, K. K. Farh, K. M. Ansel, C. J. Ye, W. J. Greenleaf, M. S. Anderson, J. A. Bluestone, H. Y. Chang, J. E. Corn, A. Marson, Discovery of stimulation-responsive immune enhancers with CRISPR activation. *Nature*. 549, 111-115 (2017). [0308] 19. Y. Liu, C. Yu, T. P. Daley, F. Wang, W. S. Cao, S. Bhate, X. Lin, C. Still 2nd, H. Liu, D. Zhao, H. Wang, X. S. Xie, S. Ding, W. H. Wong, M. Wernig, L. S. Qi, CRISPR Activation Screens Systematically Identify Factors that Drive Neuronal Fate and Reprogramming. *Cell Stem Cell*. 23, 758-771.e8 (2018). [0309] 20. X. Chen, L. Kozhaya, C. Tastan, L. Placek, M. Dogan, M. Horne, R. Abblett, E. Karhan, M. Vaeth, S. Feske, D. Unutmaz, Functional Interrogation of Primary Human T Cells via CRISPR Genetic Editing. *J. Immunol.* 201, 1586-1598 (2018). [0310] 21. R. Nasrallah, C. I. Imnianowski, L. Bossini-Castillo, F. M. Grant, M. Dogan, L. Placek, L. Kozhaya, P. Kuo, F. Sadiyah, S. K. Whiteside, M. R. Mumbach, D. Glinos, P. Vardaka, C. E. Whyte, T. Lozano, T. Fujita, H. Fujii, A. Liston, S. Andrews, A. Cozzani, I. Yang, S. Mitra, E. Lugli, H. Y. Chang, D. Unutmaz, G. Trynka, R. Roychoudhuri, A distal enhancer at risk locus I1 g3.5 promotes suppression of colitis

by Treg cells. *Nature*. 583, 447-452 (2020). [0311] 22. K. R. Sanson, R. E. Hanna, M. Hegde, K. F. Donovan, C. Strand, M. E. Sullender, E. W. Vaimberg, A. Goodale, D. E. Root, F. Piccioni, J. G. Doench, Optimized libraries for CRISPR Cas9 genetic screens with multiple modalities. *Nat. Commun.* 9, 1-15 (2018). [0312] 23. S. Konermann, M. D. Brigham, A. E. Trevino, J. Joung, O. O. Abudayyeh, C. Barcena, P. D. Hsu, N. Habib, J. S. Gootenberg, H. Nishimasu, O. Nureki, F. Zhang, Genome-scale transcriptional activation by an engineered CRISPR-Cas9 complex. *Nature*. 517, 583-588 (2014). [0313] 24. S. M. Kaech, W. Cui, Transcriptional control of effector and memory CD8⁺ T cell differentiation. *Nat. Rev. Immunol.* 12, 749-761 (2012). [0314] 25. J. Zhu, H. Yamane, W. E. Paul, Differentiation of effector CD4 T cell populations (*). *Annu. Rev. Immunol.* 28, 445-489 (2010). [0315] 26. V. Lazarevic, L. H. Glimcher, G. M. Lord, T-bet: a bridge between innate and adaptive immunity. *Nat. Rev. Immunol.* 13, 777-789 (2013). [0316] 27. C. M. Evans, R. G. Jenner, Transcription factor interplay in T helper cell differentiation. *Brief. Funct. Genomics.* 12, 499-511 (2013). [0317] 28. A. H. Courtney, W.-L. Lo, A. Weiss, TCR Signaling: Mechanisms of Initiation and Propagation. *Trends Biochem. Sci.* 43, 108-123 (2018). [0318] 29. G. Gaud, R. Lesourne, P. E. Love, Regulatory mechanisms in T cell receptor signalling. *Nat. Rev. Immunol.* 18, 485-497 (2018). [0319] 30. S. J. Holland, X. C. Liao, M. K. Mendenhall, X. Zhou, J. Pardo, P. Chu, C. Spencer, A. Fu, N. Sheng, P. Yu, E. Pali, A. Nagin, M. Shen, S. Yu, E. Chan, X. Wu, C. Li, M. Woisetschlager, G. Aversa, F. Kolbinger, M. K. Bennett, S. Molineaux, Y. Luo, D. G. [0320] Payan, H. S. Mancebo, J. Wi, Functional cloning of Src-like adapter protein-2 (SLAP-2), a novel inhibitor of antigen receptor signaling. *J. Exp. Med.* 194, 1263-1276 (2001). [0321] 31. J.-W. Shui, J. S. Boomer, J. Han, J. Xu, G. A. Dement, G. Zhou, T.-H. Tan, Hematopoietic progenitor kinase I negatively regulates T cell receptor signaling and T cell-mediated immune responses. *Nat. Immunol.* 8, 84-91 (2006). [0322] 32. T. Hart, A. H. Y. Tong, K.-. Chan, J. Van Leeuwen, A. Seetharaman, M. Aregger, M. Chandrashekar, N. Hustedt, S. Seth, A. Noonan, A. Habsid, O. Sizova, L. Nedyalkova, R. Climie, L. Tworzyanski, K. Lawson, M. A. Sartori, S. Alibeh, D. Tieu, S. Masud, P. Mero, A. Weiss, K. R. Brown, M. Ulsaj, M. Billmann, M. Rahman, M. Constanzo, C. L. [0323] Myers, B. I. Andrews, C. Boone, D. Durocher, J. Moffat, Evaluation and Design of Genome-Wide CRISPR/SpCas9 Knockout Screens. *Genome Biol.* 18, 2719-2727 (2017). [0324] 33. S. R. Ferdosi, R. Ewaisha, F. Moghadam, S. Krishna, J. (G. Park, M. R. Ebrahimkhani, S. Kiani, K. S. Anderson, Multifunctional CRISPR-Cas9 with engineered immunosilenced human T cell epitopes. *Nat. Commun.* 10, 1842 (2019). [0325] 34. H. Hu, H. Wang, Y. Xiao, J. Jin, J.-H. Chang, Q. Zou, X. Xie, X. Cheng, S.-C. Sun, Otd7b facilitates T cell activation and inflammatory responses by regulating Zap70 ubiquitination. *J. Exp. Med.* 213, 399-414 (2016). [0326] 35. E. P. Mirnitou, A. Cheng, A. Montalbano, S. Hao, M. Stoeckius, M. Legut, T. Roush, A. Herrera, E. Papalexi, Z. Ouyang, R. Satija, N. E. Sanjana, S. B. Koralov, P. Smibert, Multiplexed detection of proteins, transcriptomes, clonotypes and CRISPR perturbations in single cells. *Nat. Methods*. 16, 409-412 (2019). [0327] 36. C. Alda-Catalinas, D. Bredikhin, I. Hernando-Herraez, F. Santos, O. Kubinyecz, M. A. Eckersley-Maslin, O. Stegle, W. Reik, A Single-Cell Transcriptomics CRISPR-Activation Screen Identifies Epigenetic Regulators of the Zygotic Genome Activation Program. *Cell Syst.* 11, 25-41.e9 (2020). [0328] 37. J. M. Replogle, T. M. Norman, A. Xu, J. A. Hussmann, J. Chen, J. Z. Cogan, E. J. Meer, J. M. Terry, D. P. Riordan, N. Srinivas, I. T. Fiddes, J. G. Arthur, L. J. Alvarado, K. A. Pfeiffer, T. S. Mikkelsen, J. S. Weissman, B. Adamson, Combinatorial single-cell CRISPR screens by direct guide RNA capture and targeted sequencing. *Nat. Biotechnol.* 38, 954-961 (2020). [0329] 38. P. I. Thakore, A. M. D'Ippolito, L. Song, A. Safi, N. K. Shivakumar, A. M. Kabadi, T. E. Reddy, G. E. Crawford, C. A. Gersbach, Highly specific epigenome editing by CRISPR-Cas9 repressors for silencing of distal regulatory elements. *Nat. Methods*. 12, 1143-1149 (2015). [0330] 39. C. P. Fulco, M. Munschauer, R. Anyoha, G. Munson, S. R. Grossman, E. M. Perez, M. Kane, B. Cleary, E. S. Lander, J. M. Engreitz, Systematic mapping of functional enhancer-promoter connections with CRISPR interference. *Science*. 354, 769-773 (2016). [0331] 40. X. Xu, L. S. Qi,

A CRISPR-dCas ToolBox for Genetic Engineering and Synthetic Biology. *J. Mol. Biol.* 431, 34-47 (2019). [0332] 41. 1. Chun, K. H. Kim, Y H. Chiang, W. Xie, Y. C. G. Lee, R. Pajarillo, A. Rotolo, O. Shestova, S. J. [long, M. Abdel-Mohsen, M. Wysocka, H. J. Ballard, D. M. Barrett, A. D. Posey, D. Powell Jr, S. i Gill, S. J. Schuster, S. K. Barta, A. H. Rook, C. H. June, M. Ruella, CRISPR-Cas9 knock out of CD5 enhances the anti-tumor activity of chimeric Antigen Receptor T cells. *Blood.* 136, 51-52 (2020). [0333] 42 R. C. Lynn, E. W. Weber, E. Sotillo, D. Gennert, P. Xu, Z. Good, H. Anbunathan, J. Lattin, R. Jones, V. Tieu, S. Nagaraja, J. Granja, C. F. A. de Bourcy, R. Majzner, A. T. Satpathy, S. R. Quake, M]Vf Monje, H. Y. Chang, C. L. Mackall, c-Jun overexpression in CAR T cells induces exhaustion resistance. *Nature.* 576, 293-300 (2019). [0334] 43. P. Datlinger, A. F. Rendeiro, C. Schmidl, T. Krausgruber, P. Traxler, J. Klugharnmer, L. C. Schuster, A. Kuchler, D. Alpar, C. Bock, Pooled CRISPR screening with single-cell transcriptome readout. *Nat. Methods.* 14, 297-301 (2017). [0335] 44. J. W. Freimer, O. Shaked, S. Naqvi, N. Sinnott-Armstrong, A. Kathiria, A. F. Chen, O.1. T. Cortez, W. J. Greenleaf, J. K. Pritchard, A. Marson, Systematic discovery and perturbation of regulatory genes in human T cells reveals the architecture of immune networks. *bioRxiv* (2021), p. 2021.04.18.440363. [0336] 45. W. Li, H. Xu, T. Xiao, L. Cong, M. I. Love, F. Zhang, R. A. Irizarry, J. S. Liu, M. Brown, X. S. Liu, MAGeCK enables robust identification of essential genes from genome-scale CRISPR/Cas9 knockout screens. *Genome Biol.* 15, 554 (2014). [0337] 46. G. Korotkevich, V. Sukhov, N. Budin, B. Shpak, M. N. Artyomov, A. Sergushichev, Fast gene set enrichment analysis. *bioRxiv* (2016), , doi:10.1101/060012. [0338] 47. H. K. Finucane, B. Bulik-Sullivan, A. Gusev, Ci. Trynka, Y. Reshef, P.-R. Loh, V. Anttila, H. Xu, C. Zang, K. Farr, S. Ripke, F. R. Day, ReproGen Consortium, Schizophrenia Working Group of the Psychiatric Genomics Consortium, RACI Consortium, S. Purcell, E. Stahl, S. Lindstrom, J. R. B. Perry, Y. Okada, S. Raychaudhuri, M. J. Daly, N. Patterson, B. M. Neale, A. L. Price, Partitioning heritability by functional annotation using genome-wide association summary statistics. *Nat. Genet.* 47, 1228-1235 (2015). [0339] 48. M. A. Horlbeck, L. A. Gilhert, J. E. Villalta, B. Adamson, R. A. Pak, Y. C. Then, A. P. Fields, C. Y. Park, J. E. Corn, M. Kampmann, J. S. Weissman, Compact and highly active next-generation libraries for CRISPR-mediated gene repression and activation. *Elife.* 5 (2016), doi:10.7554/eLife.19760. [0340] 49. M. Martin, Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal.* 17, 10-12 (2011). [0341] 50. A. Dobin, C. A. Davis, F. Schlesinger, J. Drenkow, C. Zaleski, S. Jha, P. Batut, M. Chaisson, T. R. Gingeras, STAR: ultrafast universal RNA-seq aligner. *Bioinformatics.* 29, 15-21 (2013). [0342] 51. Y. Li, G. K. Smyth, W. Shi, featureCounts: an efficient general-purpose program for assigning sequence reads to genomic features. *Bioinformatics.* 30, 923-930 (2014). [0343] 52. M. E. Ritchie, B. Phipson, D. Wu, Y. Hu, C. W. Law, W. Shi, G. K. Smyth, limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* 43, e47 (2015). [0344] 53. H. Heaton, A. M. Talman, A. Knights, M. Imaz, D. J. Gaffney, R. Durbin, M. Henberg, M. K. N. Lawnczak, Souporecell: robust clustering of single-cell RNA-seq data by genotype without reference genotypes. *Nat. Methods.* 17, 615-620 (2020). [0345] 54. R. Satija, J. A. Farrell, D. Gennert, A. F. Schier, A. Regev, Spatial reconstruction of single-cell gene expression data. *Nat. Biotechnol.* 33, 495-502 (2015). [0346] 55. C. Hafemeister, R. Satija, Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression. *Genome Biol.* 20, 296 (2019). [0347] 56. Z. Steinhart, Code repository for CRISPR activation and interference screens decode stimulation responses in primary human T cells, Zenodo (2022); 10.5281/zenodo.578465 [0348] 57. B. J. Schmiedel, D. Singh, A. Madrigal, A. G. Valdovino-Gonzalez, B. M. White, J. Zapardiel Gonzalo, B. Ha, G. Altay, J. A. Greenbaum, C. G. McVicker, G. Seumois, A. Rao, M. Kronenberg, B. Peters, P. Vijayanand, Impact of Genetic Polymorphisms on Human Immune Cell Gene Expression, *Cell.* 175, 1701-1715.e16 (2018). [0349] 58. Immunological Genome Project, ImmGen at 15. *Nat. Immunol.* 21, 700-703 (2020), [0350] 59. B. P. Nicolet, A. Guislain, F. P. J. van Alphen, R. Gomez-Eerland, T. N. M. Schumacher, M. van den Biggelaar, M. C. Wolkers, CD29

identifies IFN- γ -producing human CD8⁺ T cells with an increased cytotoxic potential. Proc. Natl. Acad. Sci. U.S.A. 117, 6686-6696 (2020).

Example 3

[0351] In vitro data using the identified hits for T cell cancer therapies. For this assay, T cells from two human blood donors were virally transduced with the 1G4 anti-cancer T cell receptor as well as the respective gene from the CRISPRa screens (or “empty” virus as control) and cocultured with NYESO expressing A375 melanoma cells. A live imaging system recorded the cancer cell counts every 4h. T cells transduced with the target genes VAV1, PIK3API and CD27 showed enhanced cancer killing (FIG. 5).

[0352] All patents and publications referenced or mentioned herein are indicative of the levels of skill of those skilled in the art to which the invention pertains, and each such referenced patent or publication is hereby specifically incorporated by reference to the same extent as if it had been incorporated by reference in its entirety individually or set forth herein in its entirety. Applicants reserve the right to physically incorporate into this specification any and all materials and information from any such cited patents or publications.

[0353] The following statements are intended to describe and summarize various embodiments of the invention according to the foregoing description in the specification.

Statements:

[0354] 1. A method comprising contacting one or more test agents with one or more T cells to form an assay mixture, and detecting or quantifying interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof in the assay mixture or within one or more T cells, to generate a detected or quantified level of interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof. [0355] 2. The method of statement 1, further comprising comparing the detected or quantified level of interferon- γ production, the detected or quantified level of interleukin-2 production, the detected or quantified level of cellular proliferation, or a combination thereof with a control. [0356] 3. The method of statement 1 or 2, further comprising measuring the quantity of one or more of the regulators listed in Tables 1-7 or FIGS. 1-4 in the assay mixture or in one or more of the T cells. [0357] 4. The method of statement 1, 2, or 3, wherein one or more of the T cells initially contacted with the test agent naturally expresses any of the regulators listed in Tables 1-7 or FIGS. 1-4. [0358] 5. The method of statement 1-3 or 4, wherein one or more of the T cells initially contacted with the test agent do not express one or more of the regulators listed in any of Tables 1-7 or FIGS. 1-4. [0359] 6. The method of statement 1-4 or 5, wherein one or more of T cells initially contacted with the test agent have the potential to express one or more of the regulators but when initially mixed with a test agent the cells do not express detectable amounts of one or more of the regulators. [0360] 7. The method of statement 1-5 or 6, wherein at least one of the T cells is a mutant T cell comprising a knock-down or knockout mutation that reduces expression or activity of one or more of the regulators listed in any of Tables 1-7 or FIGS. 1-4. [0361] 8. The method of statement 7, further comprising: modifying the one or more mutant T cells to express or over-express the one or more regulators listed in Tables 1-7 or FIGS. 1-4; and detecting or quantifying interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof in a second assay mixture or within the one or more of the mutant T cells. [0362] 9. The method of statement 1-7 or 8, wherein the one or more of the T cells is in a population of T cells. [0363] 10. The method of statement 1-8 or 9, wherein one or more of T cells comprises one or more cytotoxic T cells, chimeric antigen receptor T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof, [0364] 11. The method of statement 1-9 or 10, further comprising adding at least one second cell type to the assay mixture before detecting or quantifying the interferon- γ production, interleukin-2 production, cellular proliferation, or a combination thereof. [0365] 12. The method of statement 11, wherein the second cell type is one or

more types of cancer cells, one or more types of immune cells, or a combination thereof. [0366] 13. The method of statement 12, wherein one or more of the cancer cells comprise leukemia cells, lymphoma cells, Hodgkin's disease cells, sarcomas of the soft tissue and bone, lung cancer cells, mesothelioma, esophagus cancer cells, stomach cancer cells, pancreatic cancer cells, hepatobiliary cancer cells, small intestinal cancer cells, colon cancer cells, colorectal cancer cells, rectum cancer cells, kidney cancer cells, urethral cancer cells, bladder cancer cells, prostate cancer cells, testis cancer cells, cervical cancer cells, ovarian cancer cells, breast cancer cells, endocrine system cancer cells, skin cancer cells, central nervous system cancer cells, melanoma cells of cutaneous and/or intraocular origin, cancer cells associated with AIDS, or a combination thereof. [0367] 14. The method of statement 12 or 13, wherein one or more of cancer cells comprise metastatic cancer cells. [0368] 15. The method of statement 12, 13 or 14, wherein one or more of cancer cells comprise micrometastatic tumor cells, megametastatic tumor cells, recurrent cancer cells, or a combination thereof. [0369] 16. The method of statement 12-14 or 15, wherein one or more of the immune cells comprises macrophages, natural killer cells, dendritic cells, B cells, chimeric antigen receptor cells, cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune (e.g., lymphoid and/or myeloid) cells, or a combination thereof. [0370] 17. The method of statement 12-15 or 16, further comprising measuring the cellular proliferation of at least one of the second cell types. [0371] 18. The method of statement 1-16 or 17, further comprising identifying one or more of the test agents that modulates the level of interferon- γ production, the level of interleukin-2 production, the level of cellular proliferation, or a combination thereof of one or more of the T cells, to thereby identify one or more useful test agents. [0372] 19. The method of statement 1-17 or 18, further comprising identifying one or more of the test agents that modulates expression or activity of one or more of the regulators listed in any of Tables 1-7 or FIGS. **1-4**, to thereby identify one or more useful test agents. [0373] 20. The method of statement 1 9, further comprising measuring binding of one or more useful test agents to a protein encoded by, or nucleic acid comprising, one or more the regulators listed in any of Tables 1-7 or FIGS. **1-4**. [0374] 21. The method of statement 19 or 20, further comprising administering one or more useful test agents to one or more experimental animals. [0375] 22. The method of statement 21, wherein one or more of the experimental animals has a disease, infection, or medical condition. [0376] 23. The method of statement 22, wherein the disease or condition is cancer, an immune disorder, or an immune condition. [0377] 24. The method of statement 23, wherein the immune disorder or immune condition is an autoimmune disorder, Graves disease, arthritis, psoriasis, Celiac disease, vitiligo, rheumatoid arthritis, lupus, Crohn's disease, multiple sclerosis, type 1 diabetes, alopecia, inflammatory bowel disease (IBD), Guillain-Barre syndrome, chronic inflammatory demyelinating polyneuropathy, or a combination thereof. [0378] 25. The method of statement 21-23 or 24, further comprising monitoring the one or more experimental animals for symptoms of the disease or condition, for toxic side effects of the useful test agents, or a combination thereof. [0379] 26. The method of statement 21-24 or 25, further comprising monitoring immune cell numbers and/or types in the one or more experimental animals. [0380] 27. The method of statement 22-25 or 26, further comprising identifying one or more useful test agents as a therapeutic agent useful for treatment of the disease or condition. [0381] 28. A composition comprising a useful test agent or a therapeutic agent identified by the method of any of claims **1-27**. [0382] 29. A method comprising ex vivo modification of any of the genes listed in Tables 1-7 or FIGS. **1-4** within at least one lymphoid or myeloid cell, or a combination thereof to generate at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells. [0383] 30. The method of statement 29, wherein the modification is one or more deletion, substitution or insertion into one or more genomic sites of any of the genes listed in Tables 1-7 or FIGS. **1-4**. [0384] 31. The method of statement 29 or 30, wherein the modification is one or more CRISPR-mediated modifications or activations of any of

the genes listed in Tables 1-7 or FIGS. 1-4. [0385] 32. The method of statement 29, 30 or 31, further comprising administering at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to a subject. [0386] 33. The method of statement 29, 30 or 31 further comprising incubating the at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to form a population of modified cells. [0387] 34. The method of statement 33, further comprising administering the population of modified cells to a subject. [0388] 35. The method of statement 32 or 34, wherein the subject has a disease or condition. [0389] 36. The method of statement 35, wherein the disease or condition is an immune condition or cancer. [0390] The specific methods and compositions described herein are representative of preferred embodiments and are exemplary and not intended as limitations on the scope of the invention. Other objects, aspects, and embodiments will occur to those skilled in the art upon consideration of this specification and are encompassed within the spirit of the invention as defined by the scope of the claims. It will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention.

[0391] The invention illustratively described herein suitably may be practiced in the absence of any element or elements, or limitation or limitations, which is not specifically disclosed herein as essential. The methods and processes illustratively described herein suitably may be practiced in differing orders of steps, and the methods and processes are not necessarily restricted to the orders of steps indicated herein or in the claims.

[0392] As used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “a nucleic acid” or “a protein” or “a cell” includes a plurality of such nucleic acids, proteins, or cells (for example, a solution or dried preparation of nucleic acids or expression cassettes, a solution of proteins, or a population of cells), and so forth. In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated.

[0393] Under no circumstances may the patent be interpreted to be limited to the specific examples or embodiments or methods specifically disclosed herein. Under no circumstances may the patent be interpreted to be limited by any statement made by any Examiner or any other official or employee of the Patent and Trademark Office unless such statement is specifically and without qualification or reservation expressly adopted in a responsive writing by Applicants.

[0394] The terms and expressions that have been employed are used as terms of description and not of limitation, and there is no intent in the use of such terms and expressions to exclude any equivalent of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention as claimed. Thus, it will be understood that although the present invention has been specifically disclosed by preferred embodiments and optional features, modification and variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modified cations and variations are considered to be within the scope of this invention as defined by the appended claims and statements of the invention.

[0395] The invention has been described broadly and generically herein, Each of the narrower species and subgeneric groupings falling within the generic disclosure also form part of the invention. This includes the generic description of the invention with a proviso or negative limitation removing any subject matter from the genus, regardless of whether or not the excised material is specifically recited herein. In addition, where features or aspects of the invention are described in terms of Markush groups, those skilled in the art will recognize that the invention is also thereby described in terms of any individual member or subgroup of members of the Markush group.

Claims

- 1.** A method comprising ex vivo modification of any of the genes listed in Tables 1-7 or FIGS. **1-4** within at least one lymphoid or myeloid cell, or a combination thereof to generate at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells.
- 2.** The method of claim 1, wherein the modification is one or more deletion, substitution or insertion into one or more endogenous genomic sites of any of the genes listed in Tables 1-7 or FIGS. **1-4**.
- 3.** The method of claim 1, wherein the modification is reduction of expression or translation of any of the genes listed in Tables 1-7 or FIGS. **1-4**.
- 4.** The method of claim 3, wherein the reduction of expression or translation is by an inhibitory nucleic acid.
- 5.** The method of claim 1, wherein the modification is increased expression of any of the genes listed in Tables 1-7 or FIGS. **1-4**.
- 6.** The method of claim 5, wherein the increased expression is by modification of one or more promoters of any of the genes listed in Tables 1-7 or FIGS. **1-4**.
- 7.** The method of claim 1, wherein the modification is one or more CRISPR-mediated modifications or activations of any of the genes listed in Tables 1-7 or FIGS. **1-4**.
- 8.** The method of claim 1, wherein the modification is transformation of at least one lymphoid or myeloid cell, or a combination thereof, with one or more expression cassettes comprising a promoter operably linked to a nucleic acid segment comprising a coding region of any of the genes listed in Tables 1-7 or FIGS. **1-4**.
- 9.** The method of claim 1, further comprising administering at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to a subject.
- 10.** The method of claim 1, further comprising incubating the at least one modified lymphoid cell, at least one modified myeloid cell, or a mixture of modified lymphoid and modified myeloid cells to form a population of modified cells.
- 11.** The method of claim 10, further comprising administering the population of modified cells to a subject.
- 12.** The method of claim 9, wherein the subject has a disease or condition.
- 13.** The method of claim 12, wherein the disease or condition is an immune condition or cancer.
- 14.** A method comprising contacting at least one test agent with test cells to provide a test assay mixture, and measuring: a. cellular proliferation of the test cells, cytokine release by the test cells, or a combination thereof; b. activation of the test cells; c. expression or activity of any of the regulators listed in Tables 1-7 or FIGS. **1-4** in the cells; or d. a combination thereof.
- 15.** The method of claim 14, further comprising comparing the measured results to control results.
- 16.** The method of claim 15, wherein control results are results of the test cells measured without any of the test agents.
- 17.** The method of claim 14, wherein the test cells comprise lymphoid and/or myeloid cells.
- 18.** The method of claim 14, wherein the test cells comprise cytotoxic T cells, helper T cells, regulatory T cells, naive T cells, activated T cells, CD4 T cells, CD8 T cells, gamma delta T cells, chimeric antigen receptor (CAR) cells, natural killer (NK) cells, induced pluripotent stem cell-derived immune cells, or a combination thereof.
- 19.** The method of claim 13, wherein the immune condition is an autoimmune disorder, Graves disease, arthritis, psoriasis, Celiac disease, vitiligo, rheumatoid arthritis, lupus, Crohn's disease, multiple sclerosis, type 1 diabetes, alopecia, inflammatory bowel disease (IBD), Guillain-Barre syndrome, chronic inflammatory demyelinating polyneuropathy, or a combination thereof.

20. The method of claim 13, wherein the cancer is leukemia, lymphoma, Hodgkin's disease, sarcomas of the soft tissue and bone, lung cancer, mesothelioma, esophagus cancer, stomach cancer, pancreatic cancer, hepatobiliary cancer, small intestinal cancer, colon cancer, colorectal cancer, rectum cancer, kidney cancer, urethral cancer, bladder cancer, prostate cancer, testis cancer, cervical cancer, ovarian cancer, breast cancer, endocrine system cancer, skin cancer, central nervous system cancer, melanoma, cancer associated with AIDS, or a combination thereof.
